Opening the Black Box of Financial Forecasters: A Semantic Mixture-of-Experts Yardstick for Distributional Prediction

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Abstract

We develop the Semantic Mixture of Experts (SMoE), a white-box model of the conditional distribution of daily returns in which linear-Gaussian experts with log-linear variances are weighted by a multinomial-logit gate in observable state variables, and each expert is restricted to a researcher-specified semantic group of covariates. We establish identifiability under the semantic exclusions, geometric ergodicity of the canonical SMoE autoregression, a monotone ECM algorithm with exact scores, and QMLE sandwich inference; every predictive functional has a closed form (mean, variance split into within-regime and regime-disagreement risk, probability of a rise, VaR, expected shortfall and CRPS), and every forecast an exact additive attribution. We also prove that MSE-trained price-level forecasters must lag: their maximal RMSE gain over the random walk is of order $s^2/2$ for daily drift-to-noise ratio s, i.e. well below one percent. Projecting black-box density forecasters onto the SMoE by KL minimisation yields a *yardstick*: a fidelity ratio, with an exact information-theoretic meaning, of how much of a black box's out-of-sample gain three readable regimes reproduce. On nineteen equity, commodity, bond and crypto series (2005–2026) evaluated under a leak-tested walk-forward protocol, the SMoE has the best average log-score rank among thirteen forecasters, three of them zero-shot foundation models (two sizes of Chronos-Bolt and TimesFM 2.5), and well-calibrated tails (1% VaR passes the Kupiec test on 17 of 19 series), but no model forecasts daily means or directions. A deep heterogeneous mixture-of-experts network is 92% white by the yardstick, and white-box projections of all three foundation models outperform the models themselves; a release-date test finds no sign that the foundation models had memorised the evaluation period. The white box shows that volatility, not momentum or drawdown, drives regime weights in the CSI 300 and that the short-horizon expert exhibits continuation in Chinese and reversal in developed markets. Comparing fourteen specifications, three experts forecast best out of sample, although BIC prefers two and a bootstrapped likelihood ratio four; a score test computed from the baseline fit shows that the VIX belongs in the gate of all six international equity indices and most commodity futures but of only one of the four A-share indices. With 5-minute data, a measurement-error result predicts the attenuation of the range-based daily variance loading (predicted ratio 0.39, estimated 0.40); realised

*Working paper. Code and data: see the replication package. All numbers in this paper are produced by scripts in `experiments/`; none are entered by hand.

measures improve the SMoE, but a single-regime Realized GARCH forecasts best. An adaptive conformal layer guarantees long-run VaR coverage. Trading the CSI 300 ETF on the SMoE predictive distribution cuts the maximum drawdown of buy-and-hold from −46% to −26% (index futures: from −45% to −38%), but no excess return survives deflation for the strategies tried. In a lockbox period from July 2024 to October 2026, evaluated once after all choices were frozen, the mixture models again lead on log score, with the SMoE second of thirteen behind LSTAR (average rank

3.11 against 3.05); GARCH-t matches its tail calibration (1% VaR passes the Kupiec test on 18 of 19 series for both); and there is still no reliable trading alpha.

Keywords: mixture of experts; density forecasting; interpretability; regime switching; conformal prediction; foundation models; CSI 300.
JEL: C22, C53, C58, G17.

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1 Introduction

Deep sequence models, gradient-boosted ensembles and, most recently, time-series foundation models are increasingly used to forecast financial returns. Three problems recur in this literature. The first is *evaluation*: forecasts of price levels are scored by root mean squared error, which a random walk nearly minimises, and directional accuracy is computed in ways that measure properties of the data rather than of the forecaster. The second is *opacity*: when a network does forecast well, it is rarely possible to say what it has learned in terms a risk manager, a regulator or an economist would recognise, let alone test. The third is *leakage*: complex pipelines make it easy for information from the forecast period to enter the forecast, and the resulting gains are both large and fictitious. This paper is a response to all three, and it began as a response to the third: the authors' own undergraduate thesis on a heterogeneous mixture-of-experts network for the CSI 300 contained two look-ahead leaks and several illustrative figures that were not model output (appendix C). Rebuilding that work from first principles produced the framework described here.

We make five contributions.

1. A formal account of the "lagging paradox". We show that the MSE-optimal forecast of tomorrow's price is today's price multiplied by the conditional moment generating function of the return, so any forecaster trained on price levels converges to a lagged random walk (proposition 1). The achievable reduction in price-level MSE relative to the random walk is bounded by $s^2/(1+s^2)$, where s is the daily drift-to-noise ratio (proposition 2); for any plausible s this rules out RMSE gains of more than about half a percent. A common directional-accuracy metric equals return sign persistence whenever the forecaster lags (proposition 3). On nineteen assets a 60-lag price-level regression indeed collapses to the random walk (coefficients summing to 0.994–1.003) and loses to it in every market.

2. A white-box mixture of experts for conditional return distributions. The Semantic Mixture of Experts (SMoE) forecasts the full conditional distribution of the next-day return as a mixture of linear-Gaussian experts with log-linear variances, weighted by a multinomial logit gate in observable state variables. Researchers assign each expert a *semantic group* of covariates (trend, short-horizon, crisis) which acts as an exclusion restriction and identifies the experts (proposition 5). We derive conditions for geometric ergodicity, an ECM algorithm with monotone likelihood, exact scores via Fisher's identity, QMLE asymptotics with sandwich inference, and closed forms for every predictive functional: mean, variance decomposed into within-regime and regime-disagreement risk, probability of a rise, VaR, ES and CRPS. Each forecast admits an exact additive attribution to covariates and experts, and each gate variable an exact elasticity.

3. A yardstick for black boxes. Because the SMoE is a proper likelihood model, any black-box density forecaster can be projected onto it by minimising expected Kullback–Leibler divergence. The resulting *shadow* is a white-box description of what the black box does, and the fidelity ratio Φ (the share of the black box's out-of-sample log-score gain reproduced by

its shadow) has an exact information-theoretic interpretation (proposition 10). We apply the yardstick to a deep heterogeneous MoE (the undergraduate architecture, rebuilt as a mixture density network), to gradient boosting, to a Markov-switching model and to three zero-shot foundation models (two sizes of Chronos-Bolt and TimesFM 2.5). Because a pretrained model may have seen the evaluation period, we also propose a release-date test for look-ahead through pretraining (proposition 16).

4. Specification and measurement tests for the white box. We derive a score test for omitted gate variables that needs only the restricted fit and reads as a test of whether a candidate state variable predicts the regime surprise $\tau_{tk} - \pi_{tk}$ (proposition 14), and we show how measurement error in a daily variance estimator moves weight from the daily to the weekly and monthly components of a log-HAR regression (proposition 15). Both results are put to work: fourteen specifications of the experts and the gate are compared on all nineteen series, and range-based variance covariates are compared with realised measures from 5-minute data, where the predicted attenuation of the daily loading matches the estimate.

5. Evidence from nineteen markets with a coverage guarantee and a deflated economic test. Using a leak-tested walk-forward protocol with a lockbox period, we evaluate thirteen forecasters (including two sizes of Chronos-Bolt and TimesFM 2.5) on Chinese and international equity indices, commodity and bond futures and Bitcoin. The SMoE has the best average log-score rank (2.95), is in the 10% model confidence set for 15 of 19 series on log score and 18 of 19 on volatility and tail losses, and its 1% VaR passes the Kupiec test on 17 of 19 series (GARCH-t: 10). It does not forecast means or directions better than simple benchmarks: the hit rate averages about 51%. Its white box reveals that volatility, not momentum or drawdown, drives regime weights in the CSI 300, and that the short-horizon expert exhibits continuation in Chinese indices but reversal in developed markets. An adaptive conformal layer delivers distribution-free long-run VaR coverage, which combined with the SMoE also removes violation clustering. Three experts forecast better out of sample than two, four or five, and adding the VIX to the gate is the only change of specification that helps; with realised variance from 5-minute data, a single-regime Realized GARCH forecasts the CSI 300 better in log score than any SMoE variant, which we report as a limit of the regime structure. Trading the CSI 300 ETF from the SMoE predictive distribution cuts the maximum drawdown from -46% to -26% (on index futures, from -45% to -38%), but after deflating for the strategies tried no excess return is statistically reliable. In a lockbox period (July 2024 to October 2026) that we evaluated once, after every modelling choice had been frozen, the mixture models again hold the top four log-score ranks among thirteen, with the SMoE second behind LSTAR; GARCH-t catches up on calibration (both pass the 1% Kupiec test on 18 of 19 series); there is still no reliable trading alpha; and the drawdown reduction is smaller (from -19% to -17%).

Organisation. Section 2 reviews related work. Section 3 formalises the lagging paradox. Sections 4 to 6 develop the SMoE, its estimation and its white-box interpretation, and section 7 the calibration and decision layers. Section 8 reports simulations, section 9 the data and

protocol, and sections 10 to 16 the empirical results.

Section 17 sets out a research agenda and section 18 concludes.

2 Related literature

Return predictability and its evaluation. The benchmark for any return forecaster is that simple predictors rarely beat the historical mean out of sample (Welch and Goyal, 2008), although economically meaningful gains can survive when forecasts are constrained (Campbell and Thompson, 2008). Because many candidate forecasters nest the no-predictability benchmark, inference must account for the nested structure (Clark and West, 2007); non-nested comparisons rely on Diebold and Mariano (1995) with the small-sample correction of Harvey et al. (1997), conditional comparisons on Giacomini and White (2006), and sets of models on the model confidence set of Hansen et al. (2011). Machine-learning studies such as Gu et al. (2020) document gains in the cross-section, and Kelly et al. (2024) argue that model complexity can be a virtue in return prediction. We ask a different question: how much of what a complex model predicts can be stated as a small number of readable regime rules that can be tested.

Regime models and mixtures. Markov switching (Hamilton, 1989), smooth transition autoregressions (Teräsvirta, 1994) and mixture autoregressions (Wong and Li, 2000, 2001) represent regime dependence with a handful of parameters; choosing the number of regimes is a non-standard testing problem, addressed by bootstrap likelihood ratios (McLachlan, 1987) and information criteria (Keribin, 2000). Mixtures of experts (Jacobs et al., 1991; Jordan and Jacobs, 1994) replace fixed or latent-Markov weights with weights that depend on observables; Carvalho and Tanner (2005) develop their asymptotic theory for autoregressive time series and Jiang and Tanner (1999) study identifiability. The SMoE belongs to this family. Its distinguishing features are the semantic exclusion restrictions, the log-linear variance experts driven by range-based HAR components (Parkinson, 1980; Corsi, 2009), and its use as a projection target for black-box forecasters.

Neural mixtures and foundation models. Mixture density networks (Bishop, 1994) output mixture parameters from a neural network; sparse neural mixtures of experts (Shazeer et al., 2017; Fedus et al., 2022) scale model capacity; Time-MoE (Shi et al., 2025) brings this architecture to time-series foundation models, alongside Chronos (Ansari et al., 2024) and TimesFM (Das et al., 2024); we evaluate Chronos-Bolt and TimesFM 2.5 zero-shot. Pretrained models raise a new form of look-ahead: their training corpora may contain the evaluation period. Glasserman and Lin (2024) and Sarkar and Vafa (2024) document this for language models used in finance, and Chen et al. (2026) study it for financial time-series foundation models. In finance, deep sequence models such as LSTMs have been applied widely (Fischer and Krauss, 2018; Sezer et al., 2020), often with price-level targets whose pitfalls we formalise in section 3. The undergraduate predecessor of this paper used such an architecture; we retain it as the black-box counterpart ("Deep SMoE") and evaluate it under the same protocol.

Interpretability. Post-hoc attribution (Lundberg and Lee, 2017) and saliency methods can be unreliable (Adebayo et al., 2018), which has led to calls for intrinsically interpretable models in high-stakes settings (Rudin, 2019). Concept bottlenecks (Koh et al., 2020) and attention-based designs (Lim et al., 2021) are partial answers. Distillation of neural networks into interpretable students (Hinton et al., 2015; Frosst and Hinton, 2017) and linear probes of internal representations (Alain and Bengio, 2016) are the tools we adapt. The SMoE is the student and the KL projection is the distillation objective; the fidelity ratio of proposition 10 then has an exact information-theoretic interpretation.

Realised measures. Intraday returns give realised variance (Andersen et al., 2003), its jump-robust bipower counterpart (Barndorff-Nielsen and Shephard, 2004) and realised semivariances (Patton and Sheppard, 2015). HAR regressions of these measures (Corsi, 2009; Andersen et al., 2007; Corsi et al., 2010) and the Realized GARCH model (Hansen et al., 2012) are the standard ways to use them in forecasting. Because squared daily returns are noisy proxies for the latent variance (Andersen and Bollerslev, 1998), the choice of variance measure is also a measurement-error problem, which we analyse in section 13.

Calibration and decisions. Proper scoring rules (Gneiting and Raftery, 2007), PIT-based tests (Berkowitz, 2001), coverage tests (Kupiec, 1995; Christoffersen, 1998), the joint VaR–ES loss of Patton et al. (2019) and robust volatility losses (Patton, 2011) evaluate the distribution. Adaptive conformal inference (Gibbs and Candès, 2021, 2024) provides model-free long-run coverage under distribution shift. On the decision side, volatility-managed portfolios (Moreira and Muir, 2017), the economic value of volatility timing (Fleming et al., 2001), trading with transaction costs (Gârleanu and Pedersen, 2013), and the statistics of backtests (Lo, 2002; Bailey and López de Prado, 2014; Bailey et al., 2017; White, 2000; Hansen, 2005) define how we translate forecasts into positions and how we discount the result for search.

China. The Chinese A-share market is dominated by retail trading, is subject to price limits and frequent policy interventions, and has distinct return anomalies (Liu et al., 2019; Carpenter et al., 2021). These features make regime dependence economically plausible and motivate the policy-event analysis in section 10.

3 The lagging paradox, formally

A recurring observation in applied work on deep price forecasting is that networks trained with squared error on price *levels* produce forecasts that "lag" the realised path: the predicted curve looks like the actual curve shifted one day to the right, and directional accuracy is close to or below one half even when the price-level RMSE looks impressive. We call this the *lagging paradox*. This section traces the lag to the loss function, whatever the architecture. It derives a sharp upper bound on what any price-level forecaster can gain over the random walk and shows that a widely used directional metric measures the sign persistence of returns, which says nothing about forecasting skill. The results motivate the distributional formulation in section 4.

3.1 Setting

Let $P_t > 0$ be the price at the close of day t, $r_{t+1} = \log P_{t+1} - \log P_t$ the next log return and $\mathcal{F}_t$ the information at the close of day t. Write $M_t(s) = \mathbb{E}[e^{sr_{t+1}} \mid \mathcal{F}_t]$ for the conditional moment generating function, assumed finite for $s \in \{1, 2\}$, and $\mu_t = \mathbb{E}[r_{t+1} \mid \mathcal{F}_t]$, $\sigma_t^2 = \mathrm{Var}(r_{t+1} \mid \mathcal{F}_t)$.

3.2 The MSE-optimal price forecast is a near-random walk

Proposition 1 (Optimal level forecast)**.** *Among $\mathcal{F}_t$-measurable forecasts f_t of P_{t+1}, the conditional mean squared error $\mathbb{E}[(P_{t+1} - f_t)^2 \mid \mathcal{F}_t]$ is minimised by $f_t^* = P_t\, M_t(1)$, with minimum value $P_t^2\{M_t(2) - M_t(1)^2\}$. The random-walk forecast $f_t^{\mathrm{RW}} = P_t$ has conditional MSE $P_t^2\{M_t(2) - 2M_t(1) + 1\}$, and*

$$\mathbb{E}[(P_{t+1} - P_t)^2 \mid \mathcal{F}_t] - \mathbb{E}[(P_{t+1} - f_t^*)^2 \mid \mathcal{F}_t] = P_t^2\{M_t(1) - 1\}^2. \tag{1}$$

Under conditional normality $M_t(1) = \exp(\mu_t + \sigma_t^2/2)$, so the optimal forecast moves away from the random walk by the factor $\exp(\mu_t + \sigma_t^2/2) - 1 \approx m_t$, $m_t := \mu_t + \sigma_t^2/2$. For daily equity returns m_t is of order 10^{-4}, so $\mathrm{Cov}(f_t^*, P_t)/\{\mathrm{Var}(f_t^*)\,\mathrm{Var}(P_t)\}^{1/2}$ is indistinguishable from one: *the optimal forecast lags by construction.*

Proposition 2 (Upper bound on price-level gains)**.** *Let $s_t = m_t/\sigma_t$ be the conditional drift-to-noise ratio. Under conditional normality and to second order in (μ_t, σ_t), the relative reduction in conditional MSE that any $\mathcal{F}_t$-measurable forecast can achieve over the random walk satisfies*

$$\frac{\mathbb{E}[(P_{t+1} - P_t)^2 \mid \mathcal{F}_t] - \min_f \mathbb{E}[(P_{t+1} - f)^2 \mid \mathcal{F}_t]}{\mathbb{E}[(P_{t+1} - P_t)^2 \mid \mathcal{F}_t]} \;=\; \frac{s_t^2}{1 + s_t^2} + O(\sigma_t^3). \tag{2}$$

Consequently the unconditional relative RMSE gain is at most $1 - \{1 - \mathbb{E}[w_t s_t^2/(1 + s_t^2)]\}^{1/2} \approx \frac{1}{2}\mathbb{E}[w_t s_t^2]$ with weights $w_t = P_t^2\sigma_t^2/\mathbb{E}[P_t^2\sigma_t^2]$.

A daily drift-to-noise ratio of $|s_t| = 0.1$ (an extreme degree of predictability, corresponding to an annualised conditional Sharpe ratio of about 1.6 on *every* day) bounds the achievable price-RMSE improvement at roughly 0.5%. Reported RMSE improvements of tens of percent on daily index levels are therefore incompatible with any information set available at t and indicate look-ahead in the evaluation. In section 10 we confirm empirically that a 60-lag linear model fitted by least squares to price levels converges to the random walk on all nineteen assets in our sample (sum of coefficients ≈ 1) and loses between 0.6% and 7.8% in RMSE out of sample, as the bound and estimation noise predict (table 27).

3.3 Differenced directional accuracy measures sign persistence

Directional accuracy is often computed from consecutive forecasts as $\mathrm{DA}^{\Delta} = \frac{1}{n}\sum_t \mathbb{1}\{\mathrm{sign}(f_{t+1} - f_t) = \mathrm{sign}(P_{t+1} - P_t)\}$, i.e. comparing the change in the forecast path with the change in the price path.

Proposition 3 (What DA^{Δ} measures)**.** *For the random-walk forecast $f_{t+1} = P_t$, $\mathrm{DA}^{\Delta} = \frac{1}{n}\sum_t \mathbb{1}\{\mathrm{sign}\, r_t = \mathrm{sign}\, r_{t+1}\}$, the empirical sign persistence of returns, whose expectation is*

$\frac{1}{2} + \frac{1}{2}\mathbb{E}[\operatorname{sign}(r_t)\operatorname{sign}(r_{t+1})]$ *when* $P(r_t = 0) = 0$. *For any forecast with* $f_{t+1} = P_t + \varepsilon_{t+1}$ *and* $|\varepsilon_{t+1} - \varepsilon_t| < |r_t|$ *the same identity holds term by term.*

Whenever the forecaster lags, DA^{Δ} is therefore a statistic of the data alone. The skill-relevant quantity compares the forecast with the *current* price, $\mathrm{DA} = \frac{1}{n}\sum_t \mathbb{1}\{\operatorname{sign}(f_{t+1} - P_t) = \operatorname{sign}(r_{t+1})\}$, which is undefined for the random walk and is what we report throughout.

3.4 Why the conditional distribution is the right target

Under 0–1 loss the optimal directional call is $\mathbb{1}\{P(r_{t+1} > 0 \mid \mathcal{F}_t) > 1/2\}$, i.e. the sign of the conditional *median*, not of the mean. For a Gaussian the two coincide; for a mixture they need not.

Example 1. Let $r_{t+1} \mid \mathcal{F}_t \sim 0.8\,\mathcal{N}(0.3, 1) + 0.2\,\mathcal{N}(-1.0, 3^2)$. The mean is $0.04 > 0$, but $P(r_{t+1} > 0) = 0.8\,\Phi(0.3) + 0.2\,\Phi(-1/3) = 0.568$; changing the second component to $\mathcal{N}(-1.5, 3^2)$ gives mean $-0.06 < 0$ while $P(r_{t+1} > 0) = 0.556 > 1/2$. A point forecaster trained by squared error calls "down"; the distributional forecaster calls "up" and also reports that a fifth of the probability mass sits in a high-variance regime.

These results organise the rest of the paper. First, price-level losses cannot reveal predictive skill; the target must be the return. Second, a forecaster should be judged by proper scoring rules for the whole predictive distribution, since direction, volatility and tail risk are functionals of it. Third, if regimes matter, the predictive distribution is naturally a mixture, and a mixture whose components and weights are *readable* is the white-box object we construct next.

4 The Semantic Mixture of Experts

4.1 Definition

Let $y_t \coloneqq r_{t+1}$ denote the return to be forecast at the close of day t (in percent). The researcher partitions the $\mathcal{F}_t$-measurable covariates into *semantic groups*: for each expert $k = 1, \dots, K$ a mean-covariate vector $\boldsymbol{x}_t^{(k)} \in \mathbb{R}^{p_k}$, a common variance-covariate vector $\boldsymbol{v}_t \in \mathbb{R}^q$, and a gate-covariate (state) vector $\boldsymbol{z}_t \in \mathbb{R}^m$. Write $\tilde{\boldsymbol{x}} = (1, \boldsymbol{x}^\top)^\top$ etc.

Definition 1 (SMoE)**.** The Semantic Mixture of Experts specifies

$$y_t \mid \mathcal{F}_t \;\sim\; \sum_{k=1}^{K} \pi_k(\boldsymbol{z}_t)\,\mathcal{N}\big(\mu_{kt},\, \sigma_{kt}^2\big), \qquad \mu_{kt} = \boldsymbol{\beta}_k^\top \tilde{\boldsymbol{x}}_t^{(k)}, \qquad \log \sigma_{kt}^2 = \boldsymbol{\delta}_k^\top \tilde{\boldsymbol{v}}_t, \tag{3}$$

$$\pi_k(\boldsymbol{z}) = \frac{\exp(\boldsymbol{\gamma}_k^\top \tilde{\boldsymbol{z}})}{\sum_{l=1}^{K} \exp(\boldsymbol{\gamma}_l^\top \tilde{\boldsymbol{z}})}, \qquad \boldsymbol{\gamma}_K = \mathbf{0}. \tag{4}$$

The parameter is $\boldsymbol{\theta} = (\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_K, \boldsymbol{\delta}_1, \dots, \boldsymbol{\delta}_K, \boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_{K-1})$, of dimension $\sum_k (1 + p_k) + K(1 + q) + (K - 1)(1 + m)$.

The word *semantic* refers to the exclusion restrictions: expert k's mean depends only on $\boldsymbol{x}^{(k)}$. In our application $K = 3$ and

- **Trend** expert: $\boldsymbol{x}^{(1)} = (\text{mom}_{20}, \text{mom}_{120})$, volatility-scaled cumulative returns over 20 and 120 days, the time-series momentum signals of Moskowitz et al. (2012);
- **Reversal** expert: $\boldsymbol{x}^{(2)} = (\text{rev}_1, \text{rev}_5)$, volatility-scaled 1- and 5-day returns, the short-horizon reversal signals of Lehmann (1990);
- **Crisis** expert: $\boldsymbol{x}^{(3)} = \emptyset$, a regime with its own level and variance.

Variance covariates are the logarithms of the 1-, 5- and 22-day averages of the Parkinson (1980) range variance, i.e. a log-HAR structure in the spirit of Corsi (2009); gate covariates are the log 22-day variance, the 60-day scaled momentum and the drawdown from the 250-day high. Semantic groups fix which covariates each expert may use. The signs of the responses are estimated, and in section 10 some of them differ from what the labels suggest, which is itself informative about the market. Section 12 varies the number of experts and the gate covariates, and section 13 replaces the range-based variance covariates by realised measures from 5-minute data.

4.2 Nested and related models

Proposition 4 (Nesting)**.** (3)–(4) *contains as special cases: (i) the Gaussian regression with log-linear variance ($K = 1$); (ii) the mixture autoregression with constant weights of Wong and Li (2000) ($\boldsymbol{\gamma}_k$ restricted to intercepts) and its heteroskedastic extension (Wong and Li, 2001); (iii) the logistic smooth-transition regression of Teräsvirta (1994) with regime-specific variances ($K = 2$, scalar $\boldsymbol{z}$); (iv) the mixture-of-experts of Jacobs et al. (1991) and Jordan and Jacobs (1994) with linear-Gaussian experts and the time-series mixtures of experts of Carvalho and Tanner (2005). The Markov-switching model of Hamilton (1989) is* not *nested: its weights depend on the unobserved lagged regime rather than on observables; it is included as a benchmark.*

We let the gate depend on observables so that every quantity of interest (regime probabilities, their sensitivity to state variables, and the forecast itself) is a closed-form function of data available at t. Section 6 builds the interpretation on this property.

4.3 Identifiability

Finite mixtures are invariant to relabelling, and mixtures of experts inherit this problem (Jiang and Tanner, 1999). Semantic restrictions remove it.

Assumption 1. (a) The support of $(\boldsymbol{x}_t, \boldsymbol{v}_t, \boldsymbol{z}_t)$ contains an open set, and the design matrices $\mathbb{E}[\tilde{\boldsymbol{x}}^{(k)}\tilde{\boldsymbol{x}}^{(k)\top}]$, $\mathbb{E}[\tilde{\boldsymbol{v}}\tilde{\boldsymbol{v}}^\top]$, $\mathbb{E}[\tilde{\boldsymbol{z}}\tilde{\boldsymbol{z}}^\top]$ are non-singular. (b) For every $k \neq l$ the pairs $(\mu_{kt}, \sigma^2_{kt})$ and $(\mu_{lt}, \sigma^2_{lt})$ differ on a set of positive probability. (c) Every expert with $p_k > 0$ has at least one non-zero slope, and no two experts share the same mean-covariate set.

Proposition 5 (Identifiability)**.** *Under assumption 1, if $p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x}, \boldsymbol{v}, \boldsymbol{z}) = p_{\boldsymbol{\theta}'}(y \mid \boldsymbol{x}, \boldsymbol{v}, \boldsymbol{z})$ for almost all $(y, \boldsymbol{x}, \boldsymbol{v}, \boldsymbol{z})$, then $\boldsymbol{\theta} = \boldsymbol{\theta}'$.*

The proof (appendix A) combines the identifiability of finite Gaussian mixtures (Teicher, 1963), which pins down the component set pointwise in the covariates, with the exclusion restrictions, which make any non-trivial relabelling contradict the covariates on which each component's mean depends.

4.4 Stationarity of the canonical SMoE-AR

When covariates are functions of past returns the SMoE defines a non-linear autoregression. Consider the canonical case in which the expert means are linear in the last p returns, $\mu_{kt} = \phi_{k0} + \sum_{j=1}^{p} \phi_{kj} y_{t-j}$, gates depend on the same lags, and $\sigma_{kt} \leq \bar{\sigma} < \infty$.

Proposition 6 (Geometric ergodicity)**.** *If* $\sup_{\boldsymbol{z}} \sum_{k=1}^{K} \pi_k(\boldsymbol{z}) \sum_{j=1}^{p} |\phi_{kj}| < 1$*, the Markov chain* $Y_t = (y_t, \ldots, y_{t-p+1})$ *is geometrically ergodic and has a unique stationary distribution with finite first moment.*

The condition is the mixture analogue of $\sum_j |\phi_j| < 1$; it allows an individual expert to be explosive provided the gate does not give it too much weight. The proof is a Foster–Lyapunov drift argument with $V(Y) = 1 + \sum_j c_j |y_{-j}|$ (Meyn and Tweedie, 2009). Geometric ergodicity implies β-mixing with geometric rate, which is the dependence condition used for the asymptotic theory in section 5.

4.5 The predictive distribution in closed form

Let $\bar{\mu}_t = \sum_k \pi_{kt} \mu_{kt}$ with $\pi_{kt} = \pi_k(\boldsymbol{z}_t)$.

Proposition 7 (Closed-form functionals)**.** *For the SMoE predictive distribution:*

$$\mathbb{E}[y_t \mid \mathcal{F}_t] = \bar{\mu}_t, \tag{5}$$

$$\mathrm{Var}(y_t \mid \mathcal{F}_t) = \underbrace{\textstyle\sum_k \pi_{kt} \sigma_{kt}^2}_{\textit{within-regime risk}} + \underbrace{\textstyle\sum_k \pi_{kt} (\mu_{kt} - \bar{\mu}_t)^2}_{\textit{regime-disagreement risk}}, \tag{6}$$

$$P(y_t > 0 \mid \mathcal{F}_t) = \textstyle\sum_k \pi_{kt}\, \Phi(\mu_{kt}/\sigma_{kt}), \tag{7}$$

$$\mathrm{ES}_{\alpha,t} = \alpha^{-1} \textstyle\sum_k \pi_{kt} \big\{ \mu_{kt} \Phi(z_{kt}) - \sigma_{kt} \varphi(z_{kt}) \big\}, \quad z_{kt} = (q_{\alpha,t} - \mu_{kt})/\sigma_{kt}, \tag{8}$$

$$\mathrm{CRPS}_t(y) = \textstyle\sum_k \pi_{kt} A(y - \mu_{kt}, \sigma_{kt}^2) - \frac{1}{2} \sum_{k,l} \pi_{kt} \pi_{lt} A(\mu_{kt} - \mu_{lt}, \sigma_{kt}^2 + \sigma_{lt}^2), \tag{9}$$

where $q_{\alpha,t}$ *is the unique root of* $\sum_k \pi_{kt} \Phi((q - \mu_{kt})/\sigma_{kt}) = \alpha$ *and* $A(m, s^2) = 2s\varphi(m/s) + m\{2\Phi(m/s) - 1\}$*. Third and fourth central moments are* $\sum_k \pi_k (d_k^3 + 3 d_k \sigma_k^2)$ *and* $\sum_k \pi_k (d_k^4 + 6 d_k^2 \sigma_k^2 + 3\sigma_k^4)$ *with* $d_k = \mu_k - \bar{\mu}$*.*

Equation (6) is the decomposition we use repeatedly: total predictive risk is the risk inside the regimes plus the risk of not knowing which regime prevails. The second term is invisible to any single-regime model. The CRPS formula is due to Grimit et al. (2006); all formulas are verified against Monte Carlo in the replication package.

5 Estimation and inference

5.1 Likelihood and the ECM algorithm

Given $(y_t, \boldsymbol{x}_t, \boldsymbol{v}_t, \boldsymbol{z}_t)_{t=1}^n$, the log-likelihood is

$$\ell_n(\boldsymbol{\theta}) = \sum_{t=1}^{n} \log \sum_{k=1}^{K} \pi_{kt}\, \varphi(y_t; \mu_{kt}, \sigma_{kt}^2). \tag{10}$$

Gaussian-mixture likelihoods are unbounded when a variance collapses onto a single observation (Kiefer and Wolfowitz, 1956). We therefore maximise the penalised criterion

$$\ell_n^{\lambda}(\boldsymbol{\theta}) = \ell_n(\boldsymbol{\theta}) - \frac{\lambda}{2}\Big(\sum_{k} \|\boldsymbol{\beta}_{k,-0}\|^2 + \sum_{k<K} \|\boldsymbol{\gamma}_{k,-0}\|^2 \Big) - \frac{\lambda_v}{2} \sum_{k} \|\boldsymbol{\delta}_k - \boldsymbol{\delta}_0\|^2, \tag{11}$$

where $\boldsymbol{\beta}_{k,-0}$ omits the intercept, $\boldsymbol{\delta}_0 = (\log \widehat{\mathrm{Var}}(y), 0, \ldots, 0)$, and $\lambda = \lambda_v = 1$ with standardised covariates. With $n \approx 4{,}000$ the penalty is $O(n^{-1})$ relative to the likelihood and has no effect on the asymptotics below; its role is to keep the iterates in a compact set.

Introduce latent labels $c_t \in \{1, \ldots, K\}$. The expected complete-data criterion at iterate $\boldsymbol{\theta}^{(s)}$ is $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(s)}) = \sum_t \sum_k \tau_{tk}^{(s)} \{\log \pi_{kt} + \log \varphi(y_t; \mu_{kt}, \sigma_{kt}^2)\} - \mathrm{pen}(\boldsymbol{\theta})$ with *responsibilities*

$$\tau_{tk}^{(s)} = \frac{\pi_{kt}^{(s)} \varphi(y_t; \mu_{kt}^{(s)}, \sigma_{kt}^{2(s)})}{\sum_l \pi_{lt}^{(s)} \varphi(y_t; \mu_{lt}^{(s)}, \sigma_{lt}^{2(s)})}. \tag{12}$$

Q separates into blocks for $\boldsymbol{\beta}$, $\boldsymbol{\delta}$ and $\boldsymbol{\gamma}$, which we maximise conditionally in turn (expectation–conditional maximisation, Meng and Rubin, 1993).

CM-step 1: expert means. Given $\boldsymbol{\delta}$, the $\boldsymbol{\beta}_k$ block is a weighted ridge regression with closed form

$$\boldsymbol{\beta}_k^{(s+1)} = \big(\tilde{X}_k^\top W_k \tilde{X}_k + \lambda R\big)^{-1} \tilde{X}_k^\top W_k \boldsymbol{y}, \qquad W_k = \mathrm{diag}\,\big(\tau_{tk}/\sigma_{kt}^2\big), \tag{13}$$

with $R = \mathrm{diag}(0, 1, \ldots, 1)$.

CM-step 2: expert variances. With residuals $e_{tk} = y_t - \mu_{kt}$ the $\boldsymbol{\delta}_k$ block maximises $\sum_t \tau_{tk}\{-\frac{1}{2}\eta_{tk} - \frac{1}{2} e_{tk}^2 e^{-\eta_{tk}}\} - \frac{\lambda_v}{2}\|\boldsymbol{\delta}_k - \boldsymbol{\delta}_0\|^2$, $\eta_{tk} = \boldsymbol{\delta}_k^\top \tilde{\boldsymbol{v}}_t$: a weighted Gamma regression with log link for e_{tk}^2. The gradient and Hessian are

$$g_k = \tfrac{1}{2} \sum_t \tau_{tk}(u_{tk} - 1)\tilde{\boldsymbol{v}}_t - \lambda_v(\boldsymbol{\delta}_k - \boldsymbol{\delta}_0), \quad H_k = -\tfrac{1}{2} \sum_t \tau_{tk} u_{tk} \tilde{\boldsymbol{v}}_t \tilde{\boldsymbol{v}}_t^\top - \lambda_v I, \quad u_{tk} = e_{tk}^2 e^{-\eta_{tk}}, \tag{14}$$

so the block objective is strictly concave and Newton's method with step halving increases it monotonically.

CM-step 3: gate. The $\boldsymbol{\gamma}$ block is a weighted multinomial logit with fractional responses τ_{tk}; its gradient is $\sum_t (\tau_{tk} - \pi_{kt})\tilde{\boldsymbol{z}}_t - \lambda R \boldsymbol{\gamma}_k$ and its Hessian has blocks $-\sum_t \pi_{kt}(\mathbb{1}_{k=l} - \pi_{lt})\tilde{\boldsymbol{z}}_t \tilde{\boldsymbol{z}}_t^\top - \mathbb{1}_{k=l}\lambda R$, again negative definite. We take damped Newton steps.

Proposition 8 (Monotonicity)**.** *Each ECM iteration satisfies* $\ell_n^\lambda(\boldsymbol{\theta}^{(s+1)}) \geq \ell_n^\lambda(\boldsymbol{\theta}^{(s)})$*. If* ℓ_n^λ *is bounded above on the iterates' closure, the sequence* $\ell_n^\lambda(\boldsymbol{\theta}^{(s)})$ *converges, and every limit point of* $\{\boldsymbol{\theta}^{(s)}\}$ *is a stationary point of* ℓ_n^λ*.*

The first statement is the standard EM inequality applied to an increase rather than a maximisation of Q (Dempster et al., 1977; Meng and Rubin, 1993); the second follows from Wu (1983) because Q is continuous in both arguments. Because EM converges linearly near the optimum, we finish with a quasi-Newton (L-BFGS) step that uses the exact analytic gradient of ℓ_n^λ, available from Fisher's identity (eq. (15) below); the final iterate is accepted only if it increases ℓ_n^λ, so monotonicity is preserved. We use four random initialisations in which the crisis expert's initial responsibility is tilted towards high-variance days; in walk-forward re-estimation each fit is warm-started from the previous quarter's estimate.

5.2 Scores and the information matrix

By Fisher's identity the per-observation score of the observed log-likelihood is the responsibility-weighted complete-data score (Louis, 1982):

$$\frac{\partial \ell_t}{\partial \boldsymbol{\beta}_k} = \tau_{tk} \frac{e_{tk}}{\sigma_{kt}^2} \tilde{\boldsymbol{x}}_t^{(k)}, \qquad \frac{\partial \ell_t}{\partial \boldsymbol{\delta}_k} = \frac{\tau_{tk}}{2}\big(u_{tk} - 1\big)\tilde{\boldsymbol{v}}_t, \qquad \frac{\partial \ell_t}{\partial \boldsymbol{\gamma}_k} = (\tau_{tk} - \pi_{kt})\tilde{\boldsymbol{z}}_t. \tag{15}$$

These expressions are exact (they are verified against numerical differentiation to 10^{-6} in the test suite) and make the white box transparent at the level of estimation as well: the gate learns from the gap between posterior and prior regime probabilities, $\tau_{tk} - \pi_{kt}$.

5.3 Asymptotic theory under misspecification

We do not assume the SMoE is the true data-generating process. Let $\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}\in\Theta} \mathbb{E}[\log p_{\boldsymbol{\theta}}(y_t \mid \mathcal{F}_t)]$ be the pseudo-true value, the KL projection of the true conditional law onto the SMoE family.

Assumption 2. (a) $\{(y_t, \boldsymbol{x}_t, \boldsymbol{v}_t, \boldsymbol{z}_t)\}$ is strictly stationary and β-mixing with mixing coefficients $\beta(j) = O(j^{-b})$, $b > r/(r-2)$, for some $r > 2$; (b) Θ is compact and $\boldsymbol{\theta}^*$ is the unique maximiser, in the interior of Θ; (c) $\mathbb{E}\sup_{\boldsymbol{\theta}} |\log p_{\boldsymbol{\theta}}| < \infty$ and $\mathbb{E}\|\partial \ell_t(\boldsymbol{\theta}^*)\|^r < \infty$; (d) $A = -\mathbb{E}[\partial^2 \ell_t(\boldsymbol{\theta}^*)/\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^\top]$ is non-singular.

Theorem 1 (QMLE)**.** *Under* **?? 1?? 2***, the penalised estimator* $\hat{\boldsymbol{\theta}}_n$ *satisfies* $\hat{\boldsymbol{\theta}}_n \to \boldsymbol{\theta}^*$ *almost surely and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}^*) \xrightarrow{d} \mathcal{N}\big(0,\ A^{-1} B A^{-1}\big), \qquad B = \lim_{n\to\infty} \operatorname{Var}\Big(n^{-1/2} \sum_t \partial \ell_t(\boldsymbol{\theta}^*)\Big). \tag{16}$$

The result is White's (1982) QMLE theorem for dependent data; for mixtures of experts of autoregressive time series it is established by Carvalho and Tanner (2005). The penalty is $O_p(n^{-1})$ in the score and therefore asymptotically negligible. Two remarks matter in practice. First, under correct specification the scores are a martingale difference sequence and $B = A$; under misspecification they need not be, so we estimate B with a Newey–West kernel (Newey

and West, 1987) using $L = \lfloor 4(n/100)^{2/9} \rfloor$ lags. Second, A is obtained by central differences of the exact analytic score. We report both the sandwich and the model-based (A^{-1}) standard errors; their divergence is itself a specification diagnostic (White, 1982).

5.4 Testing

Semantic restrictions and gate relevance. Linear restrictions on $\boldsymbol{\theta}$ (e.g. "the gate does not depend on volatility" ($\gamma_{k,\text{lvol}} = 0$ for all k)) are tested with robust Wald statistics $W = (R\hat{\boldsymbol{\theta}})^\top (R\hat{V}R^\top)^{-1} R\hat{\boldsymbol{\theta}} \sim \chi^2_{\text{rank}R}$. Because the null keeps K fixed, the parameters remain identified and the limit is regular. SMoE versus the constant-weight MAR is such a test.

Number of experts. Testing K against $K-1$ is non-standard: under the null the parameters of the extra expert are unidentified and the likelihood ratio does not have a χ^2 limit (Hansen, 1996; Chen et al., 2001). Section 12 therefore compares numbers of experts by out-of-sample log score, by BIC and by a parametric bootstrap of the LR statistic under the fitted null model. Testing whether an additional state variable enters the gate keeps K fixed and is regular; proposition 14 gives a score test that needs only the restricted fit.

5.5 Computational complexity

Each ECM iteration costs $O(n \sum_k p_k^2 + nKq^2 + nK^2m^2)$, i.e. linear in n. A cold fit on 4,600 daily observations takes about 25 seconds on one CPU core and a warm-started quarterly refit about 7 seconds, so the entire walk-forward exercise for one asset (roughly fifty refits) completes in minutes.

6 White-box interpretation and the yardstick

"White box" is used loosely in the literature. We mean three verifiable properties: (i) every forecast is an explicit, closed-form function of a small number of named parameters and covariates; (ii) the contribution of every covariate to every forecast functional can be computed exactly, without approximation or sampling; (iii) the parameters carry standard errors, so interpretations can be tested. The SMoE satisfies all three. This section derives the exact decompositions and then turns the SMoE into a yardstick for black-box forecasters.

6.1 Exact attribution of the predictive mean

With standardised covariates (so that $\boldsymbol{x} = \mathbf{0}$ is the average state),

$$\bar{\mu}_t = \sum_k \pi_{kt}\beta_{k0} + \sum_k \sum_{j=1}^{p_k} \underbrace{\pi_{kt}\beta_{kj}x_{tj}^{(k)}}_{c_{t,kj}}. \tag{17}$$

The terms $c_{t,kj}$ sum exactly to the forecast. Holding the gate fixed, (17) coincides with the interventional Shapley value of each covariate relative to the training mean (Lundberg and Lee, 2017; for linear models Shapley values reduce to $\beta_j(x_j - \mathbb{E}x_j)$), so the decomposition is

both exact and axiomatically justified. The gate enters through the weights π_{kt}: a covariate can matter because its expert is active, and a single linear model cannot express this kind of regime dependence.

6.2 Gate elasticities

Proposition 9 (Gate elasticity)**.** *For a gate covariate z_j that does not enter the expert means or variances,*

$$\frac{\partial \bar{\mu}_t}{\partial z_{tj}} = \sum_k \pi_{kt}\,\mu_{kt}\left(\gamma_{kj} - \bar{\gamma}_{tj}\right), \qquad \bar{\gamma}_{tj} = \sum_l \pi_{lt}\gamma_{lj}, \tag{18}$$

and analogously $\partial \pi_{kt}/\partial z_{tj} = \pi_{kt}(\gamma_{kj} - \bar{\gamma}_{tj})$ and $\partial P(y_t > 0)/\partial z_{tj} = \sum_k \pi_{kt}\Phi(\mu_{kt}/\sigma_{kt})(\gamma_{kj} - \bar{\gamma}_{tj})$.

The elasticity is a covariance, under the gate distribution, between expert forecasts and expert gate loadings: a state variable moves the forecast only to the extent that it shifts weight between experts that *disagree*. Neural gates offer no comparably direct reading of what a regime variable does.

6.3 Risk decomposition

The variance decomposition (6) separates within-regime risk from regime-disagreement risk. For a mean-variance investor (section 7.2) the second term reduces the optimal position even when every regime individually looks attractive, which formalises the intuition that uncertainty *about* the regime is a distinct source of risk.

6.4 The yardstick: projecting black boxes onto the SMoE

Let $p_T(\cdot \mid \boldsymbol{w}_t)$ be the predictive density of an arbitrary forecaster ("teacher") that may use any information $\boldsymbol{w}_t \in \mathcal{F}_t$, such as a deep network, a foundation model or an ensemble. Its white-box shadow is the information projection onto the SMoE family,

$$\boldsymbol{\theta}_T = \arg\min_{\boldsymbol{\theta}} \mathbb{E}\ \mathrm{KL}\left(p_T(\cdot \mid \boldsymbol{w}_t)\,\|\,p_{\boldsymbol{\theta}}(\cdot \mid \boldsymbol{x}_t, \boldsymbol{v}_t, \boldsymbol{z}_t)\right) = \arg\max_{\boldsymbol{\theta}} \mathbb{E}\,\mathbb{E}_{y\sim p_T(\cdot\mid \boldsymbol{w}_t)} \log p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x}_t, \boldsymbol{v}_t, \boldsymbol{z}_t), \tag{19}$$

which we estimate by maximum likelihood on M draws from the teacher at each training date; the ECM algorithm is unchanged. The shadow inherits all white-box properties: its parameters, gate elasticities and attributions describe *what the black box does* in the vocabulary of the semantic groups.

To measure how much of the black box the shadow reproduces, recall that for any predictive density p, $\mathbb{E}\log p(y_t \mid \cdot) = -H(p_0) - \mathbb{E}\ \mathrm{KL}(p_0\|p)$, where p_0 is the true conditional density.

Proposition 10 (Fidelity)**.** *Let $\mathrm{LS}(p) = \mathbb{E}\log p(y_t \mid \cdot)$ and let R be a reference forecaster. Then*

$$\Phi := \frac{\mathrm{LS}(p_{\boldsymbol{\theta}_T}) - \mathrm{LS}(p_R)}{\mathrm{LS}(p_T) - \mathrm{LS}(p_R)} = \frac{\mathbb{E}\,\mathrm{KL}(p_0\|p_R) - \mathbb{E}\,\mathrm{KL}(p_0\|p_{\boldsymbol{\theta}_T})}{\mathbb{E}\,\mathrm{KL}(p_0\|p_R) - \mathbb{E}\,\mathrm{KL}(p_0\|p_T)}, \tag{20}$$

the share of the teacher's KL improvement over the reference that is achieved by its white-box shadow. $\Phi = 1$ means the black box's entire out-of-sample advantage is expressible as a

semantic mixture; $\Phi < 1$ *quantifies what is lost;* $\Phi > 1$ *means the shadow generalises better than the teacher, i.e. the teacher's extra flexibility is overfitting.*

We complement Φ with three moment fidelities (the out-of-sample R^2 between teacher and shadow of the predictive mean, the log predictive variance and $P(y > 0)$) and with a formal test of whether the residual is valuable: the Amisano and Giacomini (2007) weighted likelihood-ratio test of teacher against shadow. A non-rejection says that, out of sample, nothing the black box knows beyond the semantic mixture is statistically detectable. For black boxes with an internal gate we also report linear *probes* (Alain and Bengio, 2016): the R^2 of each gate logit regressed on the SMoE's state variables.

7 Reliable deployment: calibration guarantees and decisions

7.1 Adaptive conformal calibration of tail quantiles

Tail quantiles are the part of a parametric forecast most exposed to misspecification. We attach a distribution-free calibration layer to the SMoE's value-at-risk using adaptive conformal inference (Gibbs and Candès, 2021). Let F_t be the predictive cdf and a the target violation rate. Issue $q_t = F_t^{-1}(\alpha_t)$ and update

$$\alpha_{t+1} = \alpha_t + \eta\,(a - \mathrm{err}_t), \qquad \mathrm{err}_t = \mathbb{1}\{y_t < q_t\}, \tag{21}$$

with the conventions $q_t = -\infty$ if $\alpha_t \le 0$ and $q_t = +\infty$ if $\alpha_t \ge 1$.

Theorem 2 (Long-run coverage)**.** *For every sequence* (y_t) *and every sequence of predictive cdfs* (F_t)*, deterministic or random, correctly specified or not,*

$$\Big|\frac{1}{T}\sum_{t=1}^{T}\mathrm{err}_t - a\Big| \le \frac{\max(\alpha_1, 1-\alpha_1) + \eta}{\eta\, T}. \tag{22}$$

The proof (appendix A) is two lines: (21) telescopes to $\alpha_{T+1} - \alpha_1 = \eta\sum_t(a - \mathrm{err}_t)$, and α_t never leaves $[-\eta, 1+\eta]$ because a level at or below zero cannot be violated and a level at or above one is always violated. The guarantee is on the long-run frequency, not on conditional coverage; we therefore report the Kupiec (1995) and Christoffersen (1998) tests before and after calibration, which test both.

7.2 From predictive distributions to positions

Let w be the position in the risky asset (fraction of capital), κ the proportional transaction cost and w_- the position carried from the previous day. A mean-variance investor with risk aversion γ solves

$$\max_{w\in[\underline{w},\overline{w}]}\; w\,\bar{\mu}_t - \frac{\gamma}{2}w^2 V_t - \kappa|w - w_-|, \qquad V_t = \mathrm{Var}(y_t \mid \mathcal{F}_t), \tag{23}$$

with V_t the full mixture variance (6).

Proposition 11 (No-trade band)**.** *Let* $w_t^* = \bar{\mu}_t/(\gamma V_t)$ *and* $b_t = \kappa/(\gamma V_t)$*. The solution of* (23) *is*

$$w_t = \Pi_{[\underline{w},\overline{w}]}\Big(\min\big\{ \max\{w_-,\, w_t^* - b_t\},\, w_t^* + b_t\big\}\Big), \tag{24}$$

where Π *is projection onto the box: the investor does not trade while* w_- *lies within* b_t *of the frictionless target, and otherwise moves to the nearest edge of the band.*

Proposition 12 (Regime-disagreement shrinkage)**.** *Write* $W_t = \sum_k \pi_{kt}\sigma_{kt}^2$ *and* $B_t = \sum_k \pi_{kt}(\mu_{kt} - \bar{\mu}_t)^2$*. The frictionless target satisfies*

$$w_t^* = \frac{\bar{\mu}_t}{\gamma W_t} \cdot \frac{1}{1 + B_t/W_t}. \tag{25}$$

The first factor is the position of an investor who knows the regime mixture but ignores the dispersion of regime means; the second is a shrinkage factor in $(0, 1]$ *that equals one if and only if all experts agree on the mean. Holding* $\bar{\mu}_t$ *and* W_t *fixed, a mean-preserving spread of the expert means lowers* $|w_t^*|$*.*

The proof is immediate from (6). The proposition is the decision-theoretic counterpart of the variance decomposition: the investor treats uncertainty *about* the regime like any other risk, and measuring that risk requires a model that represents regimes explicitly.
Two features are specific to the mixture. The band width is proportional to $1/V_t$, so the band narrows when regime disagreement inflates V_t, and regime disagreement shrinks w_t^* even if all experts forecast positive returns. Long-only ETF implementation uses $[0, 1]$; the futures implementation uses $[-1, 1]$. Proposition 11 follows from a one-line subgradient argument (appendix A). Gârleanu and Pedersen (2013) derive the dynamic multi-period analogue; the one-period band is enough for daily rebalancing with the costs we face. Economic value is measured by the Fleming et al. (2001) performance fee, by Sharpe ratios with Lo (2002) standard errors, by the deflated Sharpe ratio of Bailey and López de Prado (2014) to account for the number of strategies tried, and by the probability of backtest overfitting of Bailey et al. (2017).

8 Monte Carlo evidence

We simulate from a three-expert SMoE with the semantic structure of section 4 (two slopes per mean expert, three variance covariates and three gate covariates, 27 parameters in total) at sample sizes $n \in \{1000, 2500, 5000\}$, which bracket the daily histories in our application. Covariates are independent standard normal; true parameters are chosen so that all three regimes carry non-trivial weight and the gate depends on all three state variables (`smoe/simulate.py`). Each design is replicated $R = 200$ times and estimated exactly as in the application.

Accuracy and coverage. Table 1 averages over the 27 parameters. The mean absolute bias falls from 0.120 at $n = 1000$ to 0.023 at $n = 2500$ and 0.008 at $n = 5000$, roughly at the $\sqrt{n}$ rate. Sandwich confidence intervals are close to nominal from $n = 2500$ (average coverage 92.8%) and essentially exact at $n = 5000$ (94.4%); model-based intervals undercover slightly at

every size (80.8%, 90.4%, 93.3%), which is the usual finite-sample advantage of the sandwich in mixtures. At $n = 1000$ coverage is poor (84.6%) and 19 of 27 parameters fall below 90%; the worst is the intercept of the crisis expert's log variance, whose sampling distribution is skewed because that regime is rarely visited. Our empirical samples have n between 1,700 (IF futures) and 4,800, so inference for the smallest series should be read with this caveat.

Testing gate relevance. Table 2 reports the robust Wald test that all six gate slopes are zero. Under a constant-gate DGP the test over-rejects at $n = 1000$ (14% at the 5% level), is correctly sized at $n = 2500$ (5.0%) and conservative at $n = 5000$ (2.5%). Under the state-dependent DGP its power is 56% at $n = 1000$ and at least 99% for $n \geq 2500$. The rejections reported in tables 5 and 28 are therefore unlikely to be size distortions.

Table 1: Monte Carlo: estimator accuracy and 95% interval coverage (averaged over 27 parameters)

n	Mean |bias|	Mean RMSE	Coverage (sandwich)	Coverage (model)
1000.000	0.120	1.040	0.846	0.808
2500.000	0.023	0.263	0.928	0.904
5000.000	0.008	0.125	0.944	0.933

Table 2: Monte Carlo: size and power of the robust Wald test of gate relevance

n	DGP	Reject 5%	Reject 10%	R
1,000	constant gate (size)	0.140	0.200	200
1,000	state-dependent gate (power)	0.560	0.680	200
2,500	constant gate (size)	0.050	0.095	200
2,500	state-dependent gate (power)	0.990	1.000	200
5,000	constant gate (size)	0.025	0.055	200
5,000	state-dependent gate (power)	1.000	1.000	200

9 Data, features and evaluation protocol

9.1 Universe

We study nineteen daily series from January 2005 (or inception) to October 2026 (table 26): four Chinese equity indices (CSI 300, CSI 500, SSE 50, ChiNext); six international equity indices (S&P 500, Nasdaq 100, Hang Seng, Nikkei 225, DAX, FTSE 100); seven futures (CSI 300 index futures IF, SHFE gold and copper, DCE soybean meal, WTI crude, COMEX gold, US 10-year T-note); Bitcoin; and the CSI 300 ETF (510300), which is the tradable instrument for the strategy in section 15. Chinese series are obtained through AkShare, international series through Yahoo Finance; every file is stored with a SHA-256 hash in a manifest. Futures are main-contract continuous series and contain roll gaps; ETF prices are not dividend-adjusted, so its buy-and-hold returns are understated by the dividend yield. For CSI 300 we also have 5-minute bars from May 2009 to July 2025, from which we compute realised variance $\mathrm{RV}_t = \sum_i r_{t,i}^2$, bipower variation (Barndorff-Nielsen and Shephard, 2004), jumps and realised

semivariances (Patton and Sheppard, 2015). They serve as variance proxies for evaluation (Andersen et al., 2003) and, in section 13, as covariates. The vendor file repeats every bar on 36 days and contains one bar with a zero price; we drop duplicated time stamps and non-positive prices before computing any measure (an earlier version of our code did not, which left realised variance unchanged but set bipower variation to zero on the duplicated days). The CBOE VIX, used as a candidate gate variable in section 12, is taken from Yahoo Finance.

9.2 Features

All features at date t use prices up to the close of t; the target is $y_t = r_{t+1}$ in percent. Let v_t be the Parkinson (1980) range variance $v_t = (100 \log(H_t/L_t))^2/(4 \log 2)$ and $\hat{\sigma}_t = (\frac{1}{22} \sum_{j<22} v_{t-j})^{1/2}$. Then $\mathrm{mom}_h = \sum_{j<h} r_{t-j}/(\hat{\sigma}_t \sqrt{h})$ for $h \in \{20, 120\}$ (trend); $\mathrm{rev}_1 = r_t/\hat{\sigma}_t$, $\mathrm{rev}_5 = \sum_{j<5} r_{t-j}/(\hat{\sigma}_t \sqrt{5})$ (reversal); $\log \bar{v}_t^{(h)}$ for $h \in \{1, 5, 22\}$ (variance); and gate states $(\log \bar{v}_t^{(22)}, \mathrm{mom}_{60}, \log(P_t/\max_{j<250} P_{t-j}))$. Covariates are standardised with training-window moments at each re-estimation.

9.3 Protocol

1. **Walk-forward.** Expanding window; statistical models are re-estimated at the start of every quarter, the deep model every year. Out-of-sample forecasts start in January 2012 (later for series with short histories, which require 750 training observations).

2. **Leakage control.** A model fitted for the block starting on date d sees only rows with date $< d$, whose targets are realised by the close of the first forecast date. The test suite corrupts every price after a cut-off and verifies that all forecasts up to the cut-off are unchanged for every model (appendix C).

3. **Lockbox.** Observations from 1 July 2024 onward were not used in any modelling decision. All development results end on 30 June 2024; the lockbox period is evaluated once, after the protocol was frozen, and reported separately.

4. **Common sample.** Within each asset all models are compared on the intersection of their forecast dates.

9.4 Models

Table 3 lists the forecasters. All return a Gaussian mixture predictive distribution; Student-t GARCH densities are represented as Gaussian scale mixtures through 64-node generalised Gauss–Laguerre quadrature (relative density error below 0.11% for $\nu \geq 3$; appendix A), so every model is scored by the same closed-form functionals.

The two foundation-model families differ in architecture and training. Chronos-Bolt is an encoder–decoder T5 model released in November 2024 that outputs nine quantiles directly; TimesFM 2.5 is a decoder-only patch transformer released in September 2025 whose continuous quantile head we use for the same nine levels. Both receive the last 512 daily returns as context and forecast one step ahead without any fine-tuning. Their quantiles are projected onto a

Table 3: Forecasting models

Model	Description	Refit
HistGauss	$\mathcal{N}$(250-day mean, 250-day variance)	quarterly
GARCH-t	constant mean, GARCH(1,1), Student-t (Bollerslev, 1986)	quarterly
AR-GJR-t	AR(1) mean, GJR-GARCH(1,1) (Glosten et al., 1993), Student-t	quarterly
Gauss-HAR	SMoE with $K = 1$ (all mean covariates, log-HAR variance)	quarterly
MAR	SMoE with constant gate (Wong and Li, 2001)	quarterly
LSTAR	trend/reversal experts, logistic gate in volatility	quarterly
MS-AR	2-regime Markov-switching regression (Hamilton, 1989)	quarterly
XGB-Gauss	gradient-boosted mean, log-HAR residual variance	quarterly
SMoE	definition 1, $K = 3$	quarterly
Deep-SMoE	LSTM/MLP/CNN experts, attention gate, 3-seed ensemble	yearly
Chronos-Bolt	zero-shot foundation model (Ansari et al., 2024), small and base, Gaussian projection	none
TimesFM-2.5	zero-shot foundation model (Das et al., 2024), 200M, Gaussian projection	none
RealGARCH-t	log-linear Realized GARCH(1,1) (Hansen et al., 2012), Student-t; section 13 only	quarterly

Gaussian by least squares on the normal quantile function, so their tails beyond the 10% and 90% levels are extrapolated.

10 Results I: the CSI 300 through the white box

10.1 The lagging paradox in the data

Table 27 confirms section 3 on all nineteen series. A 60-lag linear model fitted to price levels by least squares has coefficients summing to 0.994–1.003 and its forecast is correlated with the previous close at 0.9993 or more: it *is* a random walk with estimation noise. Its out-of-sample price RMSE exceeds the random walk's on every series, by 0.6% (US T-note) to 7.8% (CSI 300 ETF), as proposition 2 and the cost of estimating 61 coefficients predict. The differenced directional accuracy of the random walk equals sign persistence exactly and lies between 0.44 and 0.50, which is why price-level models routinely report "directional accuracy below a coin flip". None of these numbers measures forecasting skill.

10.2 In-sample estimates

Table 4 reports the SMoE for CSI 300 over January 2005–June 2024 (n = 4,613). There are three main findings.

The mixture is strongly supported. The likelihood ratio against the single-regime Gaussian with the same covariates is 330.6 for 18 additional parameters and BIC improves by 179 (table 6). AIC ranks the SMoE first among the nested family; BIC prefers the more parsimonious LSTAR, which shares the trend/reversal experts but drops the crisis expert and two gate covariates.

Volatility is the regime variable; momentum and drawdown are not. The robust Wald test that all gate slopes are zero (SMoE against the constant-weight MAR) rejects at any conventional level ($W = 70.8$, 6 df). Decomposing it, volatility enters the gate ($W = 46.2$, 2 df, $p < 10^{-9}$) whereas 60-day momentum ($p = 0.14$) and drawdown ($p = 0.14$) do not (table 5). The trend expert's weight falls with volatility ($\hat{\gamma} = -1.03$, $z = -6.4$): in calm markets two

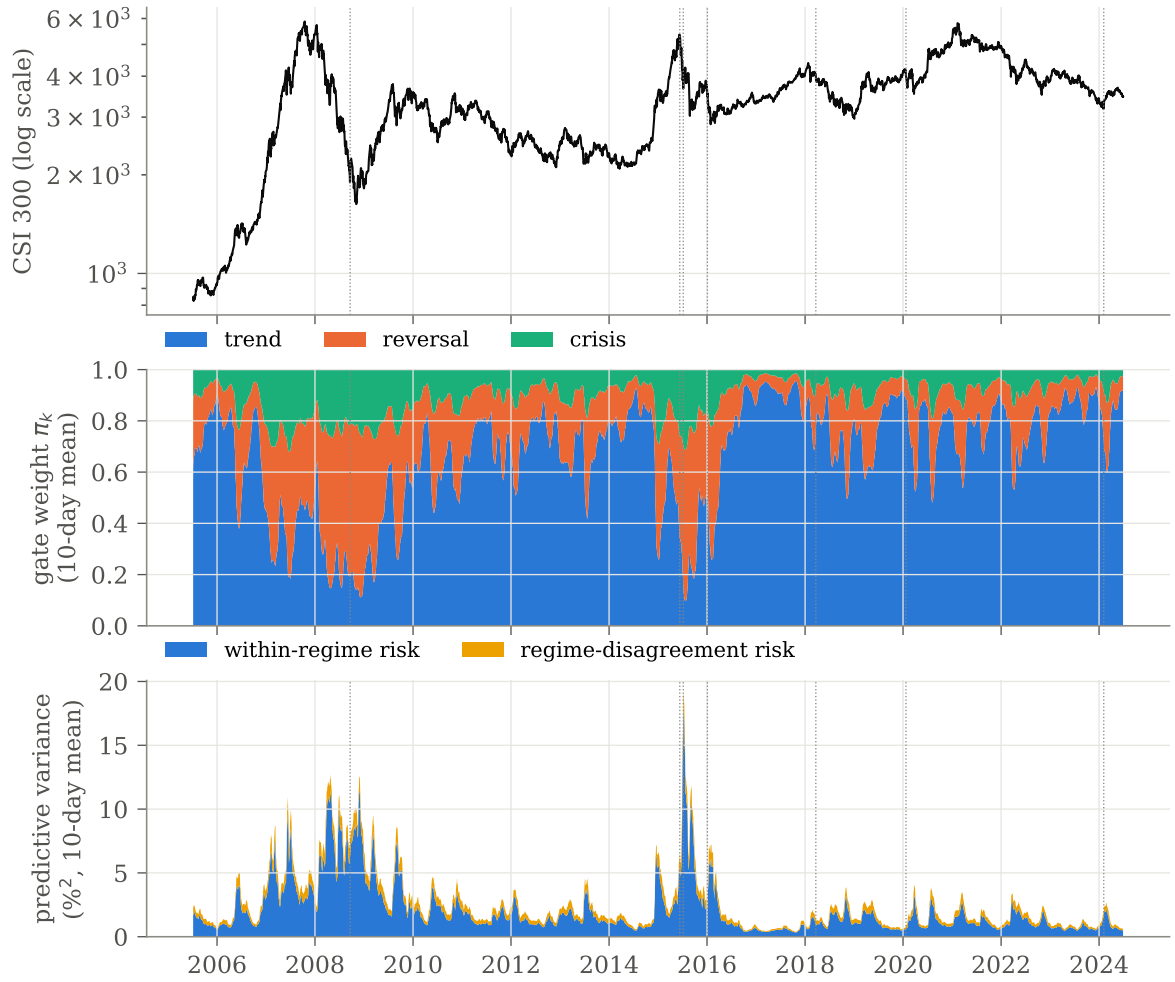


Figure 1: CSI 300: price (top), SMoE gate probabilities (middle) and the decomposition of predictive variance into within-regime and regime-disagreement risk (bottom), in sample. Dotted lines mark the policy and market events of table 29.

thirds of the probability mass sits in the trend regime, and it drains towards the other experts as volatility rises (fig. 2).

Semantic labels and estimated signs disagree, and the disagreement is informative. The trend expert loads on 120-day momentum ($\hat{\beta} = 0.106$, $z = 3.8$), as its label suggests. The "reversal" expert, however, has a *positive* coefficient on the scaled one-day return (0.278, $z = 2.2$) and a large variance: in the high-variance state, Chinese index returns continue rather than reverse. The "crisis" expert has a positive mean (1.77%, $z = 6.5$) and captures the rebound days that follow sell-offs. We return to the cross-market pattern in section 11: in developed markets the same expert has the negative sign of genuine short-term reversal.

10.3 Reading the forecasts

Figure 3 applies the closed forms of section 6 to the fitted CSI 300 model. The top panel is the exact attribution (17) of the predictive mean to the three experts. The short-horizon ("reversal") and crisis experts carry large and opposite contributions that largely cancel; in

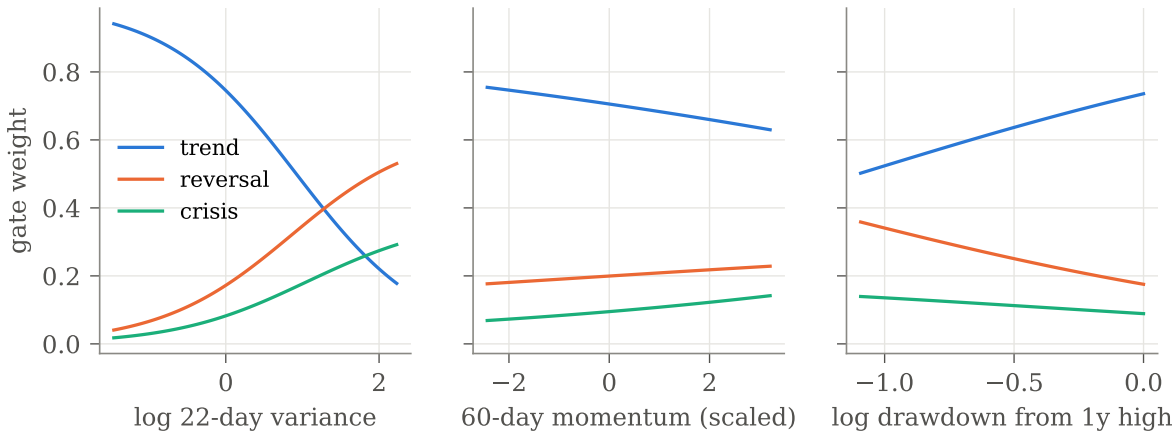


Figure 2: Gate response curves for CSI 300: π_k as a function of each state variable with the others at their sample means. Only volatility moves the gate materially, matching the Wald tests.

the 2008 and 2015 turmoil both widen to about half a percentage point a day. The predictive *mean* is therefore the small difference of two large regime-specific forecasts. This is a reason to distrust any single point forecast of the daily return, and seeing it requires a mixture with explicit components. The middle panel shows the probability of a rise, which exceeds one half on 64% of days but rarely departs from it by more than ten percentage points.

The bottom panel is the gate elasticity of the 1% VaR with respect to the volatility state, computed in closed form (appendix E). Through the gate alone (holding every expert's mean and variance fixed) a one-standard-deviation rise in log volatility lowers the 1% VaR by 0.58 percentage points on average, with a 5–95% range of 0.24 to 0.83. The effect is weakest in the middle of the 2008 and 2015 crises, when the gate has already shifted most of its weight away from the trend expert and has little left to move. Statements of this kind (exact, decomposed by channel and state-dependent) are what we mean by a white box.

10.4 Policy and market events

Because the gate is a function of observables, its reading around known events is a falsifiable description of what the model does. Around the start of the 2015 crash the 20-day average trend weight halves (from 0.42 to 0.21) while the reversal and crisis weights rise; after the rescue package of July 2015 the trend weight falls further to 0.11. The circuit-breaker episode of January 2016 shows the same pattern in smaller magnitude. Two events move the gate the other way or not at all: the 2018 tariff announcements coincided with a *higher* trend weight (the sell-off was gradual and low-volatility at first), and the 2008 stamp-duty cut left the weights unchanged. The gate therefore responds to the change in volatility that an event brings; the headline itself does not enter. This reading is available only because the gate is a known function of observables.

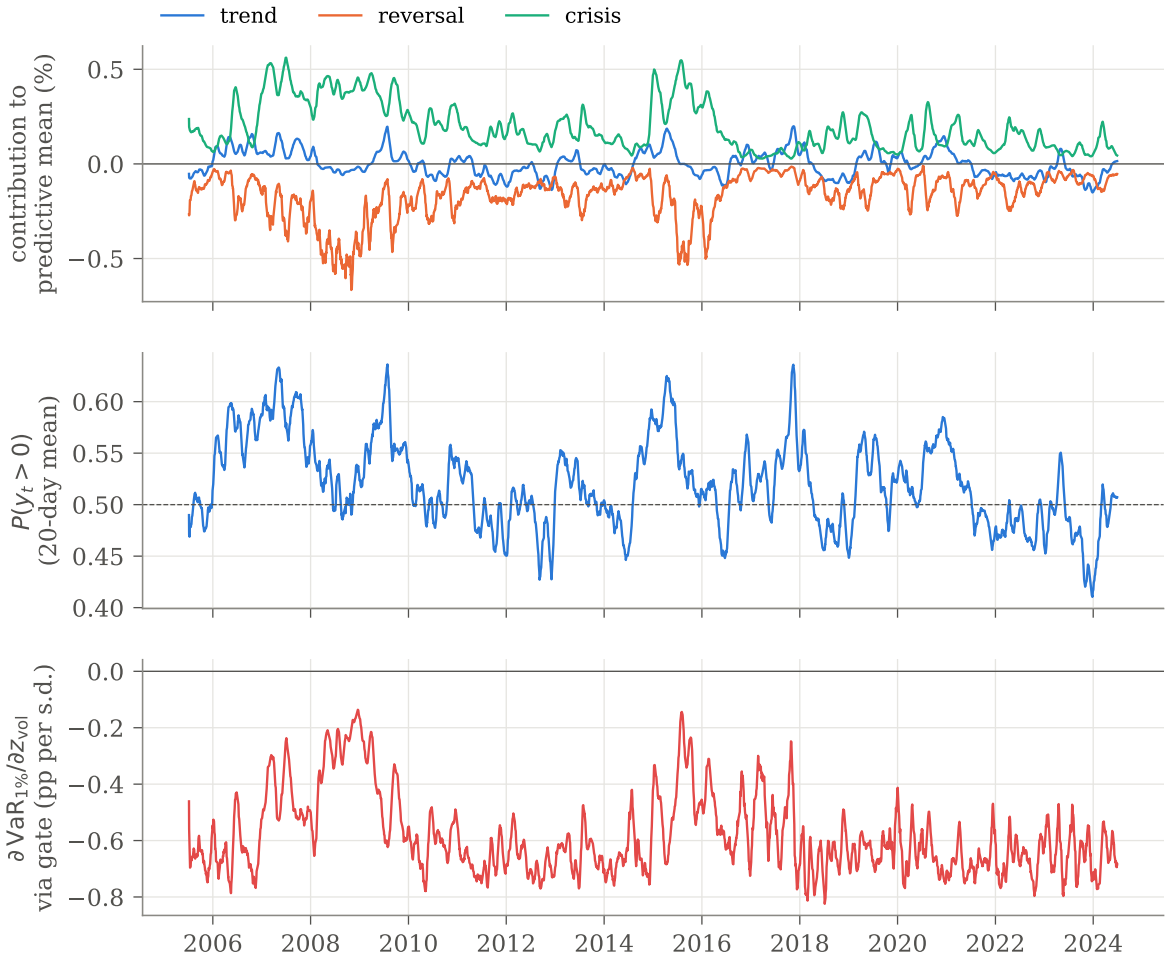


Figure 3: CSI 300 through the white box (in sample, 20-day averages). Top: exact attribution of the predictive mean to the experts. Middle: probability of a rise. Bottom: closed-form gate elasticity of the 1% VaR with respect to the volatility state, in percentage points per standard deviation.

10.5 Out-of-sample forecasts

Tables 31 to 34 report out-of-sample losses for all models and assets relative to GARCH-t. For the Chinese equity series the SMoE ties with the GARCH family. Its log-score difference from GARCH-t is -0.002 (CSI 300), -0.023 (CSI 500), $+0.003$ (SSE 50) and $+0.004$ (ChiNext); for CSI 300 it is inside the 10% model confidence set for all four losses (MCS p-values 0.64–0.89), together with the GARCH models, LSTAR and the deep SMoE. Models that are *excluded* for CSI 300 are equally informative: the historical Gaussian, the Markov-switching regression and all three zero-shot foundation models under every loss, the boosted model under every loss except CRPS, and the single-regime Gauss-HAR for log score.

The variance forecasts can be checked against realised variance from 5-minute data, which is a far more precise proxy than squared returns. Against $\mathrm{RV}_t + r^2_{\mathrm{overnight},t}$ the SMoE has the highest Mincer–Zarnowitz R^2 of all models (0.355, versus 0.244 for GARCH-t) and the second-lowest MSE. On QLIKE four models are marginally better (LSTAR, Gauss-HAR, the boosted model and the MAR, 0.293–0.299 against 0.303), all of which use the same range-based HAR

variance covariates (table 30).

Table 4: SMoE estimates for CSI 300, January 2005–June 2024

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
`beta[trend]:const`	-0.000	0.029	0.029	-0.002	0.998
`beta[trend]:mom20`	0.011	0.025	0.024	0.426	0.670
`beta[trend]:mom120`	0.106	0.028	0.027	3.809	0.000
`beta[reversal]:const`	-0.728	0.262	0.183	-2.782	0.005
`beta[reversal]:rev1`	0.278	0.126	0.120	2.208	0.027
`beta[reversal]:rev5`	0.184	0.166	0.125	1.104	0.269
`beta[crisis]:const`	1.773	0.272	0.229	6.516	0.000
`delta[trend]:const`	-0.219	0.120	0.093	-1.833	0.067
`delta[trend]:lv1`	0.054	0.070	0.064	0.769	0.442
`delta[trend]:lv5`	0.012	0.115	0.104	0.107	0.915
`delta[trend]:lv22`	0.376	0.115	0.108	3.270	0.001
`delta[reversal]:const`	1.372	0.137	0.097	10.024	0.000
`delta[reversal]:lv1`	0.226	0.087	0.088	2.591	0.010
`delta[reversal]:lv5`	0.245	0.124	0.117	1.979	0.048
`delta[reversal]:lv22`	0.159	0.116	0.102	1.372	0.170
`delta[crisis]:const`	-0.327	0.290	0.263	-1.125	0.261
`delta[crisis]:lv1`	0.422	0.187	0.209	2.254	0.024
`delta[crisis]:lv5`	0.065	0.235	0.272	0.274	0.784
`delta[crisis]:lv22`	0.331	0.206	0.225	1.603	0.109
`gamma[trend]:const`	1.970	0.407	0.339	4.845	0.000
`gamma[trend]:g_lvol`	-1.033	0.162	0.149	-6.385	0.000
`gamma[trend]:g_mom60`	-0.216	0.146	0.124	-1.481	0.139
`gamma[trend]:g_dd`	0.154	0.141	0.129	1.091	0.275
`gamma[reversal]:const`	0.725	0.481	0.362	1.505	0.132
`gamma[reversal]:g_lvol`	-0.053	0.172	0.151	-0.311	0.756
`gamma[reversal]:g_mom60`	-0.112	0.203	0.157	-0.549	0.583
`gamma[reversal]:g_dd`	-0.049	0.156	0.125	-0.316	0.752

z and p use the sandwich standard errors with Newey–West meat.

Table 5: Robust Wald tests of semantic restrictions, CSI 300

Null hypothesis	Wald	df	p
all gate slopes = 0 (SMoE vs MAR)	70.812	6	0.000
volatility does not enter the gate	46.202	2	0.000
momentum does not enter the gate	3.895	2	0.143
drawdown does not enter the gate	3.985	2	0.136
trend expert slopes = 0	21.561	2	0.000
reversal expert slopes = 0	5.616	2	0.060
all mean slopes = 0	23.956	4	0.000
equal variance dynamics across experts	16.034	6	0.014

Table 6: In-sample likelihood comparison, CSI 300

model	loglik	params	AIC	BIC
Gauss-HAR (K=1)	-7993.5	9	16,005	16,063
MAR	-7842.0	21	15,726	15,861
LSTAR	-7860.3	16	15,753	**15,856**
SMoE	-7828.2	27	**15,710**	15,884

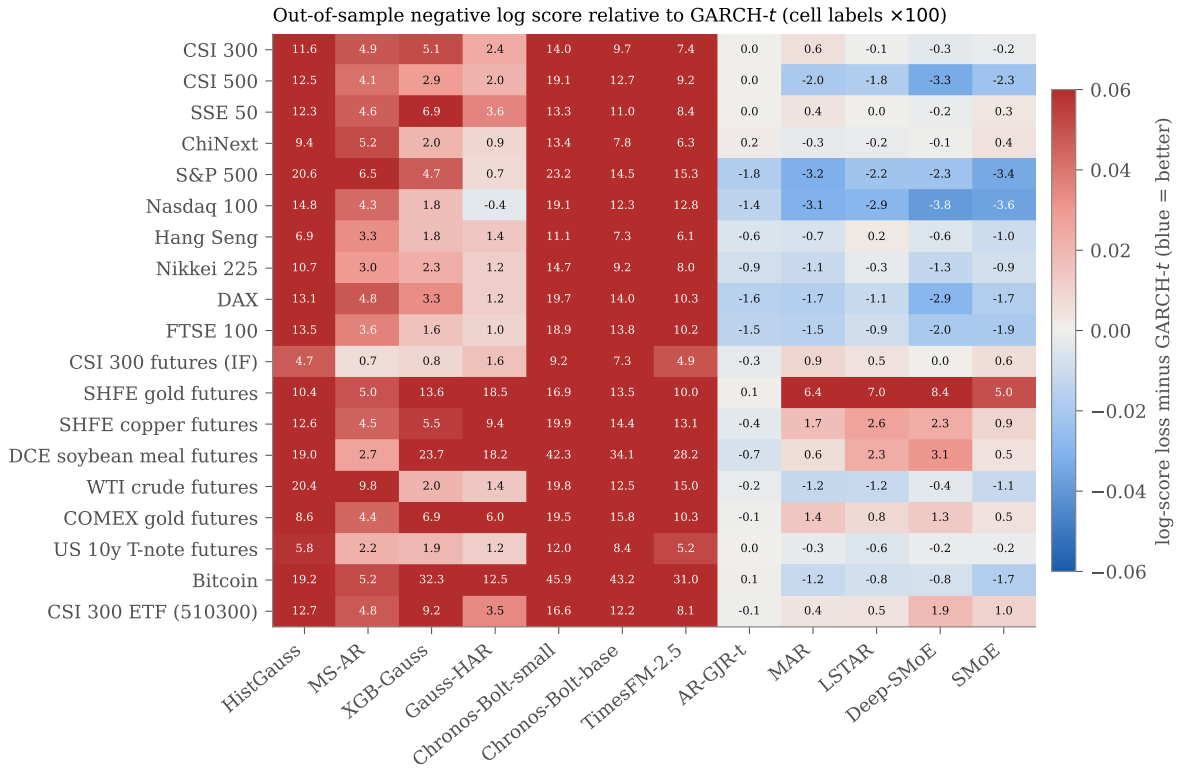


Figure 4: Out-of-sample log-score loss of every model minus that of GARCH-t, by series (blue: better than GARCH-t; red: worse; cell labels $\times 100$). The two leftmost blocks of columns are dominated by every mixture model; the SMoE, the MAR, the LSTAR and the deep SMoE are blue on most equity series.

11 Results II: nineteen markets

11.1 Density forecasts

Across the nineteen series the SMoE has the best average rank in log score (2.95 among 13 models), followed by the deep SMoE (3.3), the MAR (3.5) and AR-GJR-t (3.6). Pairwise, it beats every model outside the GARCH and mixture families on almost every asset: on log score it is significantly better (one-sided DM-HLN, 5%) than the single-regime Gauss-HAR on 17 series, the Markov-switching regression on 17, the boosted model on 17, and each of the three foundation models on all 19, and it is never significantly worse than any of them. Against GARCH-t it is better on 11 of 19 series (significantly on 8; significantly worse on one, SHFE gold), against the MAR on 14 (5 significant, 2 significantly worse) and against the deep SMoE on 10 (6 and 2). The model confidence set at 10% retains the SMoE for 15 series on log score and 18 on both QLIKE and the FZ0 loss for (VaR, ES).

We draw two conclusions. First, most of the gain of a mixture over a single Gaussian comes from the *shape* and *variance dynamics* of the predictive distribution: the constant-weight MAR already captures much of it, and the observable gate adds a small but frequently significant increment. Second, the white-box SMoE is statistically indistinguishable from its black-box counterpart in most markets, a finding we make precise with the yardstick in section 14.

11.2 Which parts of the SMoE matter?

Table 7 removes one component at a time and re-runs the walk-forward on eight series. The components rank clearly. Removing the variance dynamics (constant expert variances) does the most damage: the full model is significantly better on log score for six of eight series and on QLIKE for seven, and never worse. Dropping the short-horizon expert ($K = 2$) costs log score significantly on four series and improves it on one. Removing the semantic exclusion restrictions (letting every expert see every covariate) never improves log score and worsens it on three series: the restrictions buy identification and interpretability at no predictive cost. Finally, a gate driven by volatility alone is statistically indistinguishable from the full three-variable gate, in line with the in-sample Wald tests; a parsimonious practitioner would use it.

Table 7: Out-of-sample ablations of the SMoE on eight series (variant minus full model; positive = variant worse)

Variant	Loss	Median loss difference	Full SMoE significantly better	Full SMoE significantly worse
SMoE-K2	CRPS	-0.0002	1	0
SMoE-K2	FZ5	0.0059	2	0
SMoE-K2	NLL	0.0039	4	1
SMoE-K2	QLIKE	0.0027	3	1
SMoE-constvar	CRPS	0.0044	4	0
SMoE-constvar	FZ5	0.0210	4	0
SMoE-constvar	NLL	0.0161	6	0
SMoE-constvar	QLIKE	0.0369	7	0
SMoE-unrestricted	CRPS	0.0004	1	1
SMoE-unrestricted	FZ5	-0.0037	1	0
SMoE-unrestricted	NLL	0.0031	3	0
SMoE-unrestricted	QLIKE	-0.0024	1	0
SMoE-volgate	CRPS	-0.0000	1	0
SMoE-volgate	FZ5	-0.0034	1	2
SMoE-volgate	NLL	0.0025	1	0
SMoE-volgate	QLIKE	-0.0019	2	1

Series: CSI 300, CSI 500, SSE 50, ChiNext, S&P 500, Hang Seng, COMEX gold, Bitcoin. Counts of series with one-sided DM-HLN $p < 0.05$ in favour of the full model and two-sided $p < 0.05$ against it.

11.3 Means and directions

Consistent with section 3 and with Welch and Goyal (2008), no model forecasts daily means reliably. The median out-of-sample R^2 against a zero forecast is negative for every model, between -0.07% (AR-GJR-t) and -2.3% (TimesFM-2.5). The Clark–West test rejects at 5% for the SMoE on three series (S&P 500, Nasdaq 100, DCE soybean meal) and the hit rate averages 50.9%. On squared error and Brier score the simplest models rank best. In daily data, a distributional model earns its keep in risk forecasts; it does not forecast direction.

11.4 Calibration

Calibration is where the SMoE separates from the GARCH family. Its 1% VaR passes the Kupiec test on 17 of 19 series (GARCH-t: 10; historical Gaussian: 1) and its average 1% violation rate is 0.97%. At 5% the SMoE is slightly conservative (average violation rate 4.2%), which the conformal layer corrects (section 15.1). The PIT standard deviation is 0.97, close to

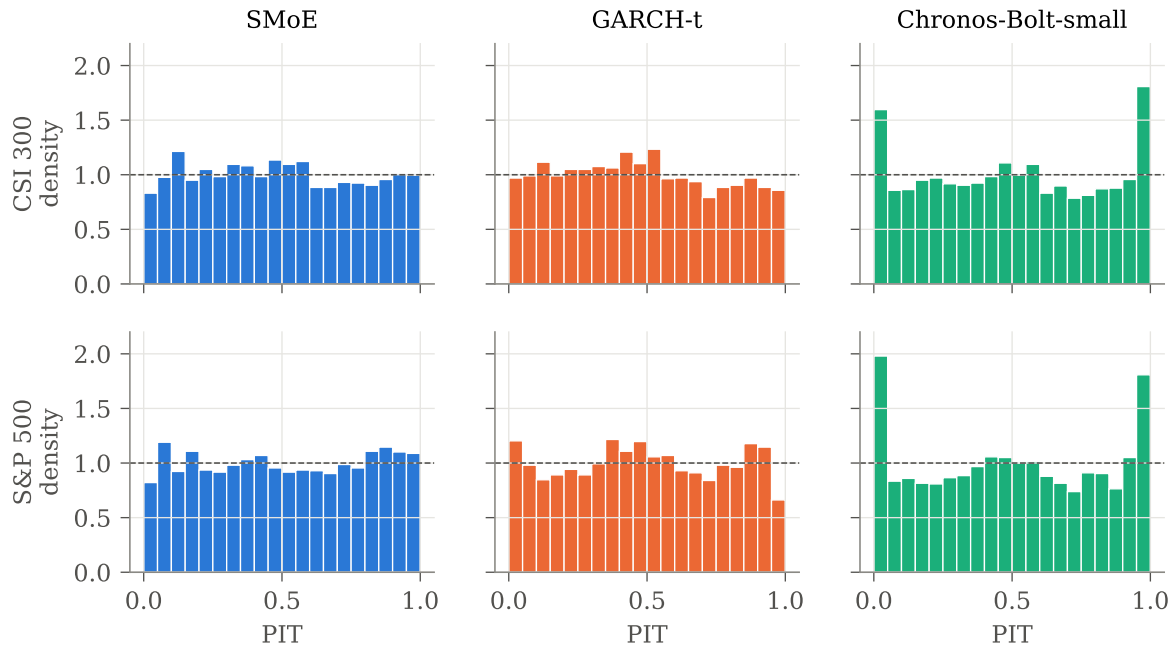


Figure 5: Probability integral transforms of out-of-sample forecasts. A calibrated forecaster has a flat histogram (dashed line). The foundation model's U shape shows predictive distributions that are too narrow.

the ideal of one. The Chronos-Bolt models have average PIT standard deviations of 1.25 and 1.32, i.e. their predictive distributions are far too narrow, which explains their poor log scores (fig. 5). TimesFM 2.5 is better calibrated (1.17) but still too narrow: its 1% VaR is violated on 3.1% of days on average and fails the Kupiec test on every series, as do both Chronos models.

11.5 Cross-market white-box fingerprints

Table 28 collects the key SMoE coefficients for every series and fig. 6 plots their robust t-statistics.

The pattern of the short-horizon expert is the clearest cross-market regularity in the paper. In the Chinese equity indices the scaled one-day return enters with a *positive* sign (CSI 300 $z = 2.2$, CSI 500 $z = 2.9$, ChiNext $z = 1.9$), while the five-day scaled return is *negative* and highly significant in the S&P 500 ($z = -5.2$), the DAX ($z = -3.3$), the Nikkei ($z = -2.3$) and the FTSE ($z = -2.1$). Short-horizon continuation in China and short-horizon reversal in developed markets is consistent with the dominance of retail order flow in the A-share market (Liu et al., 2019); in the SMoE it is not an unconditional autocorrelation but a property of a specific, high-variance regime, with a standard error. Slow momentum is equally market-specific: 120-day momentum enters the trend expert positively and significantly for CSI 300 ($z = 3.8$) and both gold contracts ($z = 3.0$ and 4.5), but negatively for the Nikkei and the DAX ($z = -2.4$ and -3.0), where the trend regime is better described by 20-day momentum.

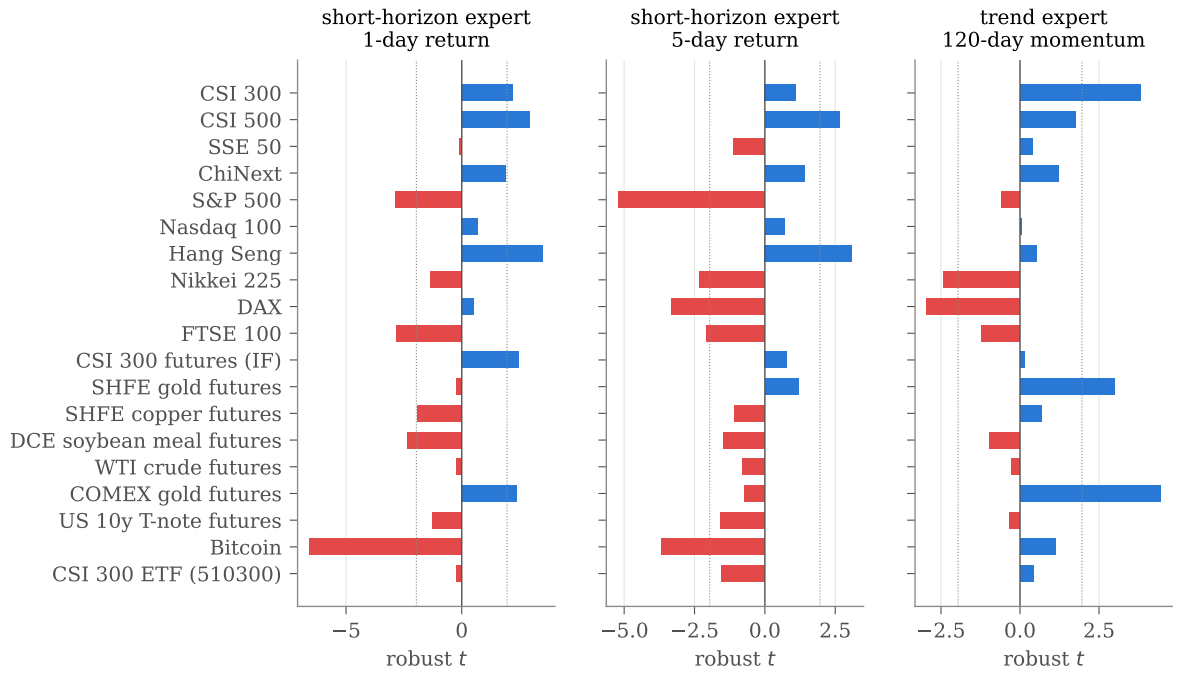


Figure 6: White-box fingerprints: robust t-statistics of the short-horizon expert's loadings on the scaled 1- and 5-day returns and of the trend expert's loading on 120-day momentum, by series. Dotted lines mark ±1.96.

12 Results III: how many experts, and what drives the gate?

The SMoE of section 4 fixes $K = 3$ and a three-variable gate by semantic design. A referee is entitled to ask whether either choice is supported by the data. This section answers with a systematic comparison: six alternative expert sets ($K = 1, \ldots, 5$), five alternative gates, in-sample information criteria, a parametric bootstrap of the likelihood ratio, and a score test that screens candidate state variables from the baseline fit alone. Three theoretical points come first, because the usual tools do not apply directly.

12.1 Theory

Experts that share a semantic group. Two of the larger specifications add a second intercept-only expert, which shares the empty covariate set with the crisis expert and therefore violates assumption 1(c). Identification then has to come from the variance functions.

Proposition 13 (Identification with a shared semantic group)**.** *Let assumption 1(a)–(b) hold and let (c) hold for every pair of experts with different mean-covariate sets. If experts k and l share a covariate set, assume $\delta_{k,0} \neq \delta_{l,0}$ and adopt the labelling convention $\delta_{k,0} < \delta_{l,0}$ (expert k is the calmer one). Then $p_{\boldsymbol{\theta}} = p_{\boldsymbol{\theta}'}$ almost everywhere implies $\boldsymbol{\theta} = \boldsymbol{\theta}'$.*

The proof (appendix A) repeats the argument for proposition 5 up to the constant permutation ς, which can now only exchange experts within a shared group; the convention rules out the exchange. In practice the convention is imposed after estimation by relabelling, and it does not affect any predictive quantity.

Testing K. Testing $K-1$ against K experts is a boundary problem with unidentified nuisance parameters. The null is reached either when the extra expert's gate weight vanishes ($\gamma_{K,0} \to -\infty$, so its mean and variance parameters are not identified) or when the extra expert coincides with an existing one (so the gate parameters separating them are not identified). In either case the information matrix is singular at the null and the likelihood ratio has no χ^2 limit; for normal mixtures it may even diverge slowly (Hartigan, 1985; Chen et al., 2001). We therefore use three tools that do not rely on a χ^2 limit. The first is out-of-sample log score, which is what the paper is about. The second is BIC, $-2\ell_n(\hat{\boldsymbol{\theta}}) + \dim(\boldsymbol{\theta}) \log n$, which is a Laplace approximation to the log marginal likelihood (Schwarz, 1978) and selects the order of a mixture consistently under conditions given by Keribin (2000). The third is the fixed-design parametric bootstrap of McLachlan (1987): hold the covariates at their observed values, draw y_t^* from the fitted null model, re-estimate both models on y^*, and compare the observed LR statistic with the bootstrap distribution. The bootstrap treats the covariates as fixed although they are functions of past returns; this is the standard conditional version of the test and it is conservative when the alternative improves the fit of the covariates' own dynamics, which the SMoE does not model.

Screening gate variables with a score test. Adding a state variable $\boldsymbol{w}_t \in \mathbb{R}^d$ to the gate changes the log-odds to $\boldsymbol{\gamma}_k^\top \tilde{\boldsymbol{z}}_t + \boldsymbol{\psi}_k^\top \boldsymbol{w}_t$. Under $H_0 : \boldsymbol{\psi} = \mathbf{0}$ the number of experts is unchanged, all parameters stay identified, and standard theory applies. The score test is especially convenient here because Fisher's identity (15) gives the new scores in closed form at the restricted estimate.

Proposition 14 (LM test for omitted gate variables)**.** *Let $\hat{\boldsymbol{\theta}}$ be the QMLE of the SMoE with gate covariates $\boldsymbol{z}_t$, and let the augmented model with gate covariates $(\boldsymbol{z}_t, \boldsymbol{w}_t)$ satisfy ?? 1?? 2 at its pseudo-true value $(\boldsymbol{\theta}^*, \boldsymbol{\psi}^*)$. The score of $\boldsymbol{\psi}_k$ at $(\hat{\boldsymbol{\theta}}, \mathbf{0})$ is*

$$\boldsymbol{s}_{\psi_k} = \sum_{t=1}^{n} \left(\hat{\tau}_{tk} - \hat{\pi}_{tk}\right) \boldsymbol{w}_t, \qquad k = 1, \ldots, K-1, \tag{26}$$

the covariance between the candidate variable and the regime surprise, the gap between posterior and prior regime probabilities. Let $\boldsymbol{s}$ stack all scores of the augmented model at $(\hat{\boldsymbol{\theta}}, \mathbf{0})$, partition A and B conformably into the $\boldsymbol{\theta}$ and $\boldsymbol{\psi}$ blocks, and set $C = [-A_{\psi\theta} A_{\theta\theta}^{-1},\ I]$. Then under $H_0 : \boldsymbol{\psi}^* = \mathbf{0}$

$$\mathrm{LM} = (C\boldsymbol{s})^\top \left(C \hat{B} C^\top\right)^{-1} (C\boldsymbol{s}) \xrightarrow{d} \chi^2_{(K-1)d}, \tag{27}$$

with $\hat{A}, \hat{B}$ the Hessian and Newey–West estimators of section 5. Under local alternatives $\boldsymbol{\psi}^* = \boldsymbol{h}/\sqrt{n}$, LM *and the robust Wald statistic from the augmented fit have the same non-central χ^2 limit.*

The test needs only the baseline fit, so a list of candidate state variables can be screened at the cost of one estimation. It also has a direct white-box reading: a state variable belongs in the gate if it predicts the regime surprise $\tau_{tk} - \pi_{tk}$, i.e. if the data assign days to regime k more (or less) often than the gate expected whenever $\boldsymbol{w}_t$ is high. This is a conditional moment test in the sense of Newey (1985) and Tauchen (1985), applied to the latent regime label. The

Table 8: Specifications compared in section 12

Name	K	Experts (mean covariates)	Gate covariates
Gauss-HAR	1	single (trend + short-horizon)	–
SMoE-K2tc	2	trend; crisis	baseline
SMoE-K2rc	2	short-horizon; crisis	baseline
SMoE-K2tr	2	trend; short-horizon	baseline
LSTAR	2	trend; short-horizon	log 22d variance
SMoE	3	trend; short-horizon; crisis	baseline
MAR	3	as SMoE	constant
SMoE-K4c	4	+ calm (intercept only)	baseline
SMoE-K4p	4	+ premium (log 22d variance, drawdown)	baseline
SMoE-K5	5	+ calm + premium	baseline
SMoE-Gvol	3	as SMoE	log 22d variance
SMoE-Gvix	3	as SMoE	baseline + log VIX_{t-1}
SMoE-Gasym	3	as SMoE	baseline + $r_t/\hat{\sigma}_t$
SMoE-Grich	3	as SMoE	baseline + VIX + $r_t/\hat{\sigma}_t$ + log 1d variance

Baseline gate: log 22-day variance, 60-day scaled momentum, drawdown from the 250-day high.

correction C matters: the restricted parameters are estimated, and the uncorrected statistic $\boldsymbol{s}_\psi^\top \hat{B}_{\psi\psi}^{-1} \boldsymbol{s}_\psi$ is not χ^2 under the null.

12.2 Design

Table 8 lists the specifications. Along the K dimension we drop either of the two covariate experts or the crisis expert ($K = 2$), add a second intercept-only "calm" expert, a "premium" expert whose mean depends on the log 22-day variance and the drawdown ($K = 4$), or both ($K = 5$). The premium expert gives the risk–return trade-off of French et al. (1987) a regime of its own: if expected returns rise with variance in some states, its variance loading is positive. Along the gate dimension we use volatility alone, add the log VIX of the previous US trading day (Whaley, 2000; Bekaert and Hoerova, 2014), add the signed last return (an asymmetric gate that can react differently to up and down days), or add all three of VIX, last return and the log one-day range variance. The VIX enters with a one-day lag for every series, so it is known at the local close in every time zone (appendix C). All specifications are estimated in the same walk-forward on all nineteen series.

12.3 Out of sample, three experts are right

Table 9 and fig. 7 summarise the walk-forward comparison. Along the K dimension the baseline is best. Every $K = 2$ specification is significantly worse than the SMoE in log score on four to seven series and significantly better on one; the median loss difference is 0.28 to 0.51 hundredths of a nat per day. Adding experts does not help either: the calm expert ($K = 4$) is never significantly better and is worse on three series, the premium expert is better on two and worse on five, and the five-expert model is better on two and worse on nine. The single-regime Gauss-HAR is worse on 17 of 19 series. Out-of-sample log score therefore traces an inverted U in K with its maximum at three, the number chosen by semantic design before any data were seen. The larger models also calibrate worse: their 1% VaR passes the Kupiec test on 13 to 15 series, against 17 for the SMoE.

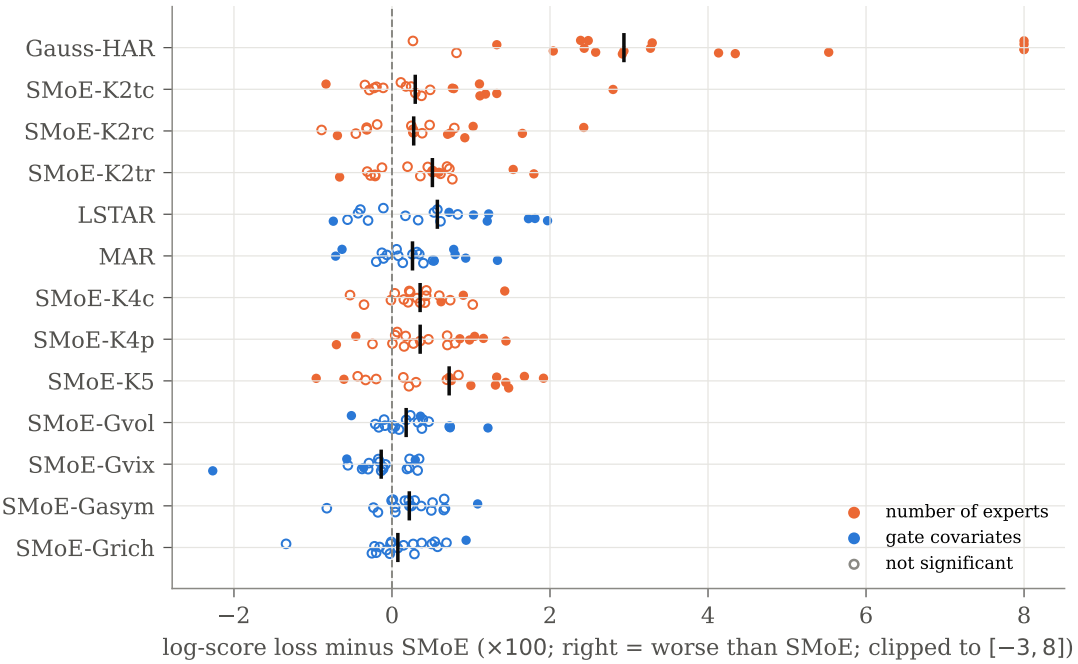


Figure 7: Out-of-sample log-score loss of each specification minus that of the SMoE, one point per series (filled: DM-HLN test significant at 5%, in either direction; vertical bars: medians). Orange: specifications that change the experts; blue: specifications that change only the gate (LSTAR also has $K = 2$).

Along the gate dimension one change helps. Adding the previous day's log VIX gives the best average rank on all four losses (log score 3.26 against 4.79 for the SMoE, among fourteen specifications), a lower log score than the SMoE on 13 of 19 series (significantly lower on three and significantly higher on one), and membership of the 10% model confidence set on all nineteen. The gain is small: the median improvement is 0.14 hundredths of a nat per day. A volatility-only gate is slightly worse than the three-variable gate (significantly on three series), the asymmetric gate that adds the signed last return is significantly different from it on only one series, and the constant-weight MAR is significantly worse on five series. The largest gate (VIX, last return and one-day variance) is no better than the gate with VIX alone.

12.4 In sample, the criteria disagree

In-sample information criteria point in opposite directions (table 11). Among the specifications that differ in K, BIC selects two experts on 14 of 19 series and three on five; AIC selects the five-expert model on nine series. The baseline SMoE is never the BIC choice among all fourteen specifications, where the VIX gate is selected most often (five series). With three to five thousand observations, BIC's penalty of $\log n \approx 8$ per parameter outweighs the improvements in fit that the walk-forward rewards, while AIC's penalty of two lets the larger models fit noise that does not survive out of sample. Neither criterion is a reliable guide to forecasting performance here, which is a further reason to choose K by out-of-sample log score.

The parametric bootstrap makes the same point from a different angle (table 10). For the CSI 300 and the S&P 500 we test $K = 1$ against $K = 3$ with 199 draws and $K = 3$ against

Table 9: Number of experts and gate covariates: out-of-sample comparison on 19 series (development period)

Model	K	Rank NLL	Rank CRPS	Rank QLIKE	Rank FZ5	NLL better	NLL worse	MCS NLL	MCS QLIKE	Kupiec 1%
Gauss-HAR	1	13.68	12.89	8.84	10.32	0	17	3	11	4
SMoE-K2tc	2	7.95	7.53	9.16	7.32	1	7	15	14	17
SMoE-K2rc	2	7.26	6.79	7.21	6.84	1	6	15	14	17
SMoE-K2tr	2	8.42	8.47	7.74	6.63	1	4	15	15	17
LSTAR	2	8.42	9.16	7.95	7.32	1	7	13	14	15
SMoE	3	4.79	5.79	7.16	8.00	–	–	18	16	17
MAR	3	6.74	5.74	6.89	7.05	2	5	16	17	16
SMoE-K4c	4	7.63	7.16	7.32	8.47	0	3	18	17	14
SMoE-K4p	4	8.05	8.68	6.21	8.16	2	5	17	17	13
SMoE-K5	5	9.05	8.68	7.26	8.68	2	9	17	17	15
SMoE-Gvol	3	6.79	7.00	6.95	6.53	1	3	17	16	15
SMoE-Gvix	3	3.26	5.16	5.84	5.63	3	1	19	17	16
SMoE-Gasym	3	7.11	6.11	8.74	7.79	0	1	18	15	15
SMoE-Grich	3	5.84	5.84	7.74	6.26	0	1	19	17	16

Average ranks among the 14 specifications (1 = best). NLL better/worse: series on which the specification is significantly better (one-sided DM-HLN, 5%) or worse (two-sided, 5%) than the SMoE in log score. MCS: series (of 19) on which the specification is in the 10% model confidence set of the 14. Kupiec 1%: series on which the 1% VaR passes at 5%.

$K = 4$ (the calm expert) with 99. The bootstrap null distributions are far to the right of the naive χ^2 limits: for $K = 1$ against $K = 3$ the 95% critical values are 39.2 and 39.2, against 28.9 for a χ^2_{18}, and for $K = 3$ against $K = 4$ they are 24.6 and 21.1, against 16.9 for a χ^2_9. This is the non-regularity described above, and a χ^2 reference would over-reject. Even against the bootstrap, both tests reject: the single regime with LR statistics of 317 and 400 ($p = 0.005$, the smallest attainable with 199 draws), and three experts in favour of four with 62.0 ($p = 0.010$) and 43.6 ($p = 0.020$). In sample, a fourth expert is therefore supported; out of sample, the same four-expert model is never significantly better than the SMoE and is worse on three series. The likelihood ratio detects a fourth regime that is real in sample but does not generalise across quarterly re-estimations.

Table 10: Parametric bootstrap of the likelihood-ratio test for the number of experts

Series	Test	LR	B	Bootstrap 95% crit.	Bootstrap mean	p
CSI 300	K1 vs K3	317.152	199	39.225	27.129	0.005
CSI 300	K3 vs K4	62.018	99	24.568	14.494	0.010
S&P 500	K1 vs K3	399.651	199	39.186	25.977	0.005
S&P 500	K3 vs K4	43.622	99	21.111	14.010	0.020

Fixed-design parametric bootstrap (McLachlan, 1987): y^* drawn from the fitted null model at the observed covariates; both models re-estimated on every draw (best of a warm start and two random starts). $p = (1 + \#\{LR^* \geq LR\})/(1 + B)$. The observed LR uses the same estimation procedure as the draws; the four-start fit of table 6 gives a slightly larger LR for the CSI 300 ($K = 1$ against $K = 3$).

12.5 Which state variables belong in the gate?

Table 12 screens three candidates with the score test of proposition 14, computed from the baseline fit alone. The VIX is rejected as irrelevant on 11 of 19 series, the signed last return on six and the one-day variance on two. The robust Wald test from the augmented VIX fit rejects on twelve, and the two statistics are of the same size series by series (for the S&P 500, 37.6 and 46.3 with two degrees of freedom; for SHFE gold, 36.8 and 49.1), as the proposition predicts when the effect is real.

Table 11: In-sample model selection by BIC (development sample)

Asset	BIC-best (all 14)	BIC-best K-spec.	ΔBIC SMoE
SHFE gold futures	SMoE-Gvix	SMoE-K2rc	107.3
Bitcoin	MAR	SMoE	24.0
ChiNext	LSTAR	SMoE-K2rc	37.9
WTI crude futures	SMoE-K2rc	SMoE-K2rc	53.4
CSI 300	SMoE-Gasym	SMoE-K2tc	46.1
CSI 500	SMoE-Gvol	SMoE	13.9
SHFE copper futures	SMoE-Gvix	SMoE-K2tr	10.6
DAX	SMoE-Gasym	SMoE-K2tr	27.0
CSI 300 ETF (510300)	LSTAR	SMoE-K2tc	50.7
FTSE 100	SMoE-Grich	SMoE-K2tc	35.3
COMEX gold futures	SMoE-Gvix	SMoE-K2tc	20.5
Hang Seng	SMoE-K2rc	SMoE-K2rc	15.5
CSI 300 futures (IF)	LSTAR	SMoE-K2rc	60.3
DCE soybean meal futures	SMoE-Gvol	SMoE	26.0
Nikkei 225	SMoE-Gvix	SMoE	23.7
Nasdaq 100	SMoE-Grich	SMoE-K2rc	45.6
S&P 500	SMoE-Gvix	SMoE	40.4
SSE 50	LSTAR	SMoE-K2tc	45.9
US 10y T-note futures	SMoE-K2rc	SMoE-K2rc	63.9

ΔBIC SMoE: BIC of the baseline SMoE minus the smallest BIC (0 when the SMoE is selected).

The VIX result is geographically structured. By the robust Wald test at 5%, the US fear gauge enters the gate of all five developed equity indices, the Hang Seng, both gold contracts, WTI crude and soybean meal, but not that of the CSI 300 ($p = 0.06$), the CSI 500, ChiNext, the IF futures, copper, the T-note or Bitcoin. Among the A-share series only the SSE 50 and, marginally, the CSI 300 ETF ($p = 0.037$, on a sample that starts in 2012) respond. In the developed equity indices the sign is uniform: at a given level of the index's own 22-day variance, a higher VIX moves weight from the crisis expert to the two covariate experts (z between 1.8 and 6.0). The VIX is forward-looking while the range-based state is not, so conditional on recent realised variance a high VIX appears to signal risk that the trend and short-horizon regimes, with their own variance dynamics, can carry. For the A-share market, the absence of a VIX effect is consistent with its partial segmentation from global risk appetite. For the CSI 300 the score test instead flags the most recent day: the one-day variance ($p = 0.004$) and the signed last return ($p = 0.033$) are both rejected as irrelevant, so regime weights in the Chinese index respond to yesterday's move as well as to the 22-day variance.

The premium expert does not find a risk–return trade-off. Its loading on the log 22-day variance is positive and significant only for the CSI 500 and WTI and negative and significant for the CSI 300 and the SSE 50. Its loading on the drawdown state, by contrast, is negative and significant on seven series (four Chinese equity indices, the FTSE, SHFE gold and the T-note). Because the drawdown state $\log(P_t/\max_{j<250} P_{t-j})$ is non-positive, a negative loading means a higher expected return deep in a drawdown. The fourth expert, given the chance, describes rebounds after large losses rather than compensation for variance.

Summary. Three experts are supported out of sample. In sample, BIC prefers fewer and the bootstrapped likelihood ratio prefers more, and neither choice forecasts better. The three-variable gate is a reasonable default, and the VIX is the only addition that both passes the

Table 12: Screening gate variables: robust LM p-values from the baseline fit (proposition 14), development sample

Asset	LM: VIX	LM: return	LM: 1d var.	LM: all three	Wald VIX p	premium: β_{lv22} (z)
CSI 300	0.083	0.033	0.004	0.003	0.060	-0.950 (-2.8)
CSI 500	0.512	0.224	0.039	0.005	0.252	0.485 (4.6)
SSE 50	0.003	0.018	0.853	0.000	0.000	-0.890 (-2.0)
ChiNext	0.385	0.822	0.493	0.739	0.421	0.351 (1.2)
S&P 500	0.000	0.081	0.205	0.000	0.000	-0.576 (-0.2)
Nasdaq 100	0.000	0.000	0.957	0.000	0.000	0.010 (0.3)
Hang Seng	0.009	0.337	0.141	0.010	0.001	-0.136 (-0.7)
Nikkei 225	0.000	0.013	0.508	0.000	0.000	0.154 (0.5)
DAX	0.027	0.002	0.377	0.004	0.017	0.290 (0.4)
FTSE 100	0.000	0.000	0.070	0.000	0.000	0.143 (1.1)
CSI 300 futures (IF)	0.181	0.167	0.738	0.236	0.074	0.014 (0.1)
SHFE gold futures	0.000	0.464	0.840	0.000	0.000	-0.072 (-1.6)
SHFE copper futures	0.189	0.139	0.191	0.120	0.051	0.067 (0.9)
DCE soybean meal futures	0.001	0.233	0.708	0.004	0.000	0.059 (1.0)
WTI crude futures	0.008	0.274	0.709	0.066	0.040	0.284 (2.3)
COMEX gold futures	0.000	0.510	0.328	0.000	0.000	0.040 (0.5)
US 10y T-note futures	0.385	0.171	0.565	0.235	0.165	0.103 (1.2)
Bitcoin	0.510	0.963	0.154	0.460	0.587	0.110 (0.6)
CSI 300 ETF (510300)	0.052	0.069	0.529	0.056	0.037	-0.329 (-1.1)

LM: p-value of the score test that the candidate is absent from the gate, computed from the baseline SMoE alone ($2d$ df). Wald VIX p: robust Wald test of the same hypothesis for VIX from the augmented fit (SMoE-Gvix). Premium: standardised loading of the premium expert's mean on log 22-day variance in SMoE-K4p, with robust z-statistic.

score test on most series and improves out-of-sample forecasts. Because we made these comparisons on the development sample after the main specification was fixed, the lockbox results of section 16 still refer to the original SMoE.

13 Results IV: variance features from 5-minute data

The SMoE's variance covariates are built from the daily high–low range because the range is available for every series and every day. For the CSI 300 we also have 5-minute bars, and realised variance from intraday returns is the textbook measure of daily variance (Andersen et al., 2003). This section asks whether the conclusions change when the range is replaced by realised measures, and it turns the comparison into a test of a measurement-error prediction that the white box makes possible.

13.1 Theory: what a better variance measure should change

Let $\ell_t = \log \sigma_t^2$ denote the latent log variance of day t's trading session. Every daily estimator we use has the multiplicative form $\hat{v}_t = \sigma_t^2 \varepsilon_t$, so

$$\log \hat{v}_t = \ell_t + \kappa + \eta_t, \qquad \mathbb{E}\eta_t = 0, \quad \operatorname{Var} \eta_t = \omega^2, \tag{28}$$

with η_t independent of $\{\ell_s\}$. Under a Brownian semimartingale with constant intraday volatility and no microstructure noise, ε_t has a known law. For the squared daily return $\varepsilon_t \sim \chi^2_1$ and $\omega^2 = \psi'(1/2) = \pi^2/2 \approx 4.93$, where ψ' is the trigamma function. For realised variance from M intraday returns $\varepsilon_t \sim \chi^2_M/M$, so $\kappa = \psi(M/2) - \log(M/2)$ and $\omega^2 = \psi'(M/2) \approx 2/M$; with $M = 48$ five-minute returns, $\omega^2 = 0.043$. For the Parkinson estimator ε_t is the squared

range of a standard Brownian motion on $[0,1]$ divided by $4\log 2$; its log-variance has no simple closed form and we compute it by simulation (table 13). Microstructure noise and the intraday volatility pattern raise the effective ω^2 of both intraday estimators (Hansen and Lunde, 2006), which is why we also estimate it from the data.

What a log-variance error costs. The cost of a variance error in log score has a simple exact form.

Lemma 1 (Cost of a log-variance error)**.** *Let $y \sim \mathcal{N}(\mu, \sigma^2)$ and let the forecast be $\mathcal{N}(\mu, s^2)$ with $u = \log(s^2/\sigma^2)$. Then*

$$\mathbb{E}\big[-\log\varphi(y;\mu,s^2)\big] - \mathbb{E}\big[-\log\varphi(y;\mu,\sigma^2)\big] = \tfrac{1}{2}\big(u + e^{-u} - 1\big) = \tfrac{u^2}{4} - \tfrac{u^3}{12} + O(u^4), \tag{29}$$

which is one half of the QLIKE loss of s^2 evaluated at the true variance. If u is random with mean b and variance v, independent of y given σ^2, the expected excess log score is $\frac{1}{4}(b^2+v)$ to second order.

The lemma converts the measurement problem into a forecasting prediction: a variance covariate that lowers the variance of the log-variance forecast error by Δv should improve the average log score by about $\Delta v/4$ nats per day. It also explains why log score and QLIKE rank variance forecasts identically to second order.

Where the measurement error goes. The variance covariates enter through the log-HAR vector $\boldsymbol{v}_t = (\log\hat{v}_t, \log\bar{v}_t^{(5)}, \log\bar{v}_t^{(22)})$. Averaging over five or twenty-two days removes most of η, so the error sits mainly in the daily component. Let $\boldsymbol{v}_t^* = (\ell_t, L_t^{(5)}, L_t^{(22)})$ be the latent counterparts with covariance Σ^*, let $\boldsymbol{b}$ and ν_1 be the coefficients and residual variance of the linear projection of ℓ_t on $(L_t^{(5)}, L_t^{(22)})$, and let $\boldsymbol{\delta} = (\delta_1, \delta_5, \delta_{22})$ be the slopes of the projection of the target log variance on $(1, \boldsymbol{v}_t^*)$.

Proposition 15 (Errors in the variance regressors)**.** *If $\boldsymbol{v}_t = \boldsymbol{v}_t^* + \kappa\mathbf{1} + \boldsymbol{\xi}_t$ with $\boldsymbol{\xi}_t$ independent of $\boldsymbol{v}^*$ and of the target, $\mathbb{E}\boldsymbol{\xi}_t = \mathbf{0}$ and $\operatorname{Var}\boldsymbol{\xi}_t = \Sigma_\xi$, the slopes of the projection on the observed covariates are $\boldsymbol{\delta}^{\text{obs}} = (\Sigma^* + \Sigma_\xi)^{-1}\Sigma^*\boldsymbol{\delta}$. If the error is confined to the daily component, $\Sigma_\xi = \omega^2 \boldsymbol{e}_1\boldsymbol{e}_1^\top$, then*

$$\delta_1^{\text{obs}} = \frac{\nu_1}{\nu_1 + \omega^2}\,\delta_1, \qquad \delta_h^{\text{obs}} = \delta_h + \frac{\omega^2}{\nu_1 + \omega^2}\,b_h\,\delta_1 \quad (h = 5, 22), \tag{30}$$

and the sum of slopes changes by $-\frac{\omega^2}{\nu_1+\omega^2}\delta_1(1 - b_5 - b_{22})$.

Three predictions follow. First, the daily loading is attenuated by $\nu_1/(\nu_1+\omega^2)$, where ν_1 is not the variance of daily log variance but the much smaller part of it that weekly and monthly averages do not explain. Even a modest ω^2 can therefore remove most of the daily loading. Second, the lost weight moves to the weekly and monthly components in proportion to how well they predict the daily one, so a noisy daily measure makes volatility look like a slower process than it is. Third, when $b_5 + b_{22} \approx 1$, as for a persistent log variance, total persistence is

almost unchanged: measurement error re-allocates weight across horizons rather than removing it. Comparing a range-based and a realised-variance fit gives the testable ratio

$$\frac{\delta_1^{\mathrm{PK}}}{\delta_1^{\mathrm{RV}}} = \frac{\nu_1 + \omega^2_{\mathrm{RV}}}{\nu_1 + \omega^2_{\mathrm{PK}}}. \tag{31}$$

The SMoE's variance equation is a log-link Gamma regression rather than a linear projection, so (30) holds there to the order of the regression-calibration approximation (Carroll et al., 2006); we check it on the $K = 1$ model and on each expert.

Specifications. We estimate six SMoE variants that differ only in the variance covariates and the volatility state of the gate. SMoE-PK is the baseline. SMoE-RV replaces the three range components by the log-HAR of realised variance and the gate's volatility state by $\log \overline{\mathrm{RV}}^{(22)}$. SMoE-TV adds the squared overnight return to realised variance, because the target is a close-to-close return and the trading-session measures miss the overnight gap. SMoE-CJ splits realised variance into a continuous part, measured by bipower variation (Barndorff-Nielsen and Shephard, 2004), and a jump term $\log(1 + J_t)$ with $J_t = \max(\mathrm{RV}_t - \mathrm{BV}_t, 0)$, in the spirit of the HAR-CJ model (Andersen et al., 2007; Corsi et al., 2010). SMoE-SV uses the negative and positive realised semivariances of day t in place of the daily component, following Patton and Sheppard (2015), who find that "bad" volatility predicts future volatility more strongly than "good" volatility. SMoE-PKRV includes both range and realised components, so it encompasses SMoE-PK and SMoE-RV. The benchmarks are the $K = 1$ versions of SMoE-PK and SMoE-RV, GARCH-t, and the log-linear Realized GARCH of Hansen et al. (2012) with Student-t errors,

$$r_t = \mu + \sqrt{h_t} z_t, \quad \log h_t = \omega + \beta \log h_{t-1} + \gamma \log \mathrm{RV}_{t-1}, \quad \log \mathrm{RV}_t = \xi + \phi \log h_t + \tau_1 z_t + \tau_2 (z_t^2 - 1) + u_t, \tag{32}$$

which is the standard way to bring realised measures into a return density and is estimated by maximising the joint likelihood of $(r_t, \log \mathrm{RV}_t)$.

We use the CSI 300 and the two instruments written on it, the IF futures and the 510300 ETF; all three use the index's realised measures. Every model is re-estimated on the same rows (days with 5-minute data) in the same quarterly walk-forward, from the first quarter with 750 training days to the end of the development period. The 5-minute data cover only 246 lockbox days, so the lockbox is not used here.

13.2 The measurement-error prediction holds

Table 13 puts numbers on proposition 15. Under Brownian motion the log Parkinson estimator has $\omega^2 = 0.337$; the value implied by the data, from the residual variance of the daily range on its weekly and monthly averages, is 0.350. The two agree closely, so the index's daily range behaves almost exactly like the range of a Brownian path. Realised variance has $\omega^2 = 0.043$. The residual variance of daily log variance given its weekly and monthly averages is only $\nu_1 = 0.146$, and this small denominator is what makes the range costly: the attenuation factor $\nu_1/(\nu_1 + \omega^2)$ is 0.30 for the range and 0.77 for realised variance. Equation (31) then predicts $\delta_1^{\mathrm{PK}}/\delta_1^{\mathrm{RV}} = 0.39$ (or 0.38 with the data-implied ω^2). In the $K = 1$ model the estimated daily

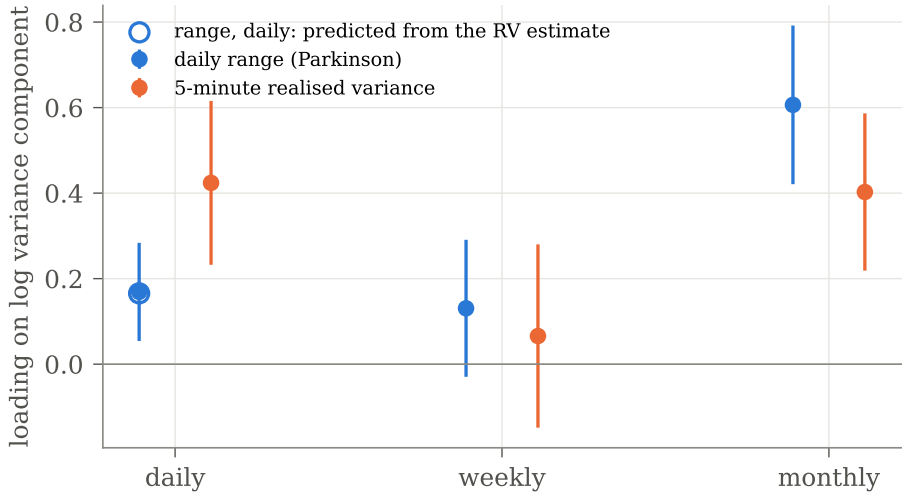


Figure 8: Log-HAR loadings of the $K = 1$ model for CSI 300 with range-based and realised daily variance (95% intervals from sandwich standard errors). The hollow circle is the range-based daily loading predicted by proposition 15 from the realised-variance estimate and the measurement-error variances of table 13.

loadings are 0.169 (s.e. 0.059) with the range and 0.424 (s.e. 0.098) with realised variance, a ratio of 0.40 (fig. 8).

The other two predictions hold in the direction stated. With the range, the weekly and monthly loadings are larger (0.131 and 0.606 against 0.066 and 0.403), so the noisy daily measure makes CSI 300 volatility look slower than it is, and the sum of the three loadings is almost unchanged (0.906 against 0.893). The proposition puts the shifted weight on the weekly component ($b_5 = 0.96$, $b_{22} \approx 0$), whereas the estimates move it mainly to the monthly one. The weekly and monthly components are highly collinear and their loadings have standard errors of 0.08 to 0.11, so this split is not well determined. Inside the three-expert SMoE the expert-specific daily loadings are too imprecise for the ratio to be informative, and we do not interpret them.

Table 13: Measurement error of daily variance estimators and the implied attenuation of the daily HAR loading (CSI 300, development sample)

Daily estimator	$\omega^2 = \operatorname{Var} \eta_t$	$\nu_1/(\nu_1 + \omega^2)$
squared return	4.935	0.029
Parkinson range (BM)	0.337	0.303
Parkinson range (implied)	0.350	0.295
5-min RV (M=48)	0.043	0.775

ω^2 for squared returns and RV from the χ^2 laws in the text; Parkinson (BM) by simulation of 20,000 Brownian paths with 2,400 steps; Parkinson (implied) from the residual variance of the daily range on its weekly and monthly averages minus ν_1. $\nu_1 = 0.146$ is the residual variance of daily log RV on its weekly and monthly averages minus ω^2_{RV}; projection coefficients $b_5 = 0.958$, $b_{22} = -0.002$. Predicted $\delta_1^{\mathrm{PK}}/\delta_1^{\mathrm{RV}}$ from (31): 0.39 (BM) and 0.38 (implied); estimated in the $K = 1$ model: 0.40.

13.3 Out-of-sample results

Table 14 reports the walk-forward comparison. Four results stand out.

Realised measures improve the SMoE everywhere, modestly. Each of the five realised-variance specifications has a better log score than SMoE-PK on each of the three series (15 of 15 comparisons). The gains range from 0.14 to 1.61 hundredths of a nat per day and are significant in 4 of the 15 comparisons. They are clearer for the variance forecast itself: on the CSI 300 every realised specification has a significantly lower QLIKE than SMoE-PK, and against the 5-minute proxy QLIKE falls from 0.331 to 0.289–0.306 and the Mincer–Zarnowitz R^2 rises from 0.19 to 0.34–0.36 (except for SMoE-CJ, whose R^2 of 0.11 illustrates how fragile that statistic is on variance levels). The refinements of SMoE-RV add little: the overnight term (TV), the jump split (CJ), the semivariances (SV) and the encompassing model (PKRV) all lie within one hundredth of a nat of SMoE-RV on every series.

The mixture still adds to a single regime. With realised covariates on both sides, SMoE-RV has a significantly better log score than the single-regime Gauss-HAR-RV on all three series (one-sided $p \approx 0.02$ in each case), and the single-regime models fail the 1% Kupiec test on more series. The size of the mixture's advantage shrinks, however, from 0.022 nats per day with range-based covariates to 0.016 with realised ones, a point we return to below.

A single-regime model with realised variance is the best density forecaster here. The log-linear Realized GARCH with Student-t errors has the best log score on all three series. It beats SMoE-PK significantly everywhere and SMoE-RV significantly on the CSI 300 ($p < 0.001$) and the ETF ($p = 0.008$), but not on the IF futures ($p = 0.17$). On QLIKE and on the FZ0 tail losses it is statistically indistinguishable from SMoE-RV, and the 10% model confidence set for log score keeps every model except the range-based Gauss-HAR on all three series. We report this result plainly: once a precise variance measure is available, a GARCH recursion that uses it with fat-tailed errors is at least as good as three readable regimes, and on two of three series better.

Lemma 1 explains most of the single-regime gains and why the mixture gains less. Table 15 compares the log-score difference predicted from the accuracy of each model's variance forecast, $\Delta \operatorname{Var}(e_t)/4$, with the difference observed. For the single-regime pair the prediction is close: -1.01 against -1.15 hundredths of a nat when the range is replaced by realised variance; for Realized GARCH it is -2.43 against -2.08. For the SMoE the observed gains are roughly a third of the prediction (-0.49 against -1.34 for SMoE-RV). The lemma assumes a Gaussian predictive; a scale mixture is less sensitive to an error in its overall variance because its components already spread probability over several variance levels. Part of the mixture's advantage in the main results is therefore insurance against a noisy variance covariate, and that part shrinks when the covariate improves. This is consistent with the smaller mixture advantage noted above.

13.4 In-sample white box

Table 16 re-estimates the CSI 300 SMoE on the days with 5-minute data (June 2009 to June 2024). Realised covariates raise the log-likelihood by 16.6 (SMoE-RV) and 21.4 (SMoE-TV)

Table 14: Realised-variance covariates out of sample (development period)

Model	CSI NLL	CSI QLIKE	CSI FZ5	IF NLL	ETF NLL	QLIKE_{RV}	MZ R^2_{RV}	Kupiec 1%
SMoE-PK	1.5727	1.5848	1.0246	1.4866	1.4796	0.331	0.189	3
SMoE-RV	−0.49	−2.93*	−1.82	−1.21	−0.97*	0.291	0.351	3
SMoE-TV	−0.14	−2.42*	−0.72	−0.92	−0.79	0.306	0.358	3
SMoE-CJ	−0.31	−2.33*	−1.37	−1.50*	−1.08*	0.300	0.112	3
SMoE-SV	−0.56	−2.12*	−1.70	−1.61*	−0.56	0.289	0.338	3
SMoE-PKRV	−0.26	−1.79*	−1.00	−0.71	−0.58	0.304	0.343	3
Gauss-HAR-PK	2.21	−0.67	0.19	0.82	2.43	0.317	0.270	2
Gauss-HAR-RV	1.06	−3.02*	−1.72	0.17	1.38	0.286	0.326	1
GARCH-t	−0.51	1.54	1.04	−0.92	−1.02*	0.350	0.237	3
RealGARCH-t	−2.08*	−3.32*	−1.98	−1.95*	−2.24*	0.272	0.401	2

First row: SMoE-PK losses. Other rows in loss columns: loss of the model minus loss of SMoE-PK, ×100 (negative = model better); *: one-sided DM-HLN $p < 0.05$ in favour of the model. QLIKE_{RV} and MZ R^2_{RV}: CSI 300 variance forecasts against $\text{RV}_{t+1} + r^2_{\text{overnight},t+1}$. Kupiec 1%: series (of 3) on which the 1% VaR passes at 5%. Common samples: CSI 300 $n = 2850$, IF $n = 904$, ETF $n = 2062$.

Table 15: Log-score gains predicted by lemma 1 from the accuracy of the variance forecast, CSI 300

Model	$\text{Var}(e_t)$	Predicted ΔNLL	Observed ΔNLL
SMoE-PK	0.465	0.000	0.000
SMoE-RV	0.412	-1.336	-0.490
SMoE-TV	0.429	-0.898	-0.141
SMoE-CJ	0.418	-1.176	-0.308
SMoE-SV	0.419	-1.148	-0.560
SMoE-PKRV	0.435	-0.757	-0.259
Gauss-HAR-PK	0.441	-0.594	2.206
Gauss-HAR-RV	0.401	-1.604	1.055
GARCH-t	0.485	0.491	-0.509
RealGARCH-t	0.368	-2.428	-2.083

$e_t = \log(\text{RV}_{t+1} + r^2_{\text{overnight},t+1}) - \log \hat{V}_t$, with $\hat{V}_t$ the forecast variance. Predicted ΔNLL = $\{\text{Var}(e_t) - \text{Var}(e_t^{\text{SMoE-PK}})\}/4$; observed ΔNLL is the out-of-sample log-score difference from SMoE-PK. Both ×100, nats per day; negative = better than SMoE-PK. The error of the proxy is common to all models and cancels.

points at the same number of parameters, and BIC selects SMoE-TV. Two of the white-box conclusions of section 10 survive the change of variance measure and one weakens. Momentum and drawdown remain absent from the gate under every specification (p between 0.24 and 0.45). The volatility effect in the gate is weaker on this shorter sample, which starts after the 2007–08 boom and bust: the Wald statistic is 6.5 ($p = 0.04$) with the range, 3.4 ($p = 0.19$) with trading-session realised variance and 14.6 ($p < 0.001$) when overnight variance is included. Volatility is still the only candidate regime variable, but how strongly it moves the gate depends on whether the overnight gap is measured. Finally, the semivariance model does not support the asymmetry that Patton and Sheppard (2015) find for US stocks: the hypothesis that negative and positive semivariances have equal loadings in all three experts is not rejected ($W = 1.42$, $p = 0.70$).

Table 16: In-sample fits with range-based and realised variance covariates, CSI 300 (June 2009–June 2024, days with 5-minute data)

Model	Log-lik.	Params	BIC	W vol. in gate	W mom.+dd in gate
SMoE-PK	-5857.5	27	11,937	6.54 (0.038)	5.53 (0.237)
SMoE-RV	-5841.0	27	11,903	3.35 (0.187)	5.31 (0.257)
SMoE-TV	-5836.1	27	11,894	14.56 (0.001)	3.86 (0.425)
SMoE-CJ	-5845.9	30	11,938	1.11 (0.574)	4.71 (0.318)
SMoE-SV	-5839.5	30	11,925	6.46 (0.040)	3.66 (0.454)
SMoE-PKRV	-5829.8	36	11,955	1.33 (0.515)	4.66 (0.324)
Gauss-HAR-PK	-5999.5	9	12,073	–	–
Gauss-HAR-RV	-5960.8	9	11,995	–	–

Robust Wald statistics (p-values) that the volatility state (2 df) and the momentum and drawdown states (4 df) are absent from the gate. Equality of the negative and positive semivariance loadings in all three experts of SMoE-SV: $W = 1.42$ (3 df, $p = 0.70$).

Implications. The regime structure and the precision of the variance measure are partly substitutes: a better measure reduces the gain from mixing without removing it. The main results, which use the range for every series, therefore understate what any model with a precise variance measure achieves, and they somewhat overstate the share of the SMoE's advantage that is due to its regimes. Where intraday data exist, the natural next step is an SMoE whose experts have Realized-GARCH-type variance recursions, which would combine the readable regimes with the variance dynamics that won here (section 17).

14 Results V: how black are the black boxes?

14.1 Design

For each teacher and series we fit, at the start of every calendar year Y, an SMoE shadow to ten draws from the teacher's *out-of-sample* predictive distribution on every earlier date, and use the shadow to forecast year Y. The first three years of each teacher's out-of-sample history serve as burn-in, so the shadow is evaluated from 2015 onwards on most series. The shadow never sees a realised return: it learns only what the teacher predicted, and it learns it with

information available at the time. The reference model for the fidelity ratio is the historical Gaussian. Table 17 summarises; fig. 9 shows every series.

14.2 The deep SMoE is 92% white

The deep heterogeneous MoE improves on the historical Gaussian by a median of 0.129 nats per day. Its SMoE shadow achieves 0.122, a median fidelity of $\Phi = 0.92$ with interquartile range $[0.89, 0.96]$. The shadow reproduces the deep model's predictive variance almost exactly (median R^2 of log variance 0.88) but not its predictive mean ($R^2 = 0.20$), which is consistent with section 11: the deep model's means carry no out-of-sample information, so there is nothing for a white box to reproduce. The remaining eight percent is not noise: the Amisano–Giacomini test rejects equal log scores of teacher and shadow on 12 of 19 series, always in the teacher's favour. The black box knows a little more than three readable regimes do, and the fidelity ratio measures how much more.

The probes give a complementary reading. The deep model's gate is *not* a function of the SMoE's state variables: linear probes of its gate logits on volatility, momentum and drawdown have R^2 between 0.04 and 0.84 with a median of 0.26, and only 9 of 57 expert-series probes exceed 0.5. The deep network therefore routes observations differently from the SMoE but ends up with almost the same predictive distribution. When two different internal mechanisms produce the same input–output map, inspecting internal weights (as the undergraduate thesis did with its gating plots) can mislead, whereas a yardstick defined on the output cannot.

14.3 Gradient boosting, Markov switching and GARCH

The boosted model's shadow reproduces all of its gain ($\Phi = 1.03$; log-variance $R^2 = 0.99$): its predictive distribution is, to a first approximation, a log-HAR variance with a noisy mean. The Markov-switching regression has $\Phi = 0.93$ with the widest dispersion, reflecting that its latent regime is filtered from past returns rather than from observables. GARCH-t has $\Phi = 0.98$, a useful sanity check: the SMoE family contains close approximations to the GARCH predictive.

14.4 Foundation models: the shadow beats the teacher

The zero-shot Chronos-Bolt forecasts are no better than the historical Gaussian on the median series (gain -0.054 for the small and -0.004 for the base model), because their predictive distributions are too narrow (PIT standard deviations 1.32 and 1.25; section 11). Their SMoE shadows, fitted to the same forecasts, gain $+0.035$ and $+0.062$. The small model's shadow beats its teacher on every series, significantly on 15 of 19; the base model's shadow beats its teacher on 17 series, significantly on 14. Scaling the foundation model improves its calibration, but not enough to close the gap to its own white-box reading.

TimesFM 2.5, a different architecture released ten months after Chronos-Bolt, is the first foundation model in our comparison that beats the historical Gaussian on the median series, by 0.012 nats per day; its PIT standard deviation of 1.17 is the closest to one among the three. The pattern is nevertheless the same. Its shadow gains 0.043, beats its teacher on 14 of 19 series and does so significantly on 10, while the teacher is never significantly better

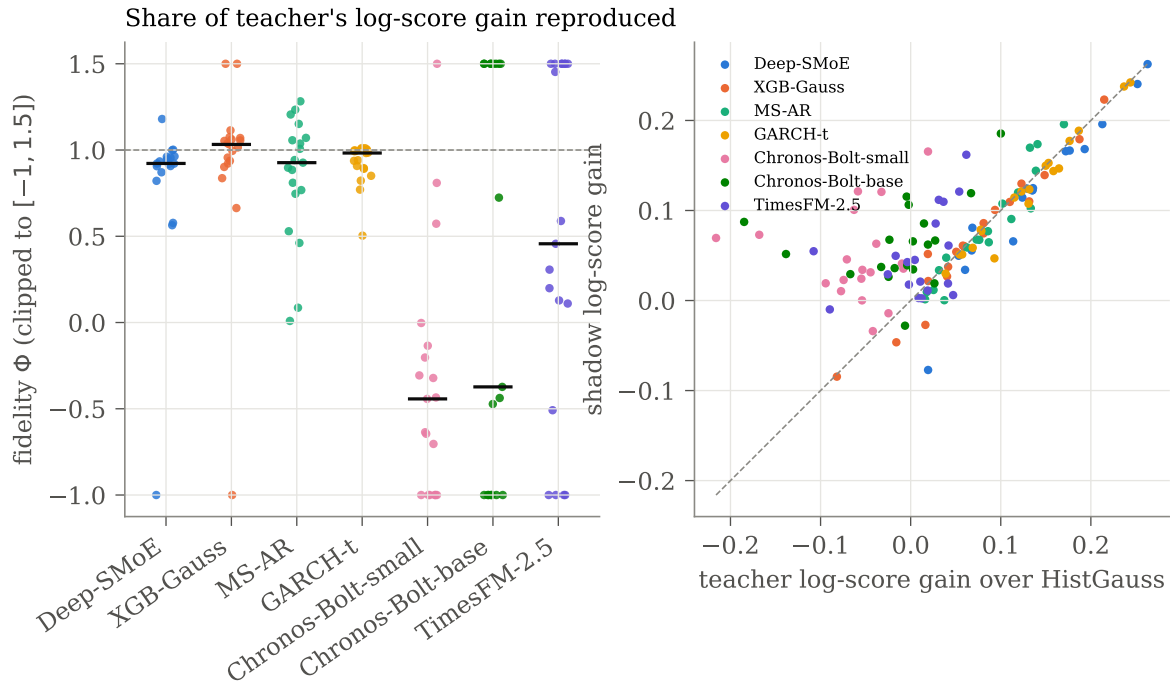


Figure 9: The white-box yardstick. Left: fidelity Φ per teacher and series (horizontal bars are medians). Right: out-of-sample log-score gain over the historical Gaussian of each teacher (x) and of its SMoE shadow (y); points above the diagonal are series on which the shadow beats its teacher.

than its shadow. Projecting a foundation model onto a structured likelihood family therefore acts as a regulariser: the shadow keeps the information in the teacher's location and relative-scale forecasts (median log-variance R^2 between 0.48 and 0.57 for the three models) while the semantic variance experts re-anchor the overall scale to the data. For the Chronos models fidelity is negative because the teacher loses to the reference, and for TimesFM the teacher's gain is too small for the ratio $\Phi = 0.46$ to be read as a share. The common finding is that none of the three foundation models produces a predictive distribution that is usable as is, while a white-box reading of each one is.

Table 17: The white-box yardstick: fidelity of SMoE shadows of black-box forecasters (out of sample)

Teacher	Series	Median LS gain	Median Φ	R^2 mean	R^2 log var	R^2 P(up)	Median KL	AG reject
Chronos-Bolt-base	19	-0.004	-0.373	0.109	0.566	0.157	0.052	14
Chronos-Bolt-small	19	-0.054	-0.443	0.143	0.478	0.244	0.063	15
Deep-SMoE	19	0.129	0.922	0.198	0.877	0.175	0.019	12
GARCH-t	19	0.131	0.983	0.003	0.804	0.249	0.016	5
MS-AR	19	0.086	0.927	0.282	0.579	0.379	0.025	9
TimesFM-2.5	19	0.012	0.457	0.254	0.494	0.310	0.050	10
XGB-Gauss	19	0.060	1.032	0.013	0.986	0.035	0.007	5

LS gain: teacher log score minus historical-Gaussian log score. Φ: share of that gain reproduced by the SMoE shadow. AG reject: number of series on which the Amisano–Giacomini test rejects equal log scores of teacher and shadow at 5%.

15 Results VI: calibration guarantees and economic value

15.1 Conformal calibration

Table 18 applies the adaptive conformal layer of theorem 2 with $\eta = 0.005$ to the VaR forecasts of five models on all nineteen series. As the theorem guarantees, after calibration every model's violation rate is within the bound of the target and the Kupiec test rejects for no series. The informative comparison is conditional coverage. Calibration fixes the *frequency* of violations but not their *clustering*: after calibration the historical Gaussian still fails the Christoffersen conditional-coverage test on 53% of series at the 5% level, whereas the calibrated SMoE fails on none. The conformal layer and the distributional model are complements: the former delivers the guarantee, the latter makes the violations unpredictable.

Table 18: VaR violations before (raw) and after adaptive conformal calibration (ACI, $\eta = 0.005$)

Model	a	Calibration	Mean rate	Max $\lvert\text{rate}-a\rvert$	Kupiec reject	Christoffersen reject
Deep-SMoE	0.0100	ACI	0.0093	0.0029	0.0000	0.0000
Deep-SMoE	0.0100	raw	0.0093	0.0074	0.2105	0.2105
Deep-SMoE	0.0500	ACI	0.0483	0.0085	0.0000	0.1053
Deep-SMoE	0.0500	raw	0.0423	0.0292	0.3684	0.3158
GARCH-t	0.0100	ACI	0.0098	0.0012	0.0000	0.0000
GARCH-t	0.0100	raw	0.0125	0.0074	0.4737	0.4211
GARCH-t	0.0500	ACI	0.0495	0.0046	0.0000	0.0526
GARCH-t	0.0500	raw	0.0543	0.0154	0.4211	0.4737
Gauss-HAR	0.0100	ACI	0.0098	0.0012	0.0000	0.0000
Gauss-HAR	0.0100	raw	0.0163	0.0142	0.7895	0.8421
Gauss-HAR	0.0500	ACI	0.0479	0.0069	0.0000	0.1053
Gauss-HAR	0.0500	raw	0.0443	0.0312	0.3684	0.4211
HistGauss	0.0100	ACI	0.0101	0.0009	0.0000	0.1579
HistGauss	0.0100	raw	0.0203	0.0187	0.9474	1.0000
HistGauss	0.0500	ACI	0.0487	0.0069	0.0000	0.5263
HistGauss	0.0500	raw	0.0482	0.0157	0.3684	0.7368
SMoE	0.0100	ACI	0.0094	0.0016	0.0000	0.0526
SMoE	0.0100	raw	0.0096	0.0074	0.1053	0.2632
SMoE	0.0500	ACI	0.0485	0.0058	0.0000	0.0000
SMoE	0.0500	raw	0.0420	0.0253	0.5263	0.3684

Rates averaged over 19 series; reject columns: share of series rejecting at 5%.

15.2 Trading the CSI 300

We trade the CSI 300 ETF (long-only, 3bp per side) and IF futures (long/short, 2bp) from the CSI 300 predictive distributions with four rules fixed in advance: the sign of $P(y > 0) - 1/2$, volatility targeting at 1% daily, volatility targeting tilted by $2P(y > 0)$, and the mean-variance rule with the no-trade band of proposition 11 ($\gamma = 5$). Decisions are taken at the close and executed at the next open (tables 19 and 20).

Three results summarise the evidence. First, on the ETF several mean-variance rules roughly halve the maximum drawdown of buy-and-hold (from -46% to between -21% and -27%) and raise the Sharpe ratio from 0.11 to 0.3–0.45. Second, none of these improvements is statistically reliable: Sharpe-ratio standard errors are about 0.3 over twelve years, and every 95% stationary-bootstrap interval for the excess return over buy-and-hold contains zero. Third, the search over models, rules and parameters has to be accounted for. Across the 360 configurations at the realistic cost level, the deflated Sharpe ratio of the best configuration is

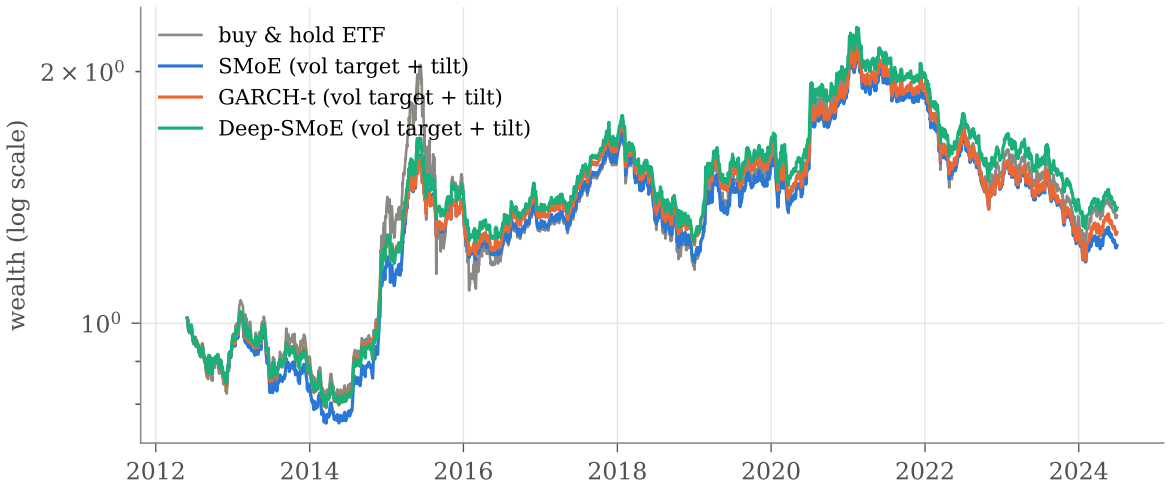


Figure 10: Wealth of the volatility-target-with-tilt rule on the CSI 300 ETF, net of costs, against buy and hold.

0.76, below the 0.95 threshold (0.56 if all 1,440 configurations across cost levels are counted as trials), and the probability of backtest overfitting is 0.51 (table 21): the configuration that looks best in one half of the sample is no better than the median configuration in the other half.

The economic conclusion matches the statistical one. Daily predictive distributions of the CSI 300 are good enough to control risk (exposure falls when regime disagreement and volatility rise, which is what reduces drawdowns), but they do not deliver reliable directional alpha. The white box shows the mechanism: positions shrink through the variance term of (6) rather than through the mean.

16 Results VII: the lockbox

All modelling choices (covariates, number of experts, penalties, refit schedules, trading rules and their parameters) were fixed on the development sample ending 30 June 2024. We then re-estimated every model from scratch on data up to 30 June 2024 and ran the same walk-forward through 28 October 2026, a period of 563 to 849 trading days per series that includes the September 2024 policy rally in China and the April 2025 tariff shock. The lockbox was evaluated once; no result in this section fed back into any choice. TimesFM 2.5 was added to the comparison after the lockbox had been opened. As a zero-shot model it has no parameters or settings that could have been tuned on lockbox data, but its inclusion changes the ranks and model confidence sets, which are relative; the results below are for all thirteen forecasters, and the twelve-model results differ only in these relative statistics.

Density forecasts. The ranking of model families survives intact (table 22). The four mixture models occupy the top four average ranks on log score (LSTAR 3.05, SMoE 3.11, MAR 3.21, deep SMoE 3.26), and the SMoE has the best average rank on QLIKE (2.95) and on the FZ0 tail loss (3.26). The SMoE is in the 10% model confidence set for every one of the nineteen

Table 19: Trading results: CSI 300 ETF (510300), long-only, 3bp, out of sample

Model	Rule	Return	Vol	Sharpe	SE	MaxDD	Avg w	Turnover	Excess CI lo	Excess CI hi
Deep-SMoE	meanvar	0.059	0.130	0.452	0.297	-0.266	0.389	41.380	-0.045	0.122
Gauss-HAR	meanvar	0.034	0.111	0.304	0.294	-0.230	0.375	14.048	-0.082	0.106
SMoE	meanvar	0.038	0.127	0.300	0.295	-0.261	0.419	15.460	-0.075	0.103
MAR	meanvar	0.030	0.116	0.259	0.297	-0.208	0.376	6.034	-0.093	0.098
Chronos-Bolt-small	meanvar	0.020	0.147	0.135	0.294	-0.421	0.455	20.223	-0.082	0.066
HistGauss	meanvar	0.020	0.150	0.134	0.295	-0.407	0.333	0.783	-0.098	0.083
GARCH-t	meanvar	0.016	0.156	0.102	0.293	-0.476	0.806	0.845	-0.065	0.052
Chronos-Bolt-small	sign	0.046	0.171	0.271	0.295	-0.397	0.541	38.255	-0.049	0.090
Deep-SMoE	sign	0.037	0.188	0.196	0.294	-0.412	0.619	68.275	-0.048	0.067
HistGauss	sign	0.030	0.179	0.171	0.295	-0.438	0.486	7.033	-0.070	0.083
GARCH-t	sign	0.025	0.226	0.112	0.293	-0.463	1.000	0.000	0.000	0.000
MAR	sign	0.003	0.178	0.018	0.293	-0.447	0.595	20.585	-0.092	0.047
Gauss-HAR	sign	-0.010	0.154	-0.067	0.292	-0.537	0.556	50.263	-0.104	0.039
SMoE	sign	-0.014	0.178	-0.081	0.292	-0.459	0.581	29.334	-0.108	0.020
HistGauss	voltarget	0.026	0.164	0.158	0.293	-0.419	0.786	0.490	-0.040	0.043
GARCH-t	voltarget	0.021	0.150	0.141	0.293	-0.441	0.795	3.262	-0.051	0.047
Chronos-Bolt-small	voltarget	0.023	0.181	0.128	0.293	-0.461	0.912	5.697	-0.037	0.037
Deep-SMoE	voltarget	0.018	0.151	0.118	0.293	-0.434	0.803	8.872	-0.057	0.042
SMoE	voltarget	0.016	0.153	0.106	0.293	-0.439	0.814	7.403	-0.059	0.040
Gauss-HAR	voltarget	0.015	0.153	0.101	0.293	-0.434	0.813	9.140	-0.061	0.041
MAR	voltarget	0.015	0.152	0.098	0.293	-0.433	0.810	10.543	-0.062	0.041
Deep-SMoE	vt_tilt	0.027	0.158	0.173	0.293	-0.425	0.813	9.437	-0.038	0.046
HistGauss	vt_tilt	0.027	0.167	0.159	0.294	-0.422	0.786	0.688	-0.037	0.041
GARCH-t	vt_tilt	0.021	0.156	0.137	0.293	-0.450	0.825	3.148	-0.049	0.046
SMoE	vt_tilt	0.018	0.157	0.117	0.293	-0.443	0.819	8.233	-0.050	0.040
MAR	vt_tilt	0.018	0.156	0.115	0.293	-0.437	0.817	11.022	-0.051	0.040
Gauss-HAR	vt_tilt	0.018	0.153	0.115	0.293	-0.435	0.812	9.188	-0.056	0.041
Chronos-Bolt-small	vt_tilt	0.021	0.184	0.114	0.293	-0.474	0.914	6.058	-0.038	0.031

Buy and hold: Sharpe 0.112, maximum drawdown -0.463. Annualised; decisions at the close, execution at the next open. Excess CI: 95% stationary-bootstrap interval of the annualised mean return in excess of buy and hold.

Table 20: Trading results: IF futures, long/short, 2bp, out of sample

Model	Rule	Return	Vol	Sharpe	SE	MaxDD	Avg w	Turnover	Excess CI lo	Excess CI hi
Deep-SMoE	meanvar	0.067	0.156	0.430	0.376	-0.261	0.094	97.980	-0.089	0.203
Chronos-Bolt-small	meanvar	0.045	0.170	0.267	0.372	-0.269	0.092	44.951	-0.099	0.200
MAR	meanvar	0.036	0.140	0.255	0.375	-0.206	0.117	13.509	-0.112	0.188
Gauss-HAR	meanvar	0.015	0.141	0.105	0.373	-0.249	0.174	46.336	-0.118	0.150
HistGauss	meanvar	0.015	0.164	0.092	0.374	-0.387	0.054	1.954	-0.161	0.183
GARCH-t	meanvar	-0.006	0.163	-0.036	0.373	-0.472	0.853	1.168	-0.043	0.014
SMoE	meanvar	-0.006	0.167	-0.038	0.373	-0.380	0.084	56.403	-0.180	0.149
Chronos-Bolt-small	sign	0.093	0.198	0.468	0.372	-0.275	0.092	78.096	-0.054	0.267
Deep-SMoE	sign	0.084	0.199	0.420	0.374	-0.273	0.243	138.900	-0.060	0.223
HistGauss	sign	0.021	0.199	0.107	0.374	-0.450	0.044	10.041	-0.166	0.208
MAR	sign	0.019	0.196	0.098	0.374	-0.430	0.199	38.490	-0.146	0.177
GARCH-t	sign	0.007	0.197	0.036	0.373	-0.454	1.000	0.000	0.000	0.000
SMoE	sign	-0.048	0.199	-0.240	0.372	-0.489	0.163	64.151	-0.226	0.120
Gauss-HAR	sign	-0.068	0.197	-0.344	0.373	-0.521	0.155	116.029	-0.238	0.084
Deep-SMoE	voltarget	0.002	0.157	0.015	0.373	-0.428	0.853	8.430	-0.031	0.021
Chronos-Bolt-small	voltarget	0.003	0.184	0.014	0.373	-0.460	0.952	4.643	-0.017	0.008
MAR	voltarget	0.000	0.159	0.003	0.373	-0.427	0.865	10.071	-0.033	0.020
Gauss-HAR	voltarget	-0.001	0.160	-0.005	0.373	-0.428	0.867	9.158	-0.033	0.018
SMoE	voltarget	-0.001	0.161	-0.009	0.373	-0.432	0.870	7.578	-0.034	0.016
GARCH-t	voltarget	-0.003	0.157	-0.018	0.373	-0.438	0.849	3.286	-0.039	0.017
HistGauss	voltarget	-0.007	0.166	-0.040	0.373	-0.422	0.855	0.417	-0.039	0.011
Deep-SMoE	vt_tilt	0.005	0.161	0.034	0.374	-0.422	0.861	7.493	-0.026	0.021
MAR	vt_tilt	0.001	0.161	0.003	0.373	-0.429	0.869	10.172	-0.033	0.020
Gauss-HAR	vt_tilt	-0.000	0.160	-0.003	0.373	-0.428	0.867	9.130	-0.034	0.018
Chronos-Bolt-small	vt_tilt	-0.002	0.184	-0.008	0.373	-0.473	0.953	4.600	-0.021	0.005
GARCH-t	vt_tilt	-0.002	0.162	-0.015	0.373	-0.448	0.874	3.094	-0.036	0.015
SMoE	vt_tilt	-0.003	0.162	-0.018	0.373	-0.435	0.872	8.386	-0.036	0.016
HistGauss	vt_tilt	-0.007	0.165	-0.045	0.373	-0.425	0.853	0.624	-0.041	0.010

Buy and hold: Sharpe 0.036, maximum drawdown -0.454. Annualised; decisions at the close, execution at the next open. Excess CI: 95% stationary-bootstrap interval of the annualised mean return in excess of buy and hold.

Table 21: Backtest overfitting diagnostics

Trials	Best (all costs)	Sharpe (all)	DSR (all)	Best at 3bp	Sharpe (3bp)	DSR (3bp)	PBO (3bp)
1,440	Deep-SMoE, meanvar, c0.0, g10.0, v0.8	0.583	0.565	Gauss-HAR, meanvar, c3.0, g10.0, v0.8	0.498	0.758	0.514

DSR: deflated Sharpe ratio, the probability that the true Sharpe ratio exceeds the expected maximum of the trials under the null; values below 0.95 are not significant. PBO: probability of backtest overfitting by combinatorially symmetric cross-validation with 16 blocks.

series under all four losses. Pairwise, it is better than GARCH-t on 14 series (significantly on 3, significantly worse on none), and significantly better than the Markov-switching regression on 12, the Gauss-HAR on 9, the boosted model on 7 and each of the three foundation models on 18 series. Against the other mixtures (MAR, deep SMoE) the lockbox is too short to separate the models: each comparison produces one significant win and one significant loss.

Calibration. The SMoE's 1% VaR passes the Kupiec test on 18 of 19 series (GARCH-t: 18; Chronos-Bolt: 1; TimesFM 2.5: 2). Its PIT standard deviation is 1.01, whereas the foundation models' remain at 1.26–1.35. After adaptive conformal calibration no model fails the Kupiec test on any series, and the SMoE and deep SMoE pass the conditional-coverage test on every series at the 5% level.

Realised variance. The 5-minute data cover only the first 246 lockbox days. On QLIKE against realised variance the mixtures again lead (MAR 0.567, LSTAR 0.598, SMoE 0.606, GARCH-t 0.663). The Mincer–Zarnowitz R^2, however, is highest for the two Chronos-Bolt models (0.55 and 0.71), followed by TimesFM 2.5 (0.18), against at most 0.13 for the others. The R^2 of a regression on 246 days is dominated by the realised variance spike of late September 2024, and the zero-shot models, which condition on the raw return path, scaled up faster after the first large moves. This is the one lockbox result in which a foundation model leads; on the robust loss it does not (TimesFM 2.5: 0.705).

Trading. The strategies do not beat buy and hold in the lockbox. On the ETF, the best configuration at 3bp is a GARCH-t mean-variance rule with Sharpe ratio 0.74 against 0.38 for buy and hold, but its standard error is 0.65, its deflated Sharpe ratio 0.65 and the probability of backtest overfitting across the 360 configurations rises to 0.77 (table 23). The SMoE-based volatility-target rules match the buy-and-hold Sharpe ratio (0.37 versus 0.38) with a smaller maximum drawdown (-17% versus -19%). The lockbox thus confirms the development-period conclusion: the predictive distributions manage risk, and the data do not support claims of timing alpha.

16.1 Do foundation models remember the test period?

A zero-shot forecaster is out of sample only if its pretraining corpus ends before the forecast date. The documented corpora of Chronos and TimesFM 2.5 do not list the series we study, but they cannot be audited series by series, and Sarkar and Vafa (2024) and Glasserman and Lin

Table 22: Development period versus lockbox: average ranks across 19 series (1 = best), model-confidence-set membership (10%) and Kupiec passes

Period	Model	Rank NLL	Rank CRPS	Rank QLIKE	Rank FZ5	MCS NLL	MCS QLIKE	MCS FZ5	Kupiec 1% pass
Development 2012–2024/06	SMoE	2.95	3.68	3.53	4.32	15	18	18	17
Development 2012–2024/06	Deep-SMoE	3.26	5.32	3.58	3.68	14	18	18	15
Development 2012–2024/06	MAR	3.47	3.47	3.63	4.00	15	18	18	16
Development 2012–2024/06	LSTAR	4.00	5.05	4.11	4.05	11	17	18	15
Development 2012–2024/06	GARCH-t	4.11	5.26	5.95	5.16	11	12	15	10
Development 2012–2024/06	AR-GJR-t	3.63	3.68	5.47	4.11	17	16	18	12
Development 2012–2024/06	Gauss-HAR	7.63	8.37	4.89	5.21	2	15	15	4
Development 2012–2024/06	MS-AR	7.89	10.26	8.16	7.42	1	7	11	16
Development 2012–2024/06	XGB-Gauss	8.63	4.68	7.16	8.42	1	11	10	2
Development 2012–2024/06	Chronos-Bolt-base	11.32	7.95	11.21	11.58	0	1	2	0
Development 2012–2024/06	TimesFM-2.5	10.21	11.26	9.63	10.05	1	5	4	0
Development 2012–2024/06	HistGauss	11.00	12.37	10.84	10.26	1	5	7	1
Lockbox 2024/07–2026/10	SMoE	3.11	3.42	2.95	3.26	19	19	19	18
Lockbox 2024/07–2026/10	Deep-SMoE	3.26	3.53	3.89	3.84	19	19	19	17
Lockbox 2024/07–2026/10	MAR	3.21	3.11	3.16	3.68	19	19	19	18
Lockbox 2024/07–2026/10	LSTAR	3.05	3.32	3.63	3.79	19	19	19	18
Lockbox 2024/07–2026/10	GARCH-t	4.37	6.21	6.63	6.21	18	19	19	18
Lockbox 2024/07–2026/10	AR-GJR-t	5.05	6.16	7.00	6.21	18	19	19	17
Lockbox 2024/07–2026/10	Gauss-HAR	7.58	6.84	4.26	4.84	15	19	19	14
Lockbox 2024/07–2026/10	MS-AR	7.21	9.11	7.95	8.05	18	19	19	15
Lockbox 2024/07–2026/10	XGB-Gauss	8.32	4.95	6.37	6.37	17	19	19	6
Lockbox 2024/07–2026/10	Chronos-Bolt-base	11.58	9.63	11.58	11.58	4	9	12	1
Lockbox 2024/07–2026/10	TimesFM-2.5	10.68	12.11	10.26	10.37	5	13	16	2
Lockbox 2024/07–2026/10	HistGauss	11.00	11.95	10.68	10.42	12	18	17	8

Ranks among the thirteen forecasters (Chronos-Bolt-small not shown). MCS and Kupiec columns count series out of 19. TimesFM-2.5 was added after the lockbox had been opened (section 16).

Table 23: Lockbox: backtest overfitting diagnostics (CSI 300 ETF, 360 configurations at 3bp)

Best at 3bp	Sharpe	DSR	PBO
GARCH-t, meanvar, c3.0, g5.0, v0.8	0.738	0.648	0.770

(2024) show that look-ahead through pretraining is real for language models used in finance. A model's release date C bounds its data cut-off from above, which gives a simple test.

Proposition 16 (Release-date test)**.** *Let $d_t = \mathrm{NLL}_t^{\mathrm{FM}} - \mathrm{NLL}_t^{\mathrm{ref}}$ be the log-score loss of a foundation model relative to a reference model estimated in real time, averaged across series on date t, and let d_t^{pl} be the same difference for a placebo model that is also estimated in real time. Suppose that, without pretraining overlap, $\mathbb{E}[d_t - d_t^{\mathrm{pl}}]$ is the same before and after C (parallel trends). Then the coefficient b in $d_t - d_t^{\mathrm{pl}} = a + b\mathbb{1}\{t \geq C\} + e_t$ is zero, and its Newey–West t-statistic is asymptotically standard normal as the numbers of pre- and post-release dates grow. If the corpus contains the evaluation series up to some cut-off $C' \leq C$, forecasts before C' are partly in-sample and $b > 0$. If $C' < C$, dates in $[C', C)$ are classified as pre-release although they are unseen, which biases b towards zero; the test remains valid but loses power.*

The proof is the usual argument for a difference in means with a HAC variance; the content is in the identifying assumption. Shifts in market conditions at C that change how all models perform relative to the SMoE are removed by the placebo, for which we use GARCH-t. We apply the test with the release dates of Chronos-Bolt (26 November 2024) and TimesFM 2.5 (15 September 2025), using the development and lockbox forecasts together.

Table 24 reports the test. All three foundation models lose to the SMoE both before and after their release dates, but by less afterwards, so every estimate of b is negative: -0.074 and -0.064 nats per day for the two Chronos-Bolt models and -0.057 for TimesFM 2.5. The placebo moves in the same direction (-0.013 and -0.016), and the difference-in-differences estimates (-0.061, -0.051 and -0.041) are negative and insignificant, with one-sided p-values for $b > 0$ between 0.82 and 0.90. There is no sign that the foundation models forecast better on

data their pretraining could have contained. The one-sided 95% upper bounds on the difference-in-differences b are 0.037 and 0.040 nats per day for Chronos-Bolt and 0.011 for TimesFM 2.5, so a memorisation gain larger than about one hundredth of a nat per day is ruled out for TimesFM, whose post-release sample is the shorter of the two. The test cannot exclude an advantage that is constant across the release date (for instance from general familiarity with financial series), and the post-release windows of 409 and 702 dates are short. Within these limits, the poor calibration of the foundation models in section 11 is not offset by any detectable memory of the evaluation period.

Table 24: Release-date test for pretraining overlap (log score relative to the SMoE, averaged across series)

Model	Comparison	Release	n_{pre}	n_{post}	Mean pre	Mean post	$\hat{b}$	z	$p\ (b > 0)$
Chronos-Bolt-small	Chronos-Bolt-small	2024-11-26	4,162	702	0.241	0.167	-0.074	-1.266	0.897
Chronos-Bolt-small	placebo GARCH-t	2024-11-26	4,162	702	0.009	-0.004	-0.013	-1.771	0.962
Chronos-Bolt-small	difference	2024-11-26	4,162	702	0.232	0.171	-0.061	-1.025	0.847
Chronos-Bolt-base	Chronos-Bolt-base	2024-11-26	4,162	702	0.203	0.140	-0.064	-1.174	0.880
Chronos-Bolt-base	placebo GARCH-t	2024-11-26	4,162	702	0.009	-0.004	-0.013	-1.771	0.962
Chronos-Bolt-base	difference	2024-11-26	4,162	702	0.194	0.144	-0.051	-0.917	0.820
TimesFM-2.5	TimesFM-2.5	2025-09-15	4,455	409	0.153	0.096	-0.057	-1.730	0.958
TimesFM-2.5	placebo GARCH-t	2025-09-15	4,455	409	0.008	-0.008	-0.016	-1.990	0.977
TimesFM-2.5	difference	2025-09-15	4,455	409	0.144	0.104	-0.041	-1.297	0.903

d_t: cross-series average of NLL(model) − NLL(SMoE) on date t, development and lockbox periods. $\hat{b}$: post-release minus pre-release mean, with Newey–West standard errors. Memorisation of the evaluation period predicts $b > 0$ for the foundation model. Placebo: GARCH-t, estimated in real time. Difference: foundation model minus placebo (difference-in-differences).

17 A research agenda

The SMoE is minimal by design. Each extension below keeps the white-box property (closed-form predictive functionals, exact attribution and likelihood-based inference) and each is a self-contained research question.

1. Multivariate SMoE for allocation. Replace the univariate experts by $\mathcal{N}_d(\boldsymbol{B}_k\tilde{\boldsymbol{x}}, \boldsymbol{\Sigma}_k(\boldsymbol{v}))$ with regime-specific covariance matrices, e.g. $\boldsymbol{\Sigma}_k = \boldsymbol{D}_k\boldsymbol{R}_k\boldsymbol{D}_k$ with log-linear $\boldsymbol{D}_k$. The predictive covariance decomposes as $\sum_k \pi_k\boldsymbol{\Sigma}_k + \sum_k \pi_k(\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}})^\top$, and the second term is a rank-$(K-1)$ "regime-disagreement" factor structure. Open questions: identification with many assets, shrinkage of $\boldsymbol{R}_k$, and whether regime-disagreement factors are priced.

2. Decision-focused estimation. Section 3 traces the lagging paradox to a mismatch between the training loss and the use of the forecast. A natural next step estimates $\boldsymbol{\theta}$ by maximising realised utility, $\max_{\boldsymbol{\theta}} \sum_t U(w_t(\boldsymbol{\theta})y_t)$ with w_t given by proposition 11, or a convex combination of utility and likelihood. The questions are statistical (consistency of a non-likelihood M-estimator whose criterion is non-smooth because of the no-trade band) and economic (does it beat predict-then-optimise out of sample?).

3. Language-model gates. Gate covariates need not be prices. Policy statements, regulatory announcements and news can be mapped by a large language model to a small set of scores

(e.g. policy stance, liquidity support, risk-off intensity) that enter $\boldsymbol{z}_t$. The SMoE then answers a sharp question (does text move regime probabilities beyond what prices already imply?) with the score test of proposition 14, which needs only the price-based fit and can screen many text scores at once, followed by a Wald test on the selected $\boldsymbol{\gamma}$ coefficients. The fidelity ratio then measures how much of an LLM-based black-box forecaster the text-augmented SMoE explains. Because language models also carry look-ahead in their pretraining, the scores should come from models with dated corpora (item 8).

4. Online Bayesian SMoE. Quarterly re-estimation is a crude response to non-stationarity. Sequential Monte Carlo or online variational EM would update $\boldsymbol{\theta}$ daily with forgetting, give posterior uncertainty about the regime parameters themselves, and allow the gate to reflect parameter uncertainty in the predictive distribution. The adaptive conformal layer of theorem 2 would then calibrate a Bayesian rather than a frequentist predictive.

5. Intraday SMoE. With 5-minute data the same model can forecast intraday returns and realised variance, with experts for opening auction, lunch break and closing auction dynamics specific to the Chinese market, and gates driven by order-flow imbalance. Microstructure noise makes the distributional target essential.

6. Realised-variance experts. Section 13 found that a single-regime Realized GARCH beats every SMoE variant in log score once 5-minute data are available, and that part of the mixture's advantage with range-based covariates is insurance against a noisy variance measure. The obvious synthesis gives each expert its own measurement equation and variance recursion, $\log \sigma_{kt}^2 = \omega_k + \beta_k \log \sigma_{k,t-1}^2 + \gamma_k \log \mathrm{RV}_{t-1}$, with the gate deciding which recursion governs tomorrow. The CM-step for the variance block is then no longer a static Gamma regression but a filtering problem, and identification of regime-specific persistence from a single realised series is open. Extending realised measures to the other eighteen series requires intraday data we do not have.

7. Yardstick for foundation models. As time-series foundation models grow, the fidelity ratio becomes a general tool: for each model and asset it reports how much of the model's out-of-sample density-forecast gain is reproducible by a three-regime, ten-covariate white box. A benchmark of Φ across model families, horizons and asset classes would make "how black is the black box?" a measurable property. Newer models with covariate and multivariate inputs, such as TimesFM 3.0, can be given the SMoE's own covariates, which turns the yardstick into a direct comparison of what the network and the white box do with the same information.

8. Point-in-time foundation models. The release-date test of section 16 can only bound pretraining look-ahead from one side, because a model's release date is an upper bound on its data cut-off. Foundation models pretrained on corpora that end at known, staggered dates would turn the test into an exact event study: the same architecture trained to vintages $C_1 < C_2 < \ldots$ and evaluated on the same returns identifies how much of an apparent forecasting gain is memory. Sarkar and Vafa (2024) make the same argument for language models.

18 Conclusion

This paper started from the question whether a heterogeneous mixture-of-experts network can forecast the CSI 300. Under a leak-free protocol no model in our comparison forecasts daily means or directions, and the accuracy figures that motivated the original work came from look-ahead. We therefore asked what a forecaster knows about the conditional distribution of returns and whether that knowledge can be written down. The Semantic Mixture of Experts answers this for itself in closed form with standard errors, and for other forecasters through the KL projection.

Five findings carry over beyond this paper. First, price-level losses cannot reveal skill, and the differenced directional-accuracy metric measures sign persistence in the data rather than anything about the model (propositions 2 and 3). Second, a three-regime white box with ten covariates matches the density forecasts of a deep heterogeneous MoE to within eight percent of its gain and dominates single-regime, Markov-switching, boosted and zero-shot foundation models; most of the value comes from variance dynamics and mixture shape, and the gate's state dependence adds a smaller, testable increment. Three experts forecast better than two, four or five, even though in-sample criteria disagree on the number. Third, the white box produces economic content that a black box hides: in the CSI 300, volatility rather than momentum or drawdown moves regime weights, the short-horizon regime continues in China but reverses in developed markets, and the US VIX enters regime weights in all six international equity indices but in only one of the four A-share indices. Fourth, how daily variance is measured matters. A noisy estimator shifts weight to longer horizons by an amount that

proposition 15 predicts, part of the mixture's advantage is insurance against that noise, and with realised variance a single-regime Realized GARCH is the best density forecaster for the CSI 300. Fifth, risk forecasts are where distributional models pay off (tail calibration, a conformal coverage guarantee and lower drawdowns), whereas claims of directional alpha should be met with deflated Sharpe ratios and backtest-overfitting probabilities before anyone acts on them.

The limitations are those of a first step. The model is univariate and daily; its gate covariates, although now screened by a score test, are drawn from a short list; realised measures are available to us only for the CSI 300; the foundation models enter zero-shot and through a Gaussian projection of their quantiles, and the release-date test bounds rather than excludes pretraining look-ahead; and our economic analysis concerns one market and two instruments. The research agenda of section 17 lists the extensions we consider most promising (multivariate and decision-focused SMoEs, language-model gates, online Bayesian updating, experts with realised-variance recursions and point-in-time foundation models). Each keeps the property that made this exercise informative: every statement the model makes comes with a closed form and a standard error that a reader can check.

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A Proofs

Proof of proposition 1. Conditional on $\mathcal{F}_t$, $P_{t+1} = P_t e^{r_{t+1}}$ and P_t is known. For any $\mathcal{F}_t$-measurable f, $\mathbb{E}[(P_{t+1} - f)^2 \mid \mathcal{F}_t] = \mathrm{Var}(P_{t+1} \mid \mathcal{F}_t) + (\mathbb{E}[P_{t+1} \mid \mathcal{F}_t] - f)^2$, minimised at $f^* = \mathbb{E}[P_{t+1} \mid \mathcal{F}_t] = P_t M_t(1)$ with value $\mathrm{Var}(P_{t+1} \mid \mathcal{F}_t) = P_t^2\{M_t(2) - M_t(1)^2\}$. For $f = P_t$ the value is $P_t^2\mathbb{E}[(e^r - 1)^2 \mid \mathcal{F}_t] = P_t^2\{M_t(2) - 2M_t(1) + 1\}$; subtracting gives (1). □

Proof of proposition 2. Divide (1) by the random-walk MSE: the ratio is $\rho_t = (M_1 - 1)^2/(M_2 - 2M_1 + 1)$. Under normality $M_1 = e^m$ and $M_2 = e^{2m+\sigma^2}$ with $m = \mu + \sigma^2/2$. Expanding, $M_1 - 1 = m + O(m^2)$ and $M_2 - 2M_1 + 1 = e^{2m}(e^{\sigma^2} - 1) + (e^m - 1)^2 = \sigma^2 + m^2 + O(\sigma^3)$ since $m = O(\sigma)$ for bounded s. Hence $\rho_t = m^2/(\sigma^2 + m^2) + O(\sigma^3) = s_t^2/(1 + s_t^2) + O(\sigma^3)$. For the unconditional statement write the optimal and RW MSEs as $\mathbb{E}[P_t^2\{\cdot\}]$; the ratio of expectations is the w_t-weighted average of ρ_t with $w_t \propto P_t^2(\sigma_t^2 + m_t^2)$, and RMSE ratios are square roots of MSE ratios. The approximation uses $\sqrt{1-x} \approx 1 - x/2$. □

Proof of proposition 3. If $f_{t+1} = P_t$ then $f_{t+1} - f_t = P_t - P_{t-1}$, whose sign equals that of r_t, while $P_{t+1} - P_t$ has the sign of r_{t+1}. The expectation follows from $\mathbb{1}\{\operatorname{sign} a = \operatorname{sign} b\} = \frac{1}{2}(1 + \operatorname{sign} a \operatorname{sign} b)$ for non-zero a, b. If $f_{t+1} = P_t + \varepsilon_{t+1}$ then $f_{t+1} - f_t = (P_t - P_{t-1}) + (\varepsilon_{t+1} - \varepsilon_t)$, which has the sign of r_t whenever $|\varepsilon_{t+1} - \varepsilon_t| < |P_t - P_{t-1}|$; the stated condition is the same inequality in log units to first order. □

Proof of proposition 5. Fix covariates $(\boldsymbol{x}, \boldsymbol{v}, \boldsymbol{z})$ in the open set of assumption 1(a). Finite mixtures of univariate normals are identifiable (Teicher, 1963): equality of the mixtures implies equality of the sets $\{(\pi_k, \mu_k, \sigma_k^2)\}_k$ and $\{(\pi'_k, \mu'_k, \sigma'^2_k)\}_k$, which by assumption 1(b) have K distinct elements on a set of positive probability. Hence there is a permutation $\varsigma_{(\boldsymbol{x},\boldsymbol{v},\boldsymbol{z})}$ with $(\pi'_{\varsigma(k)}, \mu'_{\varsigma(k)}, \sigma'^2_{\varsigma(k)}) = (\pi_k, \mu_k, \sigma_k^2)$. All component parameters are continuous in the covariates and distinct, so ς is locally constant and therefore constant on the connected open set. Suppose $\varsigma(k) = l \neq k$. Then $\boldsymbol{\beta}_k^\top \tilde{\boldsymbol{x}}^{(k)} = \boldsymbol{\beta}_l'^\top \tilde{\boldsymbol{x}}^{(l)}$ on an open set. By assumption 1(c) the two covariate sets differ, so one side varies with a covariate on which the other does not depend; non-singularity of the designs forces the corresponding slope to be zero, contradicting (c), unless both experts have no slopes, which (c) excludes for more than one expert. Hence ς is the identity. Equality of μ_k, $\log \sigma_k^2$ and of the log-odds $\log(\pi_k/\pi_K) = \boldsymbol{\gamma}_k^\top \tilde{\boldsymbol{z}}$ as linear functions on an open set, together with non-singular designs, gives $\boldsymbol{\beta}_k = \boldsymbol{\beta}'_k$, $\boldsymbol{\delta}_k = \boldsymbol{\delta}'_k$ and $\boldsymbol{\gamma}_k = \boldsymbol{\gamma}'_k$. □

Proof of proposition 6. Write $Y_t = (y_t, \ldots, y_{t-p+1})$. The chain is a φ-irreducible, aperiodic T-chain because the transition density is a positive Gaussian mixture, and compact sets are small (Meyn and Tweedie, 2009, Ch. 5–6). Let $\rho = \sup_{\boldsymbol{z}} \sum_k \pi_k(\boldsymbol{z}) \sum_j |\phi_{kj}| < 1$ and choose $V(Y) = 1 + \sum_{j=1}^p c_j |y_{t-j+1}|$ with $c_1 = 1$ and c_j decreasing geometrically so that $c_{j+1} + \rho_j \leq c_j \rho'$ for some $\rho < \rho' < 1$, where $\rho_j = \sup_{\boldsymbol{z}} \sum_k \pi_k |\phi_{kj}|$. Then $\mathbb{E}[|y_{t+1}| \mid Y_t] \leq \sum_k \pi_k(|\phi_{k0}| + \sum_j |\phi_{kj}||y_{t-j+1}|) + \bar{\sigma}\sqrt{2/\pi}$, so $\mathbb{E}[V(Y_{t+1}) \mid Y_t] \leq \rho' V(Y_t) + C$ for a constant C. The Foster–Lyapunov criterion (Meyn and Tweedie, 2009, Thm. 15.0.1) yields geometric ergodicity with $\mathbb{E}_{\text{stat}} V < \infty$. □

Proof of proposition 7. Mean, cdf and probability statements are linear in the mixture weights. The variance follows from the law of total variance with the component label as conditioning variable. For ES, write $\mathbb{E}[y\mathbb{1}\{y \leq q\}] = \sum_k \pi_k \mathbb{E}[y\mathbb{1}\{y \leq q\} \mid k]$ and use $\mathbb{E}[X\mathbb{1}\{X \leq q\}] = \mu\Phi(z) - \sigma\varphi(z)$ for $X \sim \mathcal{N}(\mu, \sigma^2)$, $z = (q - \mu)/\sigma$. For CRPS use $\text{CRPS}(F, y) = \mathbb{E}|X - y| - \frac{1}{2}\mathbb{E}|X - X'|$ with $X, X' \overset{iid}{\sim} F$, and $\mathbb{E}|Z| = A(m, s^2)$ for $Z \sim \mathcal{N}(m, s^2)$ (Grimit et al., 2006). Uniqueness of q_α holds because the mixture cdf is continuous and strictly increasing. □

Proof of proposition 8. For any $\boldsymbol{\theta}$, Jensen's inequality gives $\ell_n^\lambda(\boldsymbol{\theta}) - \ell_n^\lambda(\boldsymbol{\theta}^{(s)}) \geq Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(s)}) - Q(\boldsymbol{\theta}^{(s)} \mid \boldsymbol{\theta}^{(s)})$ (Dempster et al., 1977). Each CM step does not decrease Q: CM-step 1 maximises a strictly concave quadratic exactly; CM-steps 2 and 3 take Newton steps on strictly concave objectives with step halving that accepts only non-decreasing values. Hence $Q(\boldsymbol{\theta}^{(s+1)} \mid \boldsymbol{\theta}^{(s)}) \geq Q(\boldsymbol{\theta}^{(s)} \mid \boldsymbol{\theta}^{(s)})$ and the claim follows; the final L-BFGS step is accepted only if it increases ℓ_n^λ. Convergence of limit points to stationary points follows from Wu (1983, Thm. 1) applied to the generalised EM map. □

Proof of proposition 9. For the softmax with $\boldsymbol{\gamma}_K = \mathbf{0}$, $\partial\pi_k/\partial z_j = \pi_k(\gamma_{kj} - \sum_l \pi_l \gamma_{lj})$. Since z_j enters neither μ_k nor σ_k, $\partial\bar{\mu}/\partial z_j = \sum_k \mu_k \partial\pi_k/\partial z_j$ and $\partial P(y>0)/\partial z_j = \sum_k \Phi(\mu_k/\sigma_k)\partial\pi_k/\partial z_j$. □

Proof of proposition 10. $\mathbb{E}\log p(y_t \mid \cdot) = \mathbb{E}\int p_0 \log p = \mathbb{E}\int p_0 \log p_0 - \mathbb{E}\int p_0 \log(p_0/p) = -H(p_0) - \mathbb{E}\,\mathrm{KL}(p_0 \| p)$. The entropy cancels in differences. □

Proof of theorem 2. Summing (21) gives $\alpha_{T+1} = \alpha_1 + \eta\sum_{t=1}^T (a - \mathrm{err}_t)$. If $\alpha_t \leq 0$ then $q_t = -\infty$, $\mathrm{err}_t = 0$ and $\alpha_{t+1} = \alpha_t + \eta a > \alpha_t$; if $\alpha_t \geq 1$ then $\mathrm{err}_t = 1$ and $\alpha_{t+1} < \alpha_t$. Each step moves α by at most η, so $\alpha_t \in [-\eta, 1+\eta]$ for all t when $\alpha_1 \in (0,1)$. Therefore $|\sum_t (a - \mathrm{err}_t)| = |\alpha_{T+1} - \alpha_1|/\eta \leq (\max(\alpha_1, 1-\alpha_1) + \eta)/\eta$, and dividing by T gives the bound (cf. Gibbs and Candès, 2021, Prop. 4.1). □

Student-t densities as Gaussian mixtures. All benchmarks are scored through the closed forms of proposition 7, which requires a Gaussian mixture representation. For the Student-t benchmarks we use the scale-mixture identity $t_\nu(\mu, s) = \int_0^\infty \mathcal{N}(\mu, s^2/\lambda)\, g(\lambda)\, d\lambda$ with $\lambda \sim \mathrm{Gamma}(\nu/2, \text{rate } \nu/2)$. Substituting $u = \nu\lambda/2$,

$$\mathbb{E}[h(\lambda)] = \frac{1}{\Gamma(\nu/2)} \int_0^\infty h(2u/\nu)\, u^{\nu/2-1} e^{-u}\, du \approx \sum_{i=1}^N \omega_i\, h(2u_i/\nu), \tag{33}$$

where $(u_i, \tilde{\omega}_i)$ are the nodes and weights of N-point generalised Gauss–Laguerre quadrature with parameter $\alpha = \nu/2 - 1$ and $\omega_i = \tilde{\omega}_i/\Gamma(\nu/2)$. The rule is exact for polynomials of degree $2N-1$ in u; for the smooth integrands $h(\lambda) = \varphi(y; \mu, s^2/\lambda)$ the error decays in N at fixed y, more slowly for heavier tails. We use $N = 64$. The maximal relative density error on $|y - \mu| \leq 4s$ is 0.27% for $\nu = 2.5$, 0.11% for $\nu = 3$, 0.02% for $\nu = 4$ and below 0.005% for $\nu \geq 5$ (computed against the exact t density in the replication package), an order of magnitude below the log-score differences we interpret. An earlier version with $N = 16$ had errors of up to 2.9% at $\nu = 3$; we report this because the choice of N matters for heavy-tailed benchmarks. □

Proof of proposition 12. By (6), $V_t = W_t + B_t = W_t(1 + B_t/W_t)$, so $w_t^* = \bar{\mu}_t/(\gamma V_t)$ factorises as stated. $B_t \geq 0$ with equality iff $\mu_{kt} = \bar{\mu}_t$ for all k with $\pi_{kt} > 0$; a mean-preserving spread of (μ_{kt}) increases B_t and leaves $\bar{\mu}_t$ and W_t unchanged. □

Proof of proposition 11. The objective $J(w) = w\bar{\mu} - \frac{\gamma}{2}w^2V - \kappa|w - w_-|$ is strictly concave. Its superdifferential is $\bar{\mu} - \gamma V w - \kappa\,\partial|w - w_-|$. If $|w_- - w^*| \le b$ then 0 lies in the superdifferential at $w = w_-$ (take the subgradient $(\bar{\mu} - \gamma V w_-)/\kappa \in [-1, 1]$). If $w_- < w^* - b$ the optimum is at $w > w_-$ where the first-order condition $\bar{\mu} - \gamma V w - \kappa = 0$ gives $w = w^* - b$; symmetrically for $w_- > w^* + b$. Concavity makes projection onto the box optimal for the constrained problem. □

Proof of proposition 13. The first part of the proof of proposition 5 does not use assumption 1(c): it yields a permutation ς, constant on a connected open set of covariates, with $(\pi'_{\varsigma(k)}, \mu'_{\varsigma(k)}, \sigma'^2_{\varsigma(k)}) = (\pi_k, \mu_k, \sigma_k^2)$. For a pair of experts with different covariate sets the argument of proposition 5 still excludes $\varsigma(k) = l$. Hence ς can only permute experts within a shared group. Suppose it exchanges k and l with $\delta_{k,0} < \delta_{l,0}$. Equality of the log variances as linear functions of $\tilde{\boldsymbol{v}}$ on an open set gives $\boldsymbol{\delta}'_l = \boldsymbol{\delta}_k$ and $\boldsymbol{\delta}'_k = \boldsymbol{\delta}_l$, so $\delta'_{k,0} = \delta_{l,0} > \delta_{k,0} = \delta'_{l,0}$, which violates the convention for $\boldsymbol{\theta}'$. Thus ς is the identity and the remainder of the proof of proposition 5 applies. The convention requires $\delta_{k,0} \neq \delta_{l,0}$; if the two variance functions coincided, an ordering of the mean intercepts would serve the same purpose. □

Proof of proposition 14. The score expression (26) is (15) for the gate block with $\tilde{\boldsymbol{z}}_t$ replaced by the augmented vector, evaluated at $\boldsymbol{\psi} = \mathbf{0}$, where the prior probabilities π_{tk} do not depend on $\boldsymbol{w}_t$. Write $\boldsymbol{\vartheta} = (\boldsymbol{\theta}, \boldsymbol{\psi})$, $\boldsymbol{s}_n(\boldsymbol{\vartheta}) = \sum_t \partial\ell_t(\boldsymbol{\vartheta})$ and A for the information matrix (minus the expected Hessian) of the augmented model at $\boldsymbol{\vartheta}^* = (\boldsymbol{\theta}^*, \mathbf{0})$. Under H_0, $\boldsymbol{\vartheta}^*$ is the pseudo-true value, so $\mathbb{E}\,\partial\ell_t(\boldsymbol{\vartheta}^*) = \mathbf{0}$ and, by the central limit theorem for β-mixing sequences under assumption 2, $n^{-1/2}\boldsymbol{s}_n(\boldsymbol{\vartheta}^*) \xrightarrow{d} \mathcal{N}(\mathbf{0}, B)$. The restricted estimator satisfies $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) = A_{\theta\theta}^{-1} n^{-1/2}\boldsymbol{s}_{n,\theta}(\boldsymbol{\vartheta}^*) + o_p(1)$ by theorem 1; the penalty contributes $O_p(n^{-1/2})$. A mean-value expansion gives

$$n^{-1/2}\boldsymbol{s}_n(\hat{\boldsymbol{\theta}}, \mathbf{0}) = n^{-1/2}\boldsymbol{s}_n(\boldsymbol{\vartheta}^*) - \begin{pmatrix} A_{\theta\theta} \\ A_{\psi\theta} \end{pmatrix} \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) + o_p(1).$$

Pre-multiplying by $C = [-A_{\psi\theta}A_{\theta\theta}^{-1},\, I]$ annihilates the second term, because $-A_{\psi\theta}A_{\theta\theta}^{-1}A_{\theta\theta} + A_{\psi\theta} = 0$. Hence $C\,n^{-1/2}\boldsymbol{s}_n(\hat{\boldsymbol{\theta}}, \mathbf{0}) = C\,n^{-1/2}\boldsymbol{s}_n(\boldsymbol{\vartheta}^*) + o_p(1) \xrightarrow{d} \mathcal{N}(\mathbf{0}, CBC^\top)$. $\hat{A}$ and the Newey–West $\hat{B}$ are consistent under the moment and mixing conditions of assumption 2 (Newey and West, 1987), $C\hat{B}C^\top$ is non-singular because B is, and the continuous mapping theorem gives the $\chi^2_{(K-1)d}$ limit. Under $\boldsymbol{\psi}^* = \boldsymbol{h}/\sqrt{n}$ the same expansion adds $A_{\cdot\psi}\boldsymbol{h}$ to the limit, and $CA_{\cdot\psi} = A_{\psi\psi\cdot\theta} \coloneqq A_{\psi\psi} - A_{\psi\theta}A_{\theta\theta}^{-1}A_{\theta\psi}$, so the non-centrality is $\boldsymbol{h}^\top A_{\psi\psi\cdot\theta}(CBC^\top)^{-1}A_{\psi\psi\cdot\theta}\boldsymbol{h}$. For the Wald statistic, the $\boldsymbol{\psi}$ rows of A^{-1} equal $A_{\psi\psi\cdot\theta}^{-1}C$ by the partitioned-inverse formula, so the $\boldsymbol{\psi}$ block of the sandwich $A^{-1}BA^{-1}$ is $A_{\psi\psi\cdot\theta}^{-1}CBC^\top A_{\psi\psi\cdot\theta}^{-1}$ and $\sqrt{n}\hat{\boldsymbol{\psi}}$ has mean $\boldsymbol{h}$; the two non-centrality parameters coincide. □

Proof of lemma 1. $\mathbb{E}[-\log\varphi(y; \mu, s^2)] = \frac{1}{2}\log(2\pi s^2) + \sigma^2/(2s^2)$. Subtracting the same expression at $s^2 = \sigma^2$ gives $\frac{1}{2}\{\log(s^2/\sigma^2) + \sigma^2/s^2 - 1\} = \frac{1}{2}(u + e^{-u} - 1)$. The QLIKE loss $\sigma^2/s^2 - \log(\sigma^2/s^2) - 1$ equals $e^{-u} + u - 1$. Expanding e^{-u} gives the series. For random u, take expectations of $u^2/4$: $\mathbb{E}u^2 = b^2 + v$. □

Proof of proposition 15. The slopes of a linear projection are $\mathrm{Var}(\boldsymbol{v}_t)^{-1}\,\mathrm{Cov}(\boldsymbol{v}_t, T_t)$ for target T_t. Independence gives $\mathrm{Var}(\boldsymbol{v}_t) = \Sigma^* + \Sigma_\xi$ and $\mathrm{Cov}(\boldsymbol{v}_t, T_t) = \mathrm{Cov}(\boldsymbol{v}_t^*, T_t) = \Sigma^*\boldsymbol{\delta}$, the last equality being the normal equations of the latent projection. For $\Sigma_\xi = \omega^2 \boldsymbol{e}_1\boldsymbol{e}_1^\top$ the Sherman–Morrison formula gives $(\Sigma^* + \omega^2\boldsymbol{e}_1\boldsymbol{e}_1^\top)^{-1}\Sigma^*\boldsymbol{\delta} = \boldsymbol{\delta} - \omega^2\delta_1\Sigma^{*-1}\boldsymbol{e}_1/(1+\omega^2[\Sigma^{*-1}]_{11})$. By the partitioned-inverse formula, $\Sigma^{*-1}\boldsymbol{e}_1 = \nu_1^{-1}(1, -\boldsymbol{b}^\top)^\top$, so $[\Sigma^{*-1}]_{11} = \nu_1^{-1}$ and $\boldsymbol{\delta}^{\mathrm{obs}} = \boldsymbol{\delta} - \frac{\omega^2\delta_1}{\nu_1+\omega^2}(1, -b_5, -b_{22})^\top$, which is (30). Summing the three components gives the change in the sum of slopes. □

B Forecast-evaluation statistics

Let y_t be the realisation, F_t the predictive cdf, $\bar{\mu}_t$ its mean, V_t its variance and n the number of out-of-sample forecasts. All long-run variances use the Newey–West kernel with $\lfloor 4(n/100)^{2/9} \rfloor$ lags.

Scores

$\mathrm{NLL}_t = -\log f_t(y_t)$; CRPS_t as in (9); $\mathrm{SE}_t = (y_t - \bar{\mu}_t)^2$; Brier $(P_t(y>0) - \mathbb{1}\{y_t > 0\})^2$; $\mathrm{QLIKE}_t = \hat{\sigma}_t^2/V_t - \log(\hat{\sigma}_t^2/V_t) - 1$ with proxy $\hat{\sigma}_t^2$ (Patton, 2011); pinball loss at level a; and the FZ0 loss $-\mathbb{1}\{y_t \le q_t\}(q_t - y_t)/(ae_t) + q_t/e_t + \log(-e_t) - 1$ for the pair (VaR q_t, ES e_t) (Patton et al., 2019).

Out-of-sample R^2

$1 - \sum(y_t - \bar{\mu}_t)^2 / \sum(y_t - b_t)^2$ against $b_t = 0$ or the 250-day mean (Campbell and Thompson, 2008).

Clark–West

$f_t = (y_t - b_t)^2 - [(y_t - \bar{\mu}_t)^2 - (b_t - \bar{\mu}_t)^2]$; one-sided t-test of $\mathbb{E}f_t > 0$ (Clark and West, 2007).

Diebold–Mariano

$d_t = L_t^A - L_t^B$, $\mathrm{DM} = \bar{d}/\sqrt{\widehat{\mathrm{LRV}}/n}$, multiplied by the Harvey et al. (1997) factor and compared with t_{n-1}.

Pesaran–Timmermann

standardised difference between the hit rate and its value under independence of sign $\bar{\mu}_t$ and sign y_t (Pesaran and Timmermann, 1992).

Murphy decomposition

Brier = reliability − resolution + uncertainty over deciles of $P_t(y>0)$ (Murphy, 1973).

Berkowitz

$z_t = \Phi^{-1}(F_t(y_t))$; LR test of $(\mu, \rho, \sigma) = (0, 0, 1)$ in a Gaussian AR(1) for z_t, χ^2_3 (Berkowitz, 2001).

Coverage

violations $\mathbb{1}\{y_t < q_{a,t}\}$; Kupiec unconditional LR (χ^2_1) and Christoffersen conditional-coverage LR (χ^2_2).

Mincer–Zarnowitz
regression of the variance proxy on V_t; HAC Wald test of $(\alpha, \beta) = (0, 1)$ (Mincer and Zarnowitz, 1969).

Amisano–Giacomini
weighted log-score differences with weights emphasising the centre or the tails of the distribution (Amisano and Giacomini, 2007).

Model confidence set
$T_{\max}$ elimination with moving-block bootstrap ($B = 1000$, block length $n^{1/3}$), level 10% (Hansen et al., 2011).

All statistics are implemented in `smoe/evaluation/tests.py`; their empirical size and power are verified by simulation in the test suite.

C Leakage audit protocol

Look-ahead bias is a common reason why published backtests fail out of sample. This paper's predecessor (an undergraduate thesis) contained two forms of it, which we document here because they are generic. First, a "residual correction" added a fraction of the realised test error to the forecast, $\hat{P}_t \leftarrow \hat{P}_t + 0.6 \cdot \mathrm{MA}_3(P_t - \hat{P}_t)$, which uses P_t to forecast P_t. Second, a multi-day target $\bar{r}_{t+1:t+3}$ was included among the input features, so the last input row overlapped two of the three target days. Both produced large apparent gains that vanish without the leak. Neither survives in the present code base.

How large are the effects? Table 25 reproduces both mechanisms on clean data. With the lagged three-day target among the inputs, a gradient-boosted model reaches a directional hit rate of 88.5% for the next-day return and a cost-free long-only Sharpe ratio of 8.7; without it the same model has a hit rate of 52.7% and a Sharpe ratio of 0.43. The residual correction lowers the RMSE of a 60-lag price model from 52.6 to 44.9 index points (11% *below* the random walk, which proposition 2 shows to be impossible without future information) and lifts its directional accuracy from 49.8% to 65.1%. Both numbers are close to the headline results of the undergraduate thesis, which indicates that these two mechanisms produced them. More generally, a forecaster that beats the random walk on price levels by more than a fraction of a percent should be checked for leakage before anyone uses its results.

We impose five safeguards.

1. **Causal features by construction.** Every feature is a backward-looking rolling statistic. A unit test perturbs all prices after a random date and asserts that every feature up to that date is bit-for-bit unchanged.

2. **Targets never enter `predict`.** Models receive covariates and r_t only; the target column is not passed. Recursive models (GARCH, Markov switching) update their filters with r_t, which is known at the close of t.

Table 25: Anatomy of a leak: the two look-ahead mechanisms of the undergraduate thesis reproduced on CSI 300 (out of sample, 2012–June 2024)

Mechanism	Target	N	Hit rate	PT p	OOS R^2	Long-only Sharpe	RMSE	DA$^{\Delta}$
leak-free features	3-day mean	3,029	0.5239	0.0065	-0.0705	0.5110	–	–
leak-free features	next day	3,032	0.5270	0.0019	-0.0278	0.4307	–	–
features + lagged 3-day target (leak A)	3-day mean	3,029	0.7999	0.0000	0.6128	6.2158	–	–
features + lagged 3-day target (leak A)	next day	3,030	0.8851	0.0000	0.8425	8.6909	–	–
price model, leak-free	price level	3,032	0.4977	–	–	–	52.6446	0.4992
price model + residual correction (leak B)	price level	3,032	0.6511	–	–	–	44.8885	0.5582
random walk	price level	3,032	–	–	–	–	50.6798	0.4962

Upper block: gradient boosting on 10-day windows of the causal features, with and without the 3-day forward mean return (lagged one day, as in the original code) among the inputs. Lower block: OLS on 60 lagged price levels, with and without the residual correction $0.6 \cdot \mathrm{MA}_3(P_t - \hat{P}_t)$. Long-only Sharpe ignores costs.

3. **End-to-end corruption test.** For every model class, the full walk-forward is run twice (on the original data and on data whose prices after a cut-off are multiplied by 1.3) and all forecasts up to the cut-off must coincide to relative precision 10^{-8}. The test caught one violation during development: a Markov-switching fit drew random starting values from the global random-number generator, making forecasts non-reproducible; the generator is now seeded.

4. **Execution lag.** Positions decided at the close of t are filled at the open of $t+1$ and earn only the intraday return of $t+1$; the overnight return accrues to the previous position.

5. **Second data sources.** Realised measures at row t use only the 5-minute bars of day t. The VIX enters every series with the close of the last US trading day strictly before the local date, because the US close falls after the Asian and European closes of the same calendar day. A unit test multiplies all realised measures after a cut-off, and the VIX from the cut-off date onwards, by three and checks that no feature up to the cut-off changes.

Look-ahead through pretraining. None of these tests can detect a different channel: a pretrained foundation model whose training corpus contains the evaluation period may have stored the future in its weights. The documented corpora of Chronos and TimesFM 2.5 do not list the series we use, but they cannot be audited series by series. Section 16 therefore tests for this channel with the release dates of the models.

D Additional tables

E Derivations

This appendix collects the algebra behind sections 4 to 6 in enough detail to re-implement the estimator. Throughout, $\varphi_{tk} = \varphi(y_t; \mu_{kt}, \sigma^2_{kt})$, $\eta_{tk} = \log \sigma^2_{kt} = \boldsymbol{\delta}_k^\top \tilde{\boldsymbol{v}}_t$, $e_{tk} = y_t - \mu_{kt}$ and $u_{tk} = e^2_{tk} e^{-\eta_{tk}}$.

Table 26: Descriptive statistics of daily log returns (%), January 2005 (or inception) to June 2024

Asset	Start	N	Mean	SD	Skew	ExKurt	$\rho_1(r)$	$\rho_1(\lvert r\rvert)$	LB(10) p	ARCH(5) p
CSI 300	2005-01	4,732	0.027	1.617	-0.499	4.287	0.024	0.183	0.000	0.000
CSI 500	2005-01	4,732	0.034	1.849	-0.822	3.882	0.066	0.246	0.000	0.000
SSE 50	2005-01	4,732	0.022	1.625	-0.289	4.493	0.007	0.168	0.000	0.000
ChiNext	2010-06	3,418	0.016	1.893	-0.424	2.080	0.055	0.148	0.014	0.000
S&P 500	2005-01	4,904	0.031	1.217	-0.528	13.032	-0.124	0.322	0.000	0.000
Nasdaq 100	2005-01	4,904	0.051	1.391	-0.339	7.840	-0.105	0.273	0.000	0.000
Hang Seng	2005-01	4,800	0.005	1.480	0.066	7.667	-0.011	0.234	0.013	0.000
Nikkei 225	2005-01	4,769	0.026	1.427	-0.444	7.800	-0.034	0.251	0.166	0.000
DAX	2005-01	4,949	0.029	1.315	-0.230	8.452	-0.005	0.184	0.570	0.000
FTSE 100	2005-01	4,920	0.011	1.120	-0.410	10.358	-0.034	0.279	0.000	0.000
CSI 300 futures (IF)	2017-01	1,806	0.002	1.244	-0.366	6.277	-0.028	0.123	0.064	0.000
SHFE gold futures	2008-01	4,013	0.023	1.051	-0.344	5.157	-0.023	0.210	0.001	0.000
SHFE copper futures	2005-01	4,740	0.021	1.483	-0.223	2.951	-0.046	0.270	0.000	0.000
DCE soybean meal futures	2005-01	4,743	0.010	1.504	-0.901	12.957	-0.051	0.145	0.010	0.000
WTI crude futures	2005-01	4,900	0.013	2.772	-1.993	60.193	-0.048	0.347	0.000	0.000
COMEX gold futures	2005-01	4,898	0.035	1.120	-0.316	5.477	-0.002	0.109	0.021	0.000
US 10y T-note futures	2005-01	4,891	-0.000	0.376	-0.007	4.407	-0.009	0.178	0.076	0.000
Bitcoin	2014-09	3,574	0.138	3.688	-0.745	11.361	-0.022	0.209	0.034	0.000
CSI 300 ETF (510300)	2012-05	2,938	0.010	1.426	-0.622	8.981	0.027	0.237	0.000	0.000

LB(10): Ljung–Box test of no autocorrelation in returns; ARCH(5): Engle LM test. Every series rejects normality (Jarque–Bera $p < 10^{-10}$) and every series has ADF $p < 0.001$ for returns.

Table 27: The lagging paradox: price-level forecasts out of sample (2012–June 2024)

Asset	N	RMSE RW	RMSE AR60	Ratio	$\sum \hat{\phi}_j$	$\mathrm{corr}(\hat{P}_{t+1}, P_t)$	DA^{Δ} RW	Sign persist.	DA^{Δ} AR60	DA AR60
CSI 300	3,032	50.6798	52.6446	1.0388	0.9975	0.9999	0.4962	0.4962	0.4992	0.4977
CSI 500	3,032	95.7350	98.9242	1.0333	0.9981	0.9998	0.4929	0.4929	0.5091	0.5033
SSE 50	3,032	35.3608	36.4380	1.0305	0.9973	0.9999	0.4837	0.4837	0.4916	0.5056
ChiNext	2,789	42.2267	44.8775	1.0628	0.9983	0.9997	0.4857	0.4857	0.4903	0.4930
S&P 500	3,142	32.3878	32.9645	1.0178	0.9998	1.0000	0.4823	0.4823	0.4893	0.5153
Nasdaq 100	3,142	127.1189	130.2584	1.0247	1.0004	1.0000	0.4979	0.4979	0.5081	0.5169
Hang Seng	3,072	288.6937	293.3998	1.0163	0.9959	0.9999	0.5018	0.5018	0.5021	0.4795
Nikkei 225	3,057	267.7875	271.7946	1.0150	0.9991	1.0000	0.4895	0.4895	0.4885	0.4815
DAX	3,166	136.4540	139.2803	1.0207	0.9988	1.0000	0.4818	0.4818	0.4771	0.4763
FTSE 100	3,153	62.6282	63.8688	1.0198	0.9973	0.9998	0.4990	0.4990	0.4965	0.4957
CSI 300 futures (IF)	1,087	53.5492	56.0799	1.0473	0.9952	0.9996	0.4899	0.4899	0.4825	0.4903
SHFE gold futures	3,036	2.9505	2.9960	1.0154	0.9983	1.0000	0.4702	0.4702	0.4797	0.4908
SHFE copper futures	3,033	599.5284	608.9628	1.0157	0.9961	0.9999	0.4436	0.4436	0.4594	0.5021
DCE soybean meal futures	3,032	50.0153	50.7834	1.0154	0.9944	0.9998	0.4672	0.4672	0.4866	0.5040
WTI crude futures	3,139	1.5341	1.5541	1.0130	0.9952	0.9999	0.4834	0.4834	0.4882	0.5046
COMEX gold futures	3,140	15.4916	15.8136	1.0208	0.9991	1.0000	0.4680	0.4680	0.4718	0.4949
US 10y T-note futures	3,139	0.4144	0.4168	1.0057	0.9986	1.0000	0.4557	0.4557	0.4767	0.4989
Bitcoin	2,738	979.3071	1054.1113	1.0764	1.0028	0.9997	0.4739	0.4739	0.5057	0.5117
CSI 300 ETF (510300)	2,306	0.0579	0.0624	1.0779	0.9972	0.9993	0.4842	0.4842	0.4746	0.4783

RW: random walk $\hat{P}_{t+1} = P_t$. AR60: OLS of P_{t+1} on 60 lagged price levels, re-estimated yearly. Ratio > 1 means AR60 is worse than the random walk. DA^{Δ} compares the change in the forecast path with the change in the price path; for the RW it equals sign persistence exactly (proposition 3). DA compares $\hat{P}_{t+1} - P_t$ with $P_{t+1} - P_t$.

Table 28: White-box fingerprints: SMoE estimated on each series, 2005 (or inception) to June 2024

Asset	LR vs K=1	Gate Wald	$\bar{\pi}_{\text{trend}}$	$\bar{\pi}_{\text{crisis}}$	$t(\text{mom}_{120})$	$t(\text{mom}_{20})$	$t(\text{rev}_1)$	$t(\text{rev}_5)$	$t(\gamma_{\text{trend,vol}})$
CSI 300	330.60	70.81	0.67	0.11	3.81	0.43	2.21	1.10	-6.39
CSI 500	518.49	21.88	0.35	0.53	1.75	0.79	2.94	2.65	2.95
SSE 50	357.01	16.69	0.13	0.43	0.39	0.39	-0.10	-1.13	0.03
ChiNext	163.97	84.26	0.84	0.07	1.21	2.17	1.91	1.40	-4.24
S&P 500	455.50	10.92	0.27	0.41	-0.60	2.62	-2.87	-5.22	1.28
Nasdaq 100	313.68	14.41	0.23	0.40	0.04	-4.25	0.69	0.69	1.47
Hang Seng	270.88	23.72	0.26	0.63	0.51	0.44	3.52	3.09	-2.03
Nikkei 225	312.00	17.33	0.27	0.36	-2.43	2.78	-1.37	-2.32	0.57
DAX	362.85	16.19	0.43	0.32	-2.95	2.38	0.52	-3.31	0.39
FTSE 100	301.62	31.76	0.56	0.27	-1.21	-1.18	-2.83	-2.08	0.63
CSI 300 futures (IF)	150.29	4.53	0.37	0.55	0.15	-0.29	2.44	0.78	-0.28
SHFE gold futures	887.71	43.21	0.50	0.42	3.00	-3.52	-0.22	1.20	2.65
SHFE copper futures	683.26	43.85	0.10	0.49	0.68	2.95	-1.91	-1.09	1.18
DCE soybean meal futures	1277.48	16.36	0.52	0.04	-0.98	1.11	-2.36	-1.48	2.40
WTI crude futures	276.96	30.38	0.02	0.14	-0.27	4.01	-0.23	-0.80	1.58
COMEX gold futures	534.47	40.25	0.61	0.35	4.45	-1.17	2.40	-0.73	2.59
US 10y T-note futures	204.42	10.96	0.18	0.19	-0.33	0.93	-1.29	-1.57	-1.09
Bitcoin	1098.95	18.54	0.56	0.12	1.14	2.65	-6.59	-3.67	0.10
CSI 300 ETF (510300)	282.76	10.31	0.10	0.42	0.43	1.01	-0.23	-1.53	0.63

Robust (sandwich, Newey–West) t-statistics. Gate Wald: test that all gate slopes are zero (6 df).

Table 29: CSI 300 gate weights in the 20 trading days before and after market and policy events

Date	Event	trend (pre)	reversal (pre)	crisis (pre)	trend (post)	reversal (post)	crisis (post)
2008-09-19	stamp-duty cut, Huijin buying	0.199	0.587	0.214	0.182	0.602	0.216
2015-06-12	2015 crash begins	0.416	0.354	0.230	0.211	0.507	0.282
2015-07-08	rescue package	0.246	0.479	0.275	0.106	0.587	0.307
2016-01-04	circuit breaker	0.504	0.326	0.169	0.333	0.461	0.206
2018-03-22	trade war tariffs	0.719	0.188	0.093	0.820	0.126	0.055
2020-01-23	Wuhan lockdown	0.898	0.067	0.034	0.747	0.169	0.084
2024-02-05	national team buying	0.786	0.151	0.063	0.614	0.261	0.125

Table 30: CSI 300 variance forecasts against realised variance from 5-minute data (plus squared overnight return)

Model	N	QLIKE	MSE	MZ slope	MZ R^2
HistGauss	3,032	0.535	17.690	0.560	0.035
GARCH-t	3,032	0.334	13.655	1.053	0.244
AR-GJR-t	3,032	0.334	13.727	1.059	0.240
Gauss-HAR	3,032	0.296	12.450	1.143	0.312
MAR	3,032	0.299	11.992	1.250	0.346
LSTAR	3,032	**0.293**	**11.849**	1.201	0.350
MS-AR	3,032	0.419	15.141	1.089	0.174
XGB-Gauss	3,032	0.297	13.642	1.518	0.275
SMoE	3,032	0.303	11.975	1.333	**0.355**
Deep-SMoE	3,032	0.311	12.417	1.406	0.337
Chronos-Bolt-base	3,032	0.381	14.236	1.358	0.227
Chronos-Bolt-small	3,032	0.456	14.637	1.331	0.206
TimesFM-2.5	3,032	0.387	14.706	1.558	0.206

Table 31: Out-of-sample negative log score: difference from GARCH-t (lower is better)

Asset	HistGauss	GARCH-t	AR-GJR-t	Gauss-HAR	MAR	LSTAR	MS-AR	XGB-Gauss	Chronos-Bolt-small	TimesFM-2.5	Deep-SMoE	SMoE
CSI 300	0.1162	0.0000	0.0003	0.0235	0.0056	-0.0006	0.0486	0.0513	0.1397	0.0741	-0.0035	-0.0023
CSI 500	0.1246	0.0000	0.0002	0.0205	-0.0195	-0.0176	0.0408	0.0287	0.1907	0.0923	-0.0333	-0.0230
SSE 50	0.1231	0.0000	0.0004	0.0361	0.0042	0.0003	0.0461	0.0693	0.1332	0.0842	-0.0016	0.0034
ChiNext	0.0943	0.0000	0.0018	0.0086	-0.0030	-0.0022	0.0519	0.0201	0.1343	0.0627	-0.0009	0.0044
S&P 500	0.2059	0.0000	-0.0176	0.0072	-0.0315	-0.0221	0.0651	0.0471	0.2322	0.1533	-0.0229	-0.0341
Nasdaq 100	0.1479	0.0000	-0.0144	-0.0035	-0.0311	-0.0292	0.0428	0.0177	0.1913	0.1276	-0.0385	-0.0365
Hang Seng	0.0689	0.0000	-0.0058	0.0139	-0.0068	0.0023	0.0329	0.0176	0.1112	0.0606	-0.0056	-0.0100
Nikkei 225	0.1075	0.0000	-0.0089	0.0115	-0.0109	-0.0027	0.0296	0.0226	0.1471	0.0796	-0.0130	-0.0089
DAX	0.1310	0.0000	-0.0158	0.0121	-0.0166	-0.0115	0.0480	0.0332	0.1975	0.1033	-0.0291	-0.0172
FTSE 100	0.1348	0.0000	-0.0145	0.0097	-0.0155	-0.0091	0.0364	0.0158	0.1886	0.1020	-0.0199	-0.0194
CSI 300 futures (IF)	0.0472	0.0000	-0.0032	0.0163	0.0091	0.0049	0.0068	0.0084	0.0920	0.0492	0.0004	0.0063
SHFE gold futures	0.1044	0.0000	0.0007	0.1854	0.0637	0.0700	0.0495	0.1358	0.1691	0.1003	0.0842	0.0504
SHFE copper futures	0.1260	0.0000	-0.0036	0.0942	0.0171	0.0264	0.0452	0.0550	0.1986	0.1313	0.0231	0.0091
DCE soybean meal futures	0.1900	0.0000	-0.0072	0.1816	0.0063	0.0230	0.0266	0.2375	0.4233	0.2824	0.0308	0.0049
WTI crude futures	0.2035	0.0000	-0.0023	0.0138	-0.0123	-0.0121	0.0981	0.0202	0.1981	0.1501	-0.0036	-0.0110
COMEX gold futures	0.0863	0.0000	-0.0009	0.0604	0.0144	0.0084	0.0437	0.0686	0.1950	0.1033	0.0129	0.0051
US 10y T-note futures	0.0580	0.0000	0.0003	0.0117	-0.0026	-0.0056	0.0216	0.0193	0.1201	0.0522	-0.0015	-0.0016
Bitcoin	0.1915	0.0000	0.0009	0.1248	-0.0120	-0.0080	0.0517	0.3226	0.4593	0.3097	-0.0078	-0.0170
CSI 300 ETF (510300)	0.1268	0.0000	-0.0006	0.0345	0.0039	0.0045	0.0477	0.0924	0.1659	0.0811	0.0187	0.0102

Table 32: Out-of-sample CRPS: ratio to GARCH-t (lower is better)

Asset	HistGauss	GARCH-t	AR-GJR-t	Gauss-HAR	MAR	LSTAR	MS-AR	XGB-Gauss	Chronos-Bolt-small	TimesFM-2.5	Deep-SMoE	SMoE
CSI 300	1.026	1.000	1.000	1.001	0.999	1.001	1.017	0.996	1.008	1.015	0.997	1.000
CSI 500	1.028	1.000	1.000	1.001	0.994	0.994	1.018	0.993	1.011	1.014	0.987	0.993
SSE 50	1.028	1.000	1.000	1.005	1.000	1.002	1.015	0.999	1.007	1.015	1.000	1.002
ChiNext	1.026	1.000	1.002	1.000	0.997	0.999	1.018	1.001	1.012	1.013	1.003	1.003
S&P 500	1.056	1.000	0.992	0.996	0.991	0.995	1.022	1.001	1.026	1.033	0.997	0.991
Nasdaq 100	1.037	1.000	0.994	0.995	0.991	0.992	1.013	0.997	1.020	1.023	0.992	0.990
Hang Seng	1.012	1.000	0.997	1.003	0.998	1.001	1.011	0.998	1.010	1.012	1.000	0.997
Nikkei 225	1.024	1.000	0.996	1.001	0.998	0.997	1.011	0.998	1.014	1.016	0.997	0.996
DAX	1.034	1.000	0.994	0.996	0.992	0.993	1.015	0.996	1.016	1.021	0.991	0.992
FTSE 100	1.034	1.000	0.995	0.998	0.994	0.996	1.014	0.997	1.016	1.023	0.993	0.992
CSI 300 futures (IF)	1.008	1.000	0.999	1.008	1.002	1.002	1.005	0.999	1.005	1.012	1.005	1.000
SHFE gold futures	1.017	1.000	1.000	1.067	1.023	1.026	1.017	1.020	1.011	1.012	1.040	1.021
SHFE copper futures	1.022	1.000	0.997	1.038	1.011	1.017	1.015	1.011	1.012	1.023	1.015	1.008
DCE soybean meal futures	1.024	1.000	0.997	1.020	0.995	1.000	1.000	0.996	1.004	1.008	1.000	0.995
WTI crude futures	1.053	1.000	1.000	0.995	0.994	0.995	1.019	0.995	1.015	1.021	0.993	0.993
COMEX gold futures	1.013	1.000	0.999	1.014	1.003	1.003	1.016	1.002	1.011	1.013	1.008	1.003
US 10y T-note futures	1.011	1.000	1.000	0.999	0.998	0.997	1.005	1.000	1.008	1.008	1.000	0.998
Bitcoin	1.042	1.000	0.999	1.021	0.997	0.998	1.018	1.003	1.013	1.023	1.000	0.997
CSI 300 ETF (510300)	1.041	1.000	1.000	1.008	1.003	1.000	1.019	1.005	1.013	1.017	1.008	1.005

Table 33: Out-of-sample QLIKE (proxy y^2): ratio to GARCH-t (lower is better)

Asset	HistGauss	GARCH-t	AR-GJR-t	Gauss-HAR	MAR	LSTAR	MS-AR	XGB-Gauss	Chronos-Bolt-small	TimesFM-2.5	Deep-SMoE	SMoE
CSI 300	1.102	1.000	1.000	0.987	0.990	0.987	1.054	1.025	1.129	1.040	0.989	0.989
CSI 500	1.120	1.000	1.001	0.985	0.980	0.979	1.052	0.996	1.199	1.067	0.975	0.977
SSE 50	1.098	1.000	1.000	0.992	0.993	0.993	1.040	1.036	1.110	1.043	0.991	0.998
ChiNext	1.112	1.000	1.002	0.991	0.986	0.990	1.068	1.007	1.164	1.056	0.994	0.995
S&P 500	1.217	1.000	0.986	0.970	0.953	0.966	1.064	1.020	1.242	1.131	0.963	0.950
Nasdaq 100	1.161	1.000	0.985	0.966	0.956	0.962	1.049	0.994	1.210	1.117	0.948	0.951
Hang Seng	1.068	1.000	0.991	0.998	0.989	0.994	1.039	1.001	1.117	1.048	0.990	0.983
Nikkei 225	1.113	1.000	0.989	0.989	0.979	0.982	1.032	1.004	1.162	1.066	0.979	0.977
DAX	1.126	1.000	0.981	0.977	0.965	0.970	1.042	1.004	1.203	1.079	0.961	0.963
FTSE 100	1.134	1.000	0.976	0.976	0.962	0.968	1.034	0.983	1.198	1.079	0.963	0.961
CSI 300 futures (IF)	1.042	1.000	0.995	1.000	1.006	0.998	1.009	0.987	1.100	1.032	1.001	0.998
SHFE gold futures	1.057	1.000	1.003	1.140	1.131	1.124	1.045	1.087	1.128	1.045	1.101	1.082
SHFE copper futures	1.092	1.000	0.999	1.054	1.029	1.039	1.031	1.011	1.177	1.087	1.023	1.021
DCE soybean meal futures	1.010	1.000	0.980	1.005	0.999	1.012	0.990	1.056	1.225	1.090	0.999	1.015
WTI crude futures	1.216	1.000	0.989	0.970	0.970	0.968	1.108	0.977	1.205	1.138	0.981	0.979
COMEX gold futures	1.035	1.000	1.002	1.004	1.002	1.000	1.019	1.012	1.153	1.042	0.998	0.998
US 10y T-note futures	1.039	1.000	1.001	0.988	0.988	0.989	1.015	0.998	1.110	1.029	0.994	0.991
Bitcoin	1.018	1.000	1.024	0.957	0.955	0.955	0.994	1.140	1.280	1.127	0.954	0.971
CSI 300 ETF (510300)	1.088	1.000	1.001	0.986	0.986	0.987	1.042	1.054	1.134	1.033	1.020	0.988

Table 34: Out-of-sample FZ0 loss for (VaR, ES) at 5%: ratio to GARCH-t (lower is better)

Asset	HistGauss	GARCH-t	AR-GJR-t	Gauss-HAR	MAR	LSTAR	MS-AR	XGB-Gauss	Chronos-Bolt-small	TimesFM-2.5	Deep-SMoE	SMoE
CSI 300	1.158	1.000	0.999	0.993	1.007	0.978	1.051	1.060	1.212	1.108	1.001	0.987
CSI 500	1.115	1.000	0.993	0.984	0.971	0.972	1.001	1.046	1.321	1.109	0.959	0.971
SSE 50	1.150	1.000	1.003	1.000	1.010	0.985	1.050	1.045	1.129	1.099	0.999	1.007
ChiNext	1.089	1.000	0.992	0.996	0.994	0.998	1.039	1.024	1.154	1.064	0.996	1.004
S&P 500	1.326	1.000	0.981	0.998	0.925	0.958	1.080	1.122	1.596	1.330	0.952	0.921
Nasdaq 100	1.186	1.000	0.988	0.971	0.931	0.942	1.035	1.027	1.424	1.262	0.933	0.932
Hang Seng	1.067	1.000	0.982	0.985	0.966	0.978	1.041	1.017	1.213	1.067	0.957	0.967
Nikkei 225	1.115	1.000	0.974	0.988	0.986	0.986	1.012	1.028	1.231	1.110	0.962	0.981
DAX	1.121	1.000	0.974	0.995	0.958	0.959	1.038	1.072	1.376	1.149	0.945	0.955
FTSE 100	1.183	1.000	0.959	0.990	0.935	0.963	1.039	1.025	1.422	1.178	0.935	0.966
CSI 300 futures (IF)	1.072	1.000	0.996	0.988	1.020	1.007	0.997	1.019	1.162	1.036	0.995	1.004
SHFE gold futures	1.113	1.000	1.026	1.333	1.315	1.320	1.069	1.187	1.245	1.111	1.238	1.204
SHFE copper futures	1.163	1.000	1.004	1.088	1.066	1.080	1.026	1.027	1.266	1.169	1.031	1.023
DCE soybean meal futures	1.058	1.000	0.984	1.045	1.022	1.002	1.006	1.067	1.188	1.112	1.019	1.021
WTI crude futures	1.109	1.000	0.986	0.970	0.962	0.968	1.055	0.988	1.208	1.088	0.971	0.975
COMEX gold futures	1.036	1.000	1.007	1.017	1.017	1.007	1.017	1.101	1.345	1.137	1.012	1.015
US 10y T-note futures	0.861	1.000	0.994	1.019	1.024	1.018	0.924	0.942	0.430	0.809	0.990	0.980
Bitcoin	1.015	1.000	1.025	0.976	0.962	0.964	1.005	1.102	1.116	1.056	0.969	0.981
CSI 300 ETF (510300)	1.160	1.000	0.999	0.990	0.974	0.986	1.049	1.106	1.255	1.089	1.016	0.992

E.1 Complete-data likelihood and the Q function

With latent labels c_t, the complete-data log-likelihood is

$$\ell_n^c(\boldsymbol{\theta}) = \sum_t \sum_k \mathbb{1}\{c_t = k\}\Big[\log \pi_{kt} - \tfrac{1}{2}\log(2\pi) - \tfrac{1}{2}\eta_{tk} - \tfrac{1}{2}u_{tk}\Big]. \tag{34}$$

Taking expectations under $p_{\boldsymbol{\theta}^{(s)}}(c_t \mid y_t, \cdot)$ replaces $\mathbb{1}\{c_t = k\}$ by $\tau_{tk}^{(s)}$, and $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(s)})$ splits into

$$Q_\gamma = \sum_t \sum_k \tau_{tk} \log \pi_{kt}(\boldsymbol{\gamma}), \tag{35}$$

$$Q_{\beta,\delta} = \sum_k \sum_t \tau_{tk}\Big[-\tfrac{1}{2}\eta_{tk} - \tfrac{1}{2}(y_t - \boldsymbol{\beta}_k^\top \tilde{\boldsymbol{x}}_t^{(k)})^2 e^{-\eta_{tk}}\Big], \tag{36}$$

plus constants and the penalty. The expert blocks are separable across k, which is why each CM step is a set of K independent weighted regressions.

E.2 Mean step

For fixed $\boldsymbol{\delta}_k$, $Q_{\beta,\delta}$ is a weighted least-squares criterion in $\boldsymbol{\beta}_k$ with weights $a_{tk} = \tau_{tk} e^{-\eta_{tk}}$. Adding the ridge penalty and setting the gradient to zero,

$$\sum_t a_{tk}(y_t - \boldsymbol{\beta}_k^\top \tilde{\boldsymbol{x}}_t^{(k)})\tilde{\boldsymbol{x}}_t^{(k)} - \lambda R\boldsymbol{\beta}_k = 0 \iff \boldsymbol{\beta}_k = (\tilde{X}_k^\top A_k \tilde{X}_k + \lambda R)^{-1}\tilde{X}_k^\top A_k \boldsymbol{y},$$

which is (13). The matrix is positive definite whenever $\tilde{X}_k$ has full column rank on the support of $\tau_{\cdot k}$, so the step is the unique maximiser.

E.3 Variance step

For fixed $\boldsymbol{\beta}_k$ write $q(\boldsymbol{\delta}) = \sum_t \tau_{tk}[-\tfrac{1}{2}\eta_{tk} - \tfrac{1}{2}e_{tk}^2 e^{-\eta_{tk}}] - \tfrac{\lambda_v}{2}\|\boldsymbol{\delta} - \boldsymbol{\delta}_0\|^2$. Since $\partial\eta_{tk}/\partial\boldsymbol{\delta} = \tilde{\boldsymbol{v}}_t$,

$$\nabla q = \tfrac{1}{2}\sum_t \tau_{tk}(u_{tk} - 1)\tilde{\boldsymbol{v}}_t - \lambda_v(\boldsymbol{\delta} - \boldsymbol{\delta}_0), \qquad \nabla^2 q = -\tfrac{1}{2}\sum_t \tau_{tk} u_{tk}\tilde{\boldsymbol{v}}_t\tilde{\boldsymbol{v}}_t^\top - \lambda_v I.$$

Because $u_{tk} \geq 0$, $\nabla^2 q$ is negative definite and q strictly concave. The criterion is the weighted log-likelihood of a Gamma regression of e_{tk}^2 with shape $1/2$ and log link, so the Newton step is iteratively reweighted least squares with working weights $\tfrac{1}{2}\tau_{tk}u_{tk}$. In practice three to five damped Newton steps per CM step suffice.

E.4 Gate step

With $\pi_{kt} = \exp(\boldsymbol{\gamma}_k^\top \tilde{\boldsymbol{z}}_t)/\sum_l \exp(\boldsymbol{\gamma}_l^\top \tilde{\boldsymbol{z}}_t)$ and $\boldsymbol{\gamma}_K = \mathbf{0}$, $\partial \log \pi_{kt}/\partial\boldsymbol{\gamma}_j = (\mathbb{1}_{k=j} - \pi_{jt})\tilde{\boldsymbol{z}}_t$. Hence

$$\frac{\partial Q_\gamma}{\partial \boldsymbol{\gamma}_j} = \sum_t \sum_k \tau_{tk}(\mathbb{1}_{k=j} - \pi_{jt})\tilde{\boldsymbol{z}}_t = \sum_t (\tau_{tj} - \pi_{jt})\tilde{\boldsymbol{z}}_t,$$

using $\sum_k \tau_{tk} = 1$, and $\partial^2 Q_\gamma/\partial\boldsymbol{\gamma}_j\partial\boldsymbol{\gamma}_l^\top = -\sum_t \pi_{jt}(\mathbb{1}_{j=l} - \pi_{lt})\tilde{\boldsymbol{z}}_t\tilde{\boldsymbol{z}}_t^\top$, the negative of a sum of covariance matrices of a multinomial vector; it is negative semidefinite and, with the ridge,

negative definite.

E.5 Observed-data scores (Fisher's identity)

For any latent-variable model, $\nabla \ell_t(\boldsymbol{\theta}) = \mathbb{E}_{c_t|y_t;\boldsymbol{\theta}}[\nabla \ell_t^c(\boldsymbol{\theta})]$ (Louis, 1982). Differentiating the bracket of ℓ_n^c: $\partial/\partial\boldsymbol{\beta}_k$ gives $\mathbb{1}\{c_t = k\} e_{tk} e^{-\eta_{tk}} \tilde{\boldsymbol{x}}_t^{(k)}$; $\partial/\partial\boldsymbol{\delta}_k$ gives $\mathbb{1}\{c_t = k\} \frac{1}{2}(u_{tk} - 1)\tilde{\boldsymbol{v}}_t$; $\partial/\partial\boldsymbol{\gamma}_j$ gives $(\mathbb{1}\{c_t = j\} - \pi_{jt})\tilde{\boldsymbol{z}}_t$. Replacing indicators by τ_{tk} yields (15). At a stationary point of ℓ_n^λ the scores sum to the penalty gradient, which is the identity the L-BFGS refinement solves.

E.6 Information matrix

Louis' formula gives the observed information as $I_n = -\nabla^2 \ell_n = \mathbb{E}[-\nabla^2 \ell_n^c] - \mathrm{Var}[\nabla \ell_n^c]$, where both moments are over the posterior of $(c_1, \ldots, c_n)$. The first term is block-diagonal across $(\boldsymbol{\beta}, \boldsymbol{\delta}, \boldsymbol{\gamma})$ and available from the CM steps; the second term couples the blocks through the posterior variance $\tau_{tk}(\mathbb{1}_{k=l} - \tau_{tl})$ of the labels. We obtain $A = I_n/n$ numerically by central differences of the exact analytic score, which is simpler to implement; the analytic score itself is verified against numerical differentiation of ℓ_n in the test suite. The sandwich meat B uses the per-observation scores directly.

E.7 CRPS of a Gaussian mixture

For $F = \sum_k \pi_k \mathcal{N}(\mu_k, \sigma_k^2)$, $\mathrm{CRPS}(F, y) = \mathbb{E}|X - y| - \frac{1}{2}\mathbb{E}|X - X'|$. Conditioning on the components, $X - y \sim \mathcal{N}(\mu_k - y, \sigma_k^2)$ and $X - X' \sim \mathcal{N}(\mu_k - \mu_l, \sigma_k^2 + \sigma_l^2)$. For $Z \sim \mathcal{N}(m, s^2)$,

$$\mathbb{E}|Z| = \int |z| \varphi(z; m, s^2) dz = m\{2\Phi(m/s) - 1\} + 2s\varphi(m/s) = A(m, s^2),$$

obtained by splitting the integral at zero and using $\int_0^\infty z\varphi(z; m, s^2)dz = m\Phi(m/s) + s\varphi(m/s)$. Summing over components with weights π_k and $\pi_k\pi_l$ gives (9).

E.8 Quantiles and expected shortfall

The mixture cdf $F(q) = \sum_k \pi_k \Phi((q - \mu_k)/\sigma_k)$ is continuous and strictly increasing, so q_α is found by bisection on $[\min_k(\mu_k - 12\sigma_k), \max_k(\mu_k + 12\sigma_k)]$, an interval that contains every quantile with $\alpha \geq 10^{-30}$. For expected shortfall,

$$\mathbb{E}[Y\mathbb{1}\{Y \leq q\}] = \sum_k \pi_k \int_{-\infty}^{q} y\, \varphi(y; \mu_k, \sigma_k^2) dy = \sum_k \pi_k \{\mu_k \Phi(z_k) - \sigma_k \varphi(z_k)\}, \quad z_k = (q - \mu_k)/\sigma_k,$$

and $\mathrm{ES}_\alpha = \mathbb{E}[Y\mathbb{1}\{Y \leq q_\alpha\}]/\alpha$.

E.9 Higher moments

With $d_k = \mu_k - \bar{\mu}$ and the central moments of a normal, $\mathbb{E}[(Y - \bar{\mu})^3] = \sum_k \pi_k \mathbb{E}[(d_k + \sigma_k Z)^3] = \sum_k \pi_k (d_k^3 + 3d_k\sigma_k^2)$ and $\mathbb{E}[(Y - \bar{\mu})^4] = \sum_k \pi_k (d_k^4 + 6d_k^2\sigma_k^2 + 3\sigma_k^4)$. Skewness of the predictive distribution is therefore generated entirely by disagreement between expert means ($d_k \neq 0$)

interacting with expert variances: a crisis expert with a negative mean and a large variance produces negative skewness, which the SMoE can express and a single Gaussian cannot.

E.10 Gate elasticity of tail risk

Differentiating $F(q_\alpha) = \alpha$ with respect to a gate covariate z_j that enters only the gate,

$$\frac{\partial q_\alpha}{\partial z_j} = -\frac{\sum_k \Phi(z_k)\, \partial \pi_k / \partial z_j}{\sum_k \pi_k \varphi(z_k)/\sigma_k} = -\frac{\sum_k \pi_k (\gamma_{kj} - \bar{\gamma}_j)\Phi(z_k)}{f(q_\alpha)},$$

with $z_k = (q_\alpha - \mu_k)/\sigma_k$ and f the mixture density: VaR moves towards the tail of the experts whose gate loading on z_j is above average, at a rate inversely proportional to the density at the quantile. This closed form makes statements such as "a one-standard-deviation rise in volatility lowers the 1% VaR of the CSI 300 by x percentage points through the gate" exact.

F Out-of-sample gate weights for every series

Figures 11 and 12 show the SMoE's out-of-sample gate probabilities (20-day moving averages), i.e. the regime weights the model actually used when forecasting. They are computed from the same forecasts that enter all evaluation tables and are therefore free of look-ahead.

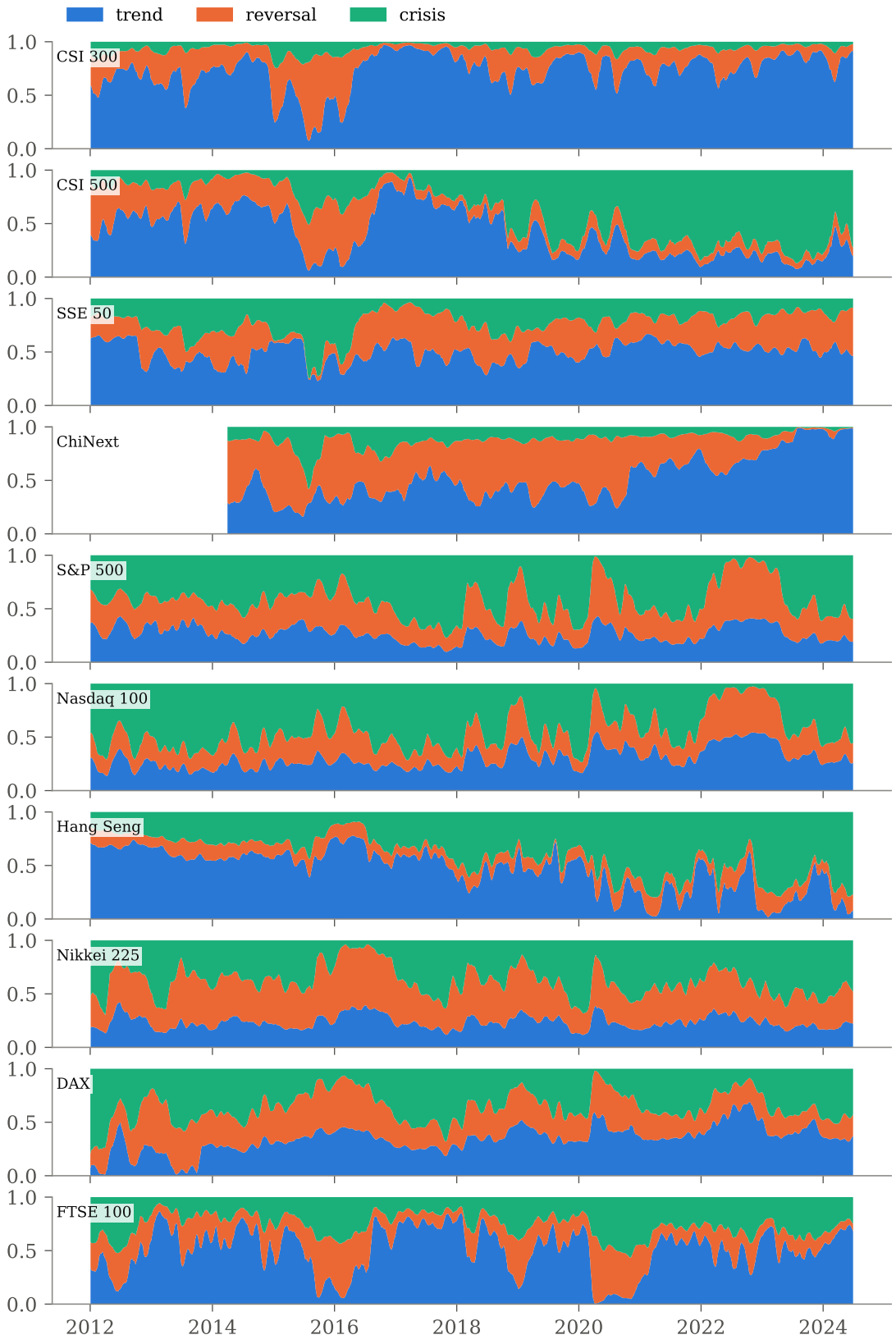


Figure 11: Out-of-sample SMoE gate weights: Chinese and international equity indices.

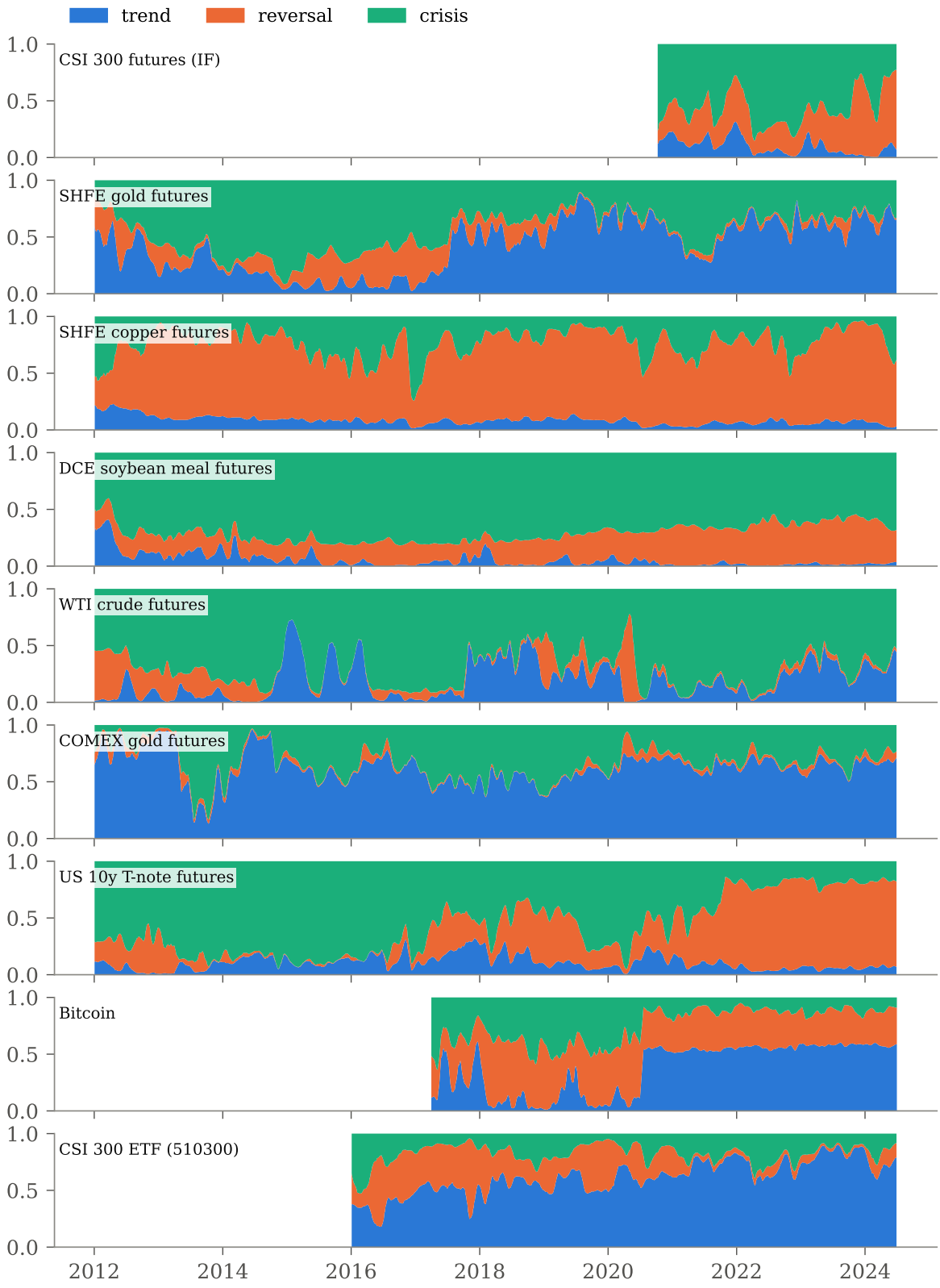


Figure 12: Out-of-sample SMoE gate weights: futures, Bitcoin and the CSI 300 ETF.

G Detailed out-of-sample results by series

For every series, the tables report mean losses (lower is better; the best model in bold), the model-confidence-set p-value for log score, the directional hit rate, VaR violation rates with the Kupiec p-value, and the Berkowitz p-value of the PIT.

Table 35: Out-of-sample results: CSI 300 ($n = 3032$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.6889	0.7112	1.7631	1.1842	0.2493	0.0000	0.5261	0.0211	0.0455	0.0000	0.0066
GARCH-t	1.5727	0.6928	1.6004	1.0229	0.2506	0.8920	0.5096	0.0122	0.0485	0.2386	0.0009
AR-GJR-t	1.5730	0.6930	1.6007	1.0216	0.2506	0.8290	0.5073	0.0122	0.0482	0.2386	0.0019
Gauss-HAR	1.5963	0.6936	**1.5790**	1.0157	0.2502	0.0090	0.5007	0.0191	0.0432	0.0000	0.6238
MAR	1.5783	0.6923	1.5837	1.0300	0.2514	0.1920	0.5125	0.0079	0.0416	0.2313	0.6746
LSTAR	1.5722	0.6935	1.5793	**0.9999**	0.2523	0.8920	0.5079	0.0082	0.0429	0.3166	0.3562
MS-AR	1.6214	0.7047	1.6868	1.0753	0.2511	0.0000	0.4990	0.0122	0.0402	0.2386	0.0000
XGB-Gauss	1.6240	**0.6904**	1.6405	1.0839	0.2492	0.0090	0.5066	0.0297	0.0676	0.0000	0.0000
Chronos-Bolt-small	1.7124	0.6982	1.8067	1.2398	0.2506	0.0000	0.5221	0.0359	0.0798	0.0000	0.0000
Chronos-Bolt-base	1.6702	0.6964	1.7234	1.1689	0.2500	0.0000	0.5254	0.0307	0.0712	0.0000	0.0000
TimesFM-2.5	1.6469	0.7029	1.6648	1.1333	0.2530	0.0000	0.4947	0.0247	0.0590	0.0000	0.0000
Deep-SMoE	**1.5693**	0.6908	1.5821	1.0240	0.2496	1.0000	0.5168	0.0102	0.0412	0.9016	0.0200
SMoE	1.5705	0.6931	1.5831	1.0095	0.2512	0.8920	0.5063	0.0079	0.0416	0.2313	0.6404

Table 36: Out-of-sample results: CSI 500 ($n = 3032$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.8334	0.8162	1.7310	1.4251	0.2490	0.0000	0.5234	0.0287	0.0564	0.0000	0.0000
GARCH-t	1.7088	0.7938	1.5461	1.2786	0.2501	0.0010	0.5323	0.0172	0.0623	0.0003	0.0000
AR-GJR-t	1.7090	0.7935	1.5473	1.2690	0.2500	0.0010	0.5257	0.0168	0.0623	0.0006	0.0000
Gauss-HAR	1.7292	0.7946	1.5230	1.2580	0.2506	0.0000	0.5056	0.0224	0.0511	0.0000	0.0011
MAR	1.6893	0.7887	1.5153	1.2421	0.2493	0.0140	0.5026	0.0073	0.0406	0.1103	0.0078
LSTAR	1.6911	0.7892	1.5133	1.2425	0.2505	0.0140	0.5026	0.0089	0.0356	0.5369	0.0017
MS-AR	1.7496	0.8083	1.6259	1.2794	0.2507	0.0000	0.5106	0.0132	0.0511	0.0921	0.0000
XGB-Gauss	1.7374	0.7881	1.5398	1.3371	0.2484	0.0010	0.5290	0.0317	0.0709	0.0000	0.0000
Chronos-Bolt-small	1.8995	0.8025	1.8541	1.6884	0.2502	0.0000	0.5211	0.0554	0.1055	0.0000	0.0000
Chronos-Bolt-base	1.8360	0.7974	1.7183	1.5786	0.2494	0.0000	0.5237	0.0439	0.0950	0.0000	0.0000
TimesFM-2.5	1.8011	0.8046	1.6493	1.4182	0.2526	0.0000	0.5099	0.0333	0.0762	0.0000	0.0000
Deep-SMoE	**1.6754**	**0.7834**	**1.5080**	**1.2259**	0.2462	1.0000	0.5386	0.0082	0.0399	0.3166	0.0000
SMoE	1.6858	0.7883	1.5103	1.2418	0.2489	0.0140	0.5109	0.0086	0.0429	0.4190	0.0641

Table 37: Out-of-sample results: SSE 50 ($n = 3032$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.7067	0.7210	1.7801	1.1624	0.2500	0.0000	0.5122	0.0201	0.0419	0.0000	0.0093
GARCH-t	1.5836	0.7010	1.6209	1.0104	0.2504	0.9110	0.4987	0.0096	0.0468	0.8082	0.0300
AR-GJR-t	1.5840	0.7013	1.6215	1.0135	0.2504	0.9110	0.4941	0.0096	0.0462	0.8082	0.0146
Gauss-HAR	1.6196	0.7044	1.6073	1.0103	0.2510	0.0010	0.4984	0.0175	0.0432	0.0002	0.7249
MAR	1.5877	0.7012	1.6101	1.0201	0.2511	0.5280	0.5053	0.0112	0.0399	0.5100	0.4513
LSTAR	1.5839	0.7022	1.6089	**0.9950**	0.2525	0.9110	0.4967	0.0106	0.0439	0.7612	0.3724
MS-AR	1.6296	0.7116	1.6852	1.0614	0.2508	0.0010	0.4911	0.0106	0.0373	0.7612	0.0000
XGB-Gauss	1.6529	**0.7000**	1.6785	1.0554	0.2508	0.0010	0.5145	0.0264	0.0663	0.0000	0.0000
Chronos-Bolt-small	1.7168	0.7061	1.7997	1.1406	0.2511	0.0000	0.5026	0.0323	0.0768	0.0000	0.0000
Chronos-Bolt-base	1.6933	0.7052	1.7524	1.1222	0.2508	0.0000	0.5023	0.0317	0.0699	0.0000	0.0000
TimesFM-2.5	1.6678	0.7113	1.6910	1.1106	0.2533	0.0000	0.4891	0.0261	0.0623	0.0000	0.0000
Deep-SMoE	**1.5820**	0.7008	**1.6068**	1.0092	0.2515	1.0000	0.5007	0.0109	0.0452	0.6296	0.1319
SMoE	1.5869	0.7022	1.6181	1.0175	0.2508	0.5710	0.5112	0.0109	0.0419	0.6296	0.2035

Table 38: Out-of-sample results: ChiNext ($n = 2306$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	2.0278	1.0167	1.5820	1.4970	0.2493	0.0000	0.5312	0.0234	0.0516	0.0000	0.0012
GARCH-t	1.9335	0.9910	1.4222	1.3746	0.2504	0.9130	0.4931	0.0139	0.0546	0.0772	0.2867
AR-GJR-t	1.9353	0.9927	1.4245	**1.3635**	0.2519	0.5930	0.4696	0.0130	0.0542	0.1651	0.3199
Gauss-HAR	1.9421	0.9912	1.4095	1.3688	0.2517	0.4580	0.4935	0.0165	0.0460	0.0042	0.4531
MAR	**1.9306**	**0.9882**	**1.4025**	1.3665	0.2519	1.0000	0.5056	0.0078	0.0442	0.2707	0.9335
LSTAR	1.9313	0.9900	1.4085	1.3725	0.2517	0.9200	0.5061	0.0108	0.0399	0.6887	0.5788
MS-AR	1.9854	1.0093	1.5185	1.4284	0.2523	0.0010	0.4653	0.0143	0.0464	0.0507	0.0374
XGB-Gauss	1.9537	0.9922	1.4316	1.4071	0.2505	0.0760	0.5191	0.0225	0.0564	0.0000	0.0000
Chronos-Bolt-small	2.0678	1.0027	1.6561	1.5865	0.2533	0.0000	0.5013	0.0390	0.0846	0.0000	0.0000
Chronos-Bolt-base	2.0112	0.9942	1.5433	1.4946	0.2509	0.0010	0.5199	0.0304	0.0733	0.0000	0.0000
TimesFM-2.5	1.9962	1.0041	1.5018	1.4623	0.2541	0.0010	0.4952	0.0269	0.0642	0.0000	0.0000
Deep-SMoE	1.9327	0.9945	1.4131	1.3695	0.2522	0.9200	0.5169	0.0095	0.0455	0.8231	0.8796
SMoE	1.9380	0.9936	1.4148	1.3798	0.2529	0.4580	0.4974	0.0091	0.0455	0.6616	0.7625

Table 39: Out-of-sample results: S&P 500 ($n = 3142$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.4202	0.5299	1.9518	0.9700	0.2486	0.0000	0.5331	0.0258	0.0522	0.0000	0.0030
GARCH-t	1.2144	0.5018	1.6039	0.7313	0.2487	0.0010	0.5411	0.0156	0.0602	0.0036	0.0337
AR-GJR-t	1.1968	0.4976	1.5811	0.7173	0.2484	0.2460	0.5423	0.0134	0.0592	0.0712	0.3116
Gauss-HAR	1.2216	0.4999	1.5554	0.7297	0.2491	0.0010	0.5236	0.0223	0.0547	0.0000	0.0068
MAR	1.1828	0.4974	1.5284	0.6767	0.2490	0.2500	0.5271	0.0108	0.0452	0.6480	0.0214
LSTAR	1.1923	0.4995	1.5492	0.7002	0.2494	0.2460	0.5220	0.0121	0.0474	0.2534	0.0056
MS-AR	1.2795	0.5126	1.7061	0.7899	0.2482	0.0000	0.5328	0.0124	0.0506	0.1903	0.0000
XGB-Gauss	1.2615	0.5022	1.6355	0.8205	0.2494	0.0000	0.5130	0.0353	0.0789	0.0000	0.0000
Chronos-Bolt-small	1.4465	0.5150	1.9913	1.1673	0.2498	0.0000	0.5395	0.0519	0.0990	0.0000	0.0000
Chronos-Bolt-base	1.3594	0.5109	1.8170	1.0271	0.2497	0.0000	0.5391	0.0471	0.0885	0.0000	0.0000
TimesFM-2.5	1.3677	0.5183	1.8147	0.9725	0.2528	0.0000	0.5178	0.0407	0.0815	0.0000	0.0000
Deep-SMoE	1.1914	0.5004	1.5450	0.6965	0.2502	0.2460	0.5029	0.0115	0.0490	0.4222	0.0094
SMoE	**1.1802**	**0.4970**	**1.5235**	**0.6739**	0.2492	1.0000	0.5216	0.0102	0.0411	0.9174	0.0459

Table 40: Out-of-sample results: Nasdaq 100 ($n = 3142$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.6258	0.6666	1.7879	1.1522	0.2477	0.0000	0.5477	0.0251	0.0560	0.0000	0.0004
GARCH-t	1.4779	0.6427	1.5404	0.9714	0.2471	0.0000	0.5538	0.0153	0.0643	0.0058	0.0801
AR-GJR-t	1.4635	0.6385	1.5176	0.9593	0.2475	0.0030	0.5557	0.0156	0.0595	0.0036	0.7861
Gauss-HAR	1.4744	0.6394	1.4874	0.9435	0.2490	0.0000	0.5356	0.0242	0.0598	0.0000	0.0133
MAR	1.4468	0.6368	1.4721	**0.9040**	0.2474	0.1130	0.5363	0.0111	0.0525	0.5285	0.0198
LSTAR	1.4486	0.6375	1.4821	0.9155	0.2481	0.1130	0.5318	0.0130	0.0528	0.1009	0.0113
MS-AR	1.5207	0.6509	1.6154	1.0051	0.2476	0.0000	0.5363	0.0169	0.0544	0.0004	0.0027
XGB-Gauss	1.4956	0.6406	1.5305	0.9977	0.2486	0.0000	0.5465	0.0315	0.0764	0.0000	0.0000
Chronos-Bolt-small	1.6692	0.6556	1.8643	1.3830	0.2486	0.0000	0.5420	0.0519	0.1015	0.0000	0.0000
Chronos-Bolt-base	1.6013	0.6514	1.7290	1.2519	0.2484	0.0000	0.5465	0.0446	0.0913	0.0000	0.0000
TimesFM-2.5	1.6055	0.6575	1.7200	1.2258	0.2501	0.0000	0.5391	0.0420	0.0878	0.0000	0.0000
Deep-SMoE	**1.4394**	0.6376	**1.4600**	0.9067	0.2481	1.0000	0.5169	0.0121	0.0554	0.2534	0.0125
SMoE	1.4414	**0.6361**	1.4652	0.9050	0.2478	0.6980	0.5232	0.0111	0.0500	0.5285	0.0399

Table 41: Out-of-sample results: Hang Seng ($n = 3072$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.6371	0.6787	1.6393	1.0697	0.2500	0.0010	0.5062	0.0208	0.0475	0.0000	0.1119
GARCH-t	1.5682	0.6706	1.5350	1.0025	0.2500	0.1420	0.5107	0.0153	0.0605	0.0062	0.0031
AR-GJR-t	1.5624	**0.6687**	1.5218	0.9848	0.2499	0.8240	0.5111	0.0130	0.0576	0.1078	0.0331
Gauss-HAR	1.5821	0.6724	1.5312	0.9878	0.2502	0.0010	0.5026	0.0160	0.0472	0.0023	0.0001
MAR	1.5613	0.6692	1.5173	0.9683	0.2502	0.8240	0.4984	0.0094	0.0426	0.7529	0.0028
LSTAR	1.5704	0.6710	1.5263	0.9804	0.2508	0.0200	0.5003	0.0114	0.0452	0.4477	0.0011
MS-AR	1.6011	0.6780	1.5955	1.0435	0.2499	0.0010	0.5049	0.0153	0.0625	0.0062	0.0585
XGB-Gauss	1.5858	0.6695	1.5361	1.0200	0.2501	0.0020	0.5238	0.0218	0.0638	0.0000	0.0000
Chronos-Bolt-small	1.6793	0.6775	1.7152	1.2155	0.2513	0.0000	0.5085	0.0407	0.0905	0.0000	0.0000
Chronos-Bolt-base	1.6414	0.6750	1.6411	1.1414	0.2507	0.0000	0.5052	0.0329	0.0801	0.0000	0.0000
TimesFM-2.5	1.6288	0.6786	1.6080	1.0696	0.2525	0.0010	0.5081	0.0290	0.0693	0.0000	0.0000
Deep-SMoE	1.5626	0.6704	1.5189	**0.9596**	0.2504	0.8240	0.5026	0.0065	0.0420	0.0379	0.0000
SMoE	**1.5582**	0.6687	**1.5092**	0.9697	0.2505	1.0000	0.5059	0.0094	0.0417	0.7529	0.0011

Table 42: Out-of-sample results: Nikkei 225 ($n = 3057$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.6697	0.6845	1.7130	1.1422	0.2501	0.0000	0.5067	0.0206	0.0556	0.0000	0.0324
GARCH-t	1.5622	0.6685	1.5393	1.0243	0.2491	0.0850	0.5345	0.0150	0.0582	0.0091	0.1068
AR-GJR-t	1.5533	**0.6658**	1.5216	0.9977	0.2492	0.8860	0.5335	0.0141	0.0569	0.0333	0.2601
Gauss-HAR	1.5737	0.6691	1.5225	1.0120	0.2501	0.0000	0.5018	0.0170	0.0474	0.0004	0.2177
MAR	1.5513	0.6671	1.5068	1.0095	0.2506	0.8860	0.5005	0.0101	0.0425	0.9378	0.2270
LSTAR	1.5595	0.6668	1.5116	1.0097	0.2501	0.0850	0.5067	0.0121	0.0429	0.2579	0.2206
MS-AR	1.5918	0.6762	1.5893	1.0367	0.2493	0.0000	0.5293	0.0114	0.0491	0.4312	0.0000
XGB-Gauss	1.5848	0.6673	1.5448	1.0528	0.2500	0.0000	0.5129	0.0252	0.0648	0.0000	0.0000
Chronos-Bolt-small	1.7093	0.6779	1.7884	1.2607	0.2506	0.0000	0.5132	0.0461	0.0926	0.0000	0.0000
Chronos-Bolt-base	1.6538	0.6738	1.6806	1.1772	0.2502	0.0000	0.5146	0.0386	0.0831	0.0000	0.0000
TimesFM-2.5	1.6418	0.6791	1.6401	1.1366	0.2523	0.0000	0.5110	0.0307	0.0729	0.0000	0.0000
Deep-SMoE	**1.5492**	0.6663	1.5072	**0.9851**	0.2507	1.0000	0.4920	0.0098	0.0432	0.9172	0.0874
SMoE	1.5533	0.6660	**1.5038**	1.0052	0.2501	0.8860	0.5005	0.0121	0.0429	0.2579	0.3680

Table 43: Out-of-sample results: DAX ($n = 3166$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.5740	0.6248	1.7686	1.0643	0.2499	0.0040	0.5123	0.0202	0.0581	0.0000	0.8871
GARCH-t	1.4430	0.6045	1.5704	0.9493	0.2488	0.0110	0.5376	0.0142	0.0654	0.0250	0.0007
AR-GJR-t	1.4272	0.6006	1.5406	0.9245	0.2487	0.5040	0.5379	0.0139	0.0600	0.0374	0.0252
Gauss-HAR	1.4551	0.6022	1.5348	0.9445	0.2489	0.0110	0.5379	0.0218	0.0603	0.0000	0.2470
MAR	1.4263	0.5995	1.5160	0.9090	0.2480	0.5040	0.5306	0.0133	0.0461	0.0786	0.2814
LSTAR	1.4315	0.6004	1.5237	0.9106	0.2484	0.1730	0.5363	0.0139	0.0524	0.0374	0.0229
MS-AR	1.4910	0.6137	1.6368	0.9856	0.2490	0.0040	0.5275	0.0148	0.0553	0.0106	0.0000
XGB-Gauss	1.4762	0.6023	1.5766	1.0174	0.2493	0.0110	0.5357	0.0322	0.0771	0.0000	0.0000
Chronos-Bolt-small	1.6405	0.6142	1.8885	1.3063	0.2512	0.0010	0.5171	0.0493	0.1036	0.0000	0.0000
Chronos-Bolt-base	1.5828	0.6119	1.7752	1.1897	0.2509	0.0110	0.5167	0.0436	0.0970	0.0000	0.0000
TimesFM-2.5	1.5463	0.6173	1.6939	1.0907	0.2529	0.0010	0.5123	0.0366	0.0821	0.0000	0.0000
Deep-SMoE	**1.4139**	**0.5992**	**1.5084**	**0.8970**	0.2493	1.0000	0.5069	0.0136	0.0499	0.0548	0.9739
SMoE	1.4258	0.5998	1.5124	0.9067	0.2484	0.5040	0.5268	0.0130	0.0442	0.1106	0.3683

Table 44: Out-of-sample results: FTSE 100 ($n = 3153$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.3456	0.4961	1.7771	0.8558	0.2502	0.0020	0.5075	0.0206	0.0533	0.0000	0.8978
GARCH-t	1.2107	0.4800	1.5670	0.7237	0.2498	0.0020	0.5281	0.0174	0.0571	0.0001	0.0005
AR-GJR-t	1.1962	0.4774	1.5300	0.6938	0.2498	0.6460	0.5230	0.0165	0.0558	0.0008	0.0007
Gauss-HAR	1.2204	0.4793	1.5295	0.7165	0.2501	0.0020	0.5144	0.0212	0.0552	0.0000	0.2965
MAR	1.1952	0.4771	1.5074	**0.6767**	0.2500	0.6460	0.5135	0.0133	0.0457	0.0745	0.1201
LSTAR	1.2016	0.4780	1.5163	0.6972	0.2507	0.0270	0.5103	0.0136	0.0441	0.0517	0.0097
MS-AR	1.2471	0.4867	1.6200	0.7519	0.2504	0.0020	0.5075	0.0127	0.0517	0.1456	0.0000
XGB-Gauss	1.2265	0.4788	1.5399	0.7416	0.2503	0.0020	0.5179	0.0282	0.0650	0.0000	0.0000
Chronos-Bolt-small	1.3993	0.4875	1.8778	1.0288	0.2502	0.0010	0.5312	0.0460	0.0990	0.0000	0.0000
Chronos-Bolt-base	1.3482	0.4868	1.7698	0.9652	0.2515	0.0020	0.5081	0.0412	0.0885	0.0000	0.0000
TimesFM-2.5	1.3127	0.4909	1.6906	0.8523	0.2536	0.0020	0.4951	0.0352	0.0736	0.0000	0.0000
Deep-SMoE	**1.1908**	0.4767	1.5095	0.6769	0.2504	1.0000	0.5154	0.0121	0.0473	0.2619	0.0201
SMoE	1.1913	**0.4761**	**1.5060**	0.6989	0.2491	0.9370	0.5103	0.0127	0.0409	0.1456	0.0001

Table 45: Out-of-sample results: CSI 300 futures (IF) ($n = 844$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.5261	0.6068	1.5864	0.9338	0.2498	0.1600	0.5213	0.0130	0.0367	0.3975	0.3007
GARCH-t	1.4789	0.6020	1.5230	0.8709	0.2521	0.9260	0.4502	0.0047	0.0474	0.0869	0.0114
AR-GJR-t	**1.4756**	**0.6012**	1.5159	0.8676	0.2517	1.0000	0.4799	0.0047	0.0438	0.0869	0.0119
Gauss-HAR	1.4952	0.6069	1.5224	**0.8608**	0.2511	0.3240	0.4834	0.0083	0.0355	0.6077	0.0001
MAR	1.4880	0.6030	1.5314	0.8880	0.2522	0.8010	0.5118	0.0083	0.0391	0.6077	0.0066
LSTAR	1.4838	0.6030	1.5202	0.8766	0.2524	0.9260	0.5095	0.0071	0.0403	0.3734	0.0026
MS-AR	1.4857	0.6052	1.5374	0.8680	0.2526	0.9260	0.4562	0.0071	0.0403	0.3734	0.0005
XGB-Gauss	1.4873	0.6015	**1.5030**	0.8873	0.2520	0.9260	0.4680	0.0201	0.0581	0.0092	0.0071
Chronos-Bolt-small	1.5709	0.6052	1.6750	1.0121	0.2496	0.0550	0.5415	0.0344	0.0746	0.0000	0.0000
Chronos-Bolt-base	1.5514	0.6050	1.6334	0.9831	0.2504	0.0990	0.5166	0.0320	0.0640	0.0000	0.0000
TimesFM-2.5	1.5281	0.6095	1.5713	0.9022	0.2545	0.1140	0.5000	0.0237	0.0509	0.0007	0.0002
Deep-SMoE	1.4793	0.6053	1.5245	0.8668	0.2537	0.9260	0.4585	0.0047	0.0367	0.0869	0.0000
SMoE	1.4852	0.6021	1.5203	0.8747	0.2521	0.9260	0.5166	0.0083	0.0379	0.6077	0.0029

Table 46: Out-of-sample results: SHFE gold futures ($n = 3036$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.2709	0.4624	1.9341	0.6888	0.2497	0.0090	0.5175	0.0181	0.0418	0.0001	0.3901
GARCH-t	**1.1664**	**0.4547**	**1.8290**	**0.6188**	0.2502	1.0000	0.5122	0.0096	0.0448	0.8026	0.0653
AR-GJR-t	1.1671	0.4548	1.8349	0.6348	0.2500	0.6560	0.5158	0.0099	0.0478	0.9475	0.0022
Gauss-HAR	1.3518	0.4850	2.0852	0.8251	0.2495	0.0000	0.5175	0.0063	0.0188	0.0260	0.0000
MAR	1.2302	0.4653	2.0683	0.8137	0.2497	0.0570	0.5198	0.0023	0.0181	0.0000	0.0000
LSTAR	1.2365	0.4665	2.0554	0.8167	0.2496	0.0390	0.5161	0.0026	0.0175	0.0000	0.0000
MS-AR	1.2160	0.4624	1.9108	0.6616	0.2500	0.0780	0.5184	0.0076	0.0329	0.1608	0.0000
XGB-Gauss	1.3023	0.4636	1.9885	0.7344	0.2502	0.0780	0.5099	0.0125	0.0336	0.1799	0.0000
Chronos-Bolt-small	1.3356	0.4595	2.0630	0.7701	0.2516	0.0020	0.5105	0.0356	0.0800	0.0000	0.0000
Chronos-Bolt-base	1.3016	0.4573	2.0012	0.7388	0.2503	0.0060	0.5217	0.0352	0.0708	0.0000	0.0000
TimesFM-2.5	1.2668	0.4602	1.9118	0.6875	0.2531	0.0390	0.4944	0.0257	0.0629	0.0000	0.0000
Deep-SMoE	1.2507	0.4728	2.0143	0.7658	0.2505	0.0390	0.5168	0.0053	0.0208	0.0040	0.0000
SMoE	1.2168	0.4641	1.9782	0.7452	0.2501	0.0780	0.5072	0.0043	0.0247	0.0004	0.0000

Table 47: Out-of-sample results: SHFE copper futures ($n = 3033$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.5387	0.5893	1.9274	0.9991	0.2503	0.0040	0.5124	0.0165	0.0419	0.0010	0.2716
GARCH-t	1.4127	0.5764	1.7654	**0.8592**	0.2502	0.0460	0.5051	0.0082	0.0439	0.3158	0.0003
AR-GJR-t	**1.4091**	**0.5749**	**1.7630**	0.8627	0.2492	1.0000	0.5325	0.0073	0.0439	0.1100	0.0000
Gauss-HAR	1.5069	0.5983	1.8611	0.9344	0.2507	0.0020	0.5058	0.0066	0.0227	0.0445	0.0000
MAR	1.4298	0.5828	1.8169	0.9156	0.2500	0.0180	0.5054	0.0020	0.0185	0.0000	0.0000
LSTAR	1.4391	0.5863	1.8342	0.9279	0.2501	0.0180	0.5064	0.0016	0.0221	0.0000	0.0000
MS-AR	1.4579	0.5848	1.8201	0.8818	0.2494	0.0090	0.5223	0.0069	0.0359	0.0715	0.0000
XGB-Gauss	1.4677	0.5826	1.7852	0.8828	0.2508	0.0180	0.5061	0.0135	0.0386	0.0646	0.0053
Chronos-Bolt-small	1.6114	0.5831	2.0784	1.0881	0.2523	0.0020	0.4988	0.0369	0.0798	0.0000	0.0000
Chronos-Bolt-base	1.5572	0.5805	1.9676	1.0020	0.2514	0.0090	0.4906	0.0310	0.0712	0.0000	0.0000
TimesFM-2.5	1.5440	0.5895	1.9188	1.0047	0.2562	0.0050	0.4810	0.0267	0.0653	0.0000	0.0000
Deep-SMoE	1.4358	0.5850	1.8059	0.8862	0.2518	0.0180	0.4870	0.0026	0.0237	0.0000	0.0000
SMoE	1.4218	0.5811	1.8026	0.8792	0.2499	0.0460	0.5163	0.0026	0.0267	0.0000	0.0000

Table 48: Out-of-sample results: DCE soybean meal futures ($n = 3032$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.8308	0.7416	2.2110	1.3053	0.2506	0.0000	0.4921	0.0165	0.0363	0.0010	0.0958
GARCH-t	1.6408	0.7243	2.1896	1.2332	0.2501	0.3640	0.5007	0.0102	0.0425	0.9016	0.3011
AR-GJR-t	**1.6336**	0.7221	**2.1458**	**1.2132**	0.2503	1.0000	0.5165	0.0099	0.0412	0.9533	0.0032
Gauss-HAR	1.8224	0.7391	2.2002	1.2890	0.2498	0.0000	0.5102	0.0129	0.0247	0.1292	0.0000
MAR	1.6471	**0.7207**	2.1881	1.2603	0.2498	0.2920	0.5175	0.0096	0.0303	0.8082	0.0000
LSTAR	1.6638	0.7243	2.2153	1.2359	0.2499	0.0060	0.5168	0.0089	0.0369	0.5369	0.0000
MS-AR	1.6674	0.7247	2.1680	1.2412	0.2500	0.0060	0.5122	0.0106	0.0465	0.7612	0.0606
XGB-Gauss	1.8783	0.7212	2.3112	1.3158	0.2504	0.0000	0.5106	0.0175	0.0511	0.0002	0.0000
Chronos-Bolt-small	2.0641	0.7271	2.6815	1.4648	0.2521	0.0000	0.4888	0.0379	0.0834	0.0000	0.0000
Chronos-Bolt-base	1.9815	0.7232	2.5204	1.4002	0.2512	0.0000	0.4980	0.0307	0.0749	0.0000	0.0000
TimesFM-2.5	1.9232	0.7302	2.3857	1.3710	0.2556	0.0000	0.4911	0.0297	0.0752	0.0000	0.0000
Deep-SMoE	1.6716	0.7241	2.1884	1.2570	0.2499	0.0010	0.5181	0.0106	0.0310	0.7612	0.0000
SMoE	1.6457	0.7208	2.2229	1.2591	0.2497	0.3640	0.5214	0.0112	0.0353	0.5100	0.0000

Table 49: Out-of-sample results: WTI crude futures ($n = 3139$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	2.3302	1.3277	1.8845	1.8180	0.2494	0.0030	0.5148	0.0210	0.0529	0.0000	0.0000
GARCH-t	2.1266	1.2609	1.5498	1.6397	0.2497	0.0970	0.5237	0.0153	0.0615	0.0057	0.2029
AR-GJR-t	2.1244	1.2608	1.5320	1.6164	0.2497	0.2830	0.5126	0.0140	0.0561	0.0329	0.1832
Gauss-HAR	2.1405	1.2551	1.5033	1.5901	0.2503	0.0600	0.4951	0.0156	0.0548	0.0035	0.7228
MAR	**2.1144**	1.2535	1.5026	**1.5774**	0.2501	1.0000	0.4960	0.0118	0.0519	0.3277	0.2341
LSTAR	2.1146	1.2549	**1.5005**	1.5875	0.2510	0.9380	0.5021	0.0115	0.0545	0.4191	0.4062
MS-AR	2.2247	1.2853	1.7167	1.7306	0.2504	0.0180	0.5094	0.0134	0.0605	0.0703	0.3736
XGB-Gauss	2.1468	1.2542	1.5142	1.6196	0.2502	0.0970	0.4884	0.0185	0.0624	0.0000	0.0002
Chronos-Bolt-small	2.3247	1.2794	1.8679	1.9808	0.2516	0.0010	0.5151	0.0468	0.1026	0.0000	0.0000
Chronos-Bolt-base	2.2517	1.2708	1.7180	1.8661	0.2511	0.0030	0.5091	0.0395	0.0930	0.0000	0.0000
TimesFM-2.5	2.2767	1.2868	1.7640	1.7840	0.2528	0.0150	0.5193	0.0293	0.0819	0.0000	0.0000
Deep-SMoE	2.1231	1.2522	1.5198	1.5923	0.2501	0.2830	0.5097	0.0105	0.0491	0.7746	0.0057
SMoE	2.1156	**1.2522**	1.5176	1.5991	0.2503	0.9290	0.5078	0.0124	0.0545	0.1884	0.5042

Table 50: Out-of-sample results: COMEX gold futures ($n = 3140$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.3927	0.5190	1.8251	0.8357	0.2499	0.0060	0.5073	0.0175	0.0487	0.0001	0.1982
GARCH-t	1.3064	0.5120	1.7634	**0.8069**	0.2509	0.4100	0.5099	0.0096	0.0525	0.8003	0.0001
AR-GJR-t	**1.3055**	**0.5117**	1.7671	0.8127	0.2508	1.0000	0.5096	0.0108	0.0551	0.6454	0.0003
Gauss-HAR	1.3668	0.5194	1.7696	0.8204	0.2504	0.0020	0.5083	0.0134	0.0443	0.0706	0.0000
MAR	1.3208	0.5134	1.7666	0.8204	0.2508	0.0360	0.5025	0.0057	0.0373	0.0090	0.0000
LSTAR	1.3148	0.5137	1.7638	0.8123	0.2515	0.0620	0.5083	0.0067	0.0401	0.0472	0.0000
MS-AR	1.3501	0.5202	1.7964	0.8204	0.2510	0.0060	0.5045	0.0124	0.0478	0.1890	0.0000
XGB-Gauss	1.3750	0.5132	1.7854	0.8888	0.2509	0.0060	0.5067	0.0293	0.0662	0.0000	0.0000
Chronos-Bolt-small	1.5014	0.5175	2.0335	1.0856	0.2517	0.0000	0.5006	0.0439	0.0898	0.0000	0.0000
Chronos-Bolt-base	1.4640	0.5153	1.9636	1.0306	0.2504	0.0000	0.5134	0.0401	0.0831	0.0000	0.0000
TimesFM-2.5	1.4097	0.5187	1.8382	0.9176	0.2533	0.0020	0.4895	0.0309	0.0713	0.0000	0.0000
Deep-SMoE	1.3194	0.5163	**1.7601**	0.8169	0.2522	0.0360	0.4885	0.0051	0.0411	0.0023	0.0000
SMoE	1.3115	0.5138	1.7603	0.8190	0.2514	0.1680	0.5064	0.0080	0.0417	0.2340	0.0000

Table 51: Out-of-sample results: US 10y T-note futures ($n = 3139$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	0.2949	0.1798	1.8470	-0.2916	0.2500	0.0020	0.5164	0.0201	0.0507	0.0000	0.0486
GARCH-t	0.2369	0.1779	1.7784	-0.3385	0.2501	0.8370	0.4922	0.0118	0.0519	0.3277	0.1887
AR-GJR-t	0.2372	0.1780	1.7804	-0.3363	0.2500	0.7660	0.5183	0.0105	0.0507	0.7746	0.3424
Gauss-HAR	0.2486	0.1778	1.7574	-0.3449	0.2500	0.0220	0.5056	0.0185	0.0500	0.0000	0.6497
MAR	0.2342	0.1776	**1.7572**	**-0.3465**	0.2506	0.8370	0.4906	0.0118	0.0507	0.3277	0.1225
LSTAR	**0.2313**	**0.1774**	1.7585	-0.3445	0.2498	1.0000	0.5151	0.0108	0.0500	0.6441	0.8743
MS-AR	0.2584	0.1789	1.8055	-0.3128	0.2498	0.0040	0.5199	0.0105	0.0465	0.7746	0.0000
XGB-Gauss	0.2562	0.1779	1.7747	-0.3187	0.2503	0.0070	0.5037	0.0226	0.0634	0.0000	0.0000
Chronos-Bolt-small	0.3570	0.1794	1.9738	-0.1457	0.2517	0.0020	0.5043	0.0350	0.0809	0.0000	0.0000
Chronos-Bolt-base	0.3211	0.1788	1.9005	-0.1841	0.2513	0.0030	0.4976	0.0303	0.0723	0.0000	0.0000
TimesFM-2.5	0.2891	0.1795	1.8295	-0.2737	0.2524	0.0020	0.5100	0.0239	0.0577	0.0000	0.0000
Deep-SMoE	0.2353	0.1780	1.7679	-0.3351	0.2502	0.8370	0.5078	0.0102	0.0462	0.9131	0.4282
SMoE	0.2353	0.1777	1.7630	-0.3316	0.2503	0.8370	0.5084	0.0115	0.0503	0.4191	0.7393

Table 52: Out-of-sample results: Bitcoin ($n = 2373$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	2.7138	1.8764	2.1217	2.1861	0.2510	0.0020	0.5074	0.0190	0.0480	0.0001	0.2306
GARCH-t	2.5223	1.8006	2.0842	2.1531	0.2507	0.1360	0.5133	0.0152	0.0615	0.0187	0.0007
AR-GJR-t	2.5232	1.7996	2.1334	2.2063	0.2500	0.1360	0.5179	0.0156	0.0657	0.0114	0.0000
Gauss-HAR	2.6471	1.8377	1.9943	2.1006	0.2516	0.0020	0.4905	0.0169	0.0396	0.0022	0.0001
MAR	2.5103	**1.7949**	1.9906	**2.0714**	0.2518	0.6450	0.4749	0.0093	0.0459	0.7178	0.0078
LSTAR	2.5142	1.7972	1.9908	2.0747	0.2518	0.6450	0.4716	0.0080	0.0451	0.3120	0.0007
MS-AR	2.5740	1.8323	2.0713	2.1632	0.2503	0.0070	0.5145	0.0097	0.0489	0.8797	0.0058
XGB-Gauss	2.8448	1.8065	2.3754	2.3735	0.2515	0.0020	0.5053	0.0442	0.0784	0.0000	0.0000
Chronos-Bolt-small	2.9816	1.8231	2.6688	2.4024	0.2524	0.0020	0.4964	0.0426	0.0864	0.0000	0.0000
Chronos-Bolt-base	2.9539	1.8228	2.6021	2.4062	0.2511	0.0020	0.5086	0.0400	0.0822	0.0000	0.0000
TimesFM-2.5	2.8320	1.8412	2.3480	2.2736	0.2562	0.0070	0.4876	0.0388	0.0780	0.0000	0.0000
Deep-SMoE	2.5144	1.7999	**1.9886**	2.0865	0.2508	0.6450	0.5027	0.0139	0.0523	0.0709	0.0380
SMoE	**2.5052**	1.7954	2.0247	2.1128	0.2506	1.0000	0.4935	0.0105	0.0514	0.7951	0.3948

Table 53: Out-of-sample results: CSI 300 ETF (510300) ($n = 2062$)

Model	NLL	CRPS	QLIKE	FZ5	Brier	MCS p (NLL)	Hit	Viol 1%	Viol 5%	Kupiec 1% p	Berkowitz p
HistGauss	1.5962	0.6381	1.9110	1.0724	0.2495	0.0000	0.5257	0.0184	0.0407	0.0006	0.7073
GARCH-t	1.4695	0.6127	1.7564	0.9246	0.2506	0.6270	0.4942	0.0073	0.0500	0.1911	0.2139
AR-GJR-t	**1.4689**	**0.6125**	1.7574	0.9237	0.2505	1.0000	0.4922	0.0082	0.0485	0.4086	0.1714
Gauss-HAR	1.5040	0.6176	**1.7319**	0.9156	0.2507	0.0200	0.4932	0.0145	0.0461	0.0519	0.0641
MAR	1.4733	0.6143	1.7324	**0.9010**	0.2513	0.6270	0.4845	0.0078	0.0490	0.2872	0.0551
LSTAR	1.4740	0.6126	1.7333	0.9119	0.2501	0.6270	0.5024	0.0102	0.0480	0.9332	0.1588
MS-AR	1.5172	0.6246	1.8305	0.9704	0.2512	0.0200	0.4922	0.0082	0.0417	0.4086	0.0000
XGB-Gauss	1.5618	0.6157	1.8520	1.0223	0.2513	0.0200	0.4903	0.0330	0.0761	0.0000	0.0000
Chronos-Bolt-small	1.6354	0.6205	1.9926	1.1605	0.2521	0.0010	0.5087	0.0393	0.0761	0.0000	0.0000
Chronos-Bolt-base	1.5917	0.6183	1.9055	1.0927	0.2513	0.0010	0.5179	0.0369	0.0718	0.0000	0.0000
TimesFM-2.5	1.5505	0.6235	1.8135	1.0073	0.2543	0.0010	0.4952	0.0252	0.0587	0.0000	0.0000
Deep-SMoE	1.4882	0.6176	1.7918	0.9392	0.2516	0.0290	0.4758	0.0097	0.0441	0.8903	0.0004
SMoE	1.4796	0.6161	1.7351	0.9171	0.2515	0.0570	0.4855	0.0102	0.0490	0.9332	0.0636

H SMoE parameter estimates for every series

Full-sample (development period) SMoE estimates with sandwich and model-based standard errors. Parameter names follow definition 1: `beta[k]` are expert-mean coefficients, `delta[k]` expert log-variance coefficients on the log-HAR range variances, and `gamma[k]` gate log-odds relative to the crisis expert. Covariates are standardised, so slopes are per standard deviation.

Table 54: SMoE estimates: CSI 300 (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
`beta[trend]:const`	-0.000	0.029	0.029	-0.002	0.998
`beta[trend]:mom20`	0.011	0.025	0.024	0.426	0.670
`beta[trend]:mom120`	0.106	0.028	0.027	3.809	0.000
`beta[reversal]:const`	-0.728	0.262	0.183	-2.782	0.005
`beta[reversal]:rev1`	0.278	0.126	0.120	2.208	0.027
`beta[reversal]:rev5`	0.184	0.166	0.125	1.104	0.269
`beta[crisis]:const`	1.773	0.272	0.229	6.516	0.000
`delta[trend]:const`	-0.219	0.120	0.093	-1.833	0.067
`delta[trend]:lv1`	0.054	0.070	0.064	0.769	0.442
`delta[trend]:lv5`	0.012	0.115	0.104	0.107	0.915
`delta[trend]:lv22`	0.376	0.115	0.108	3.270	0.001
`delta[reversal]:const`	1.372	0.137	0.097	10.024	0.000
`delta[reversal]:lv1`	0.226	0.087	0.088	2.591	0.010
`delta[reversal]:lv5`	0.245	0.124	0.117	1.979	0.048
`delta[reversal]:lv22`	0.159	0.116	0.102	1.372	0.170
`delta[crisis]:const`	-0.327	0.290	0.263	-1.125	0.261
`delta[crisis]:lv1`	0.422	0.187	0.209	2.254	0.024
`delta[crisis]:lv5`	0.065	0.235	0.272	0.274	0.784
`delta[crisis]:lv22`	0.331	0.206	0.225	1.603	0.109
`gamma[trend]:const`	1.970	0.407	0.339	4.845	0.000
`gamma[trend]:g_lvol`	-1.033	0.162	0.149	-6.385	0.000
`gamma[trend]:g_mom60`	-0.216	0.146	0.124	-1.481	0.139
`gamma[trend]:g_dd`	0.154	0.141	0.129	1.091	0.275
`gamma[reversal]:const`	0.725	0.481	0.362	1.505	0.132
`gamma[reversal]:g_lvol`	-0.053	0.172	0.151	-0.311	0.756
`gamma[reversal]:g_mom60`	-0.112	0.203	0.157	-0.549	0.583
`gamma[reversal]:g_dd`	-0.049	0.156	0.125	-0.316	0.752

Table 55: SMoE estimates: CSI 500 (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.706	0.110	0.088	6.414	0.000
beta[trend]:mom20	0.071	0.090	0.058	0.788	0.431
beta[trend]:mom120	0.098	0.056	0.048	1.753	0.080
beta[reversal]:const	-1.506	0.188	0.190	-8.004	0.000
beta[reversal]:rev1	0.504	0.171	0.181	2.942	0.003
beta[reversal]:rev5	0.407	0.153	0.153	2.655	0.008
beta[crisis]:const	-0.113	0.198	0.104	-0.569	0.569
delta[trend]:const	-0.160	0.261	0.154	-0.612	0.541
delta[trend]:lv1	0.022	0.130	0.114	0.167	0.868
delta[trend]:lv5	0.320	0.174	0.148	1.836	0.066
delta[trend]:lv22	0.475	0.115	0.118	4.136	0.000
delta[reversal]:const	1.549	0.124	0.099	12.454	0.000
delta[reversal]:lv1	0.445	0.147	0.144	3.034	0.002
delta[reversal]:lv5	0.169	0.168	0.178	1.006	0.314
delta[reversal]:lv22	0.257	0.150	0.145	1.715	0.086
delta[crisis]:const	0.636	0.075	0.071	8.489	0.000
delta[crisis]:lv1	0.229	0.129	0.085	1.777	0.076
delta[crisis]:lv5	0.368	0.096	0.101	3.842	0.000
delta[crisis]:lv22	0.508	0.093	0.092	5.445	0.000
gamma[trend]:const	-0.433	0.956	0.510	-0.453	0.650
gamma[trend]:g_lvol	0.554	0.188	0.165	2.946	0.003
gamma[trend]:g_mom60	0.410	0.192	0.142	2.128	0.033
gamma[trend]:g_dd	0.162	0.174	0.170	0.934	0.350
gamma[reversal]:const	-1.454	0.488	0.314	-2.980	0.003
gamma[reversal]:g_lvol	0.396	0.329	0.243	1.203	0.229
gamma[reversal]:g_mom60	0.559	0.161	0.155	3.480	0.001
gamma[reversal]:g_dd	-0.006	0.242	0.213	-0.026	0.980

Table 56: SMoE estimates: SSE 50 (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.153	0.357	0.104	-0.428	0.669
beta[trend]:mom20	0.449	1.148	0.197	0.391	0.696
beta[trend]:mom120	0.290	0.752	0.134	0.385	0.700
beta[reversal]:const	0.012	0.118	0.031	0.098	0.922
beta[reversal]:rev1	-0.004	0.040	0.024	-0.097	0.923
beta[reversal]:rev5	-0.068	0.060	0.028	-1.135	0.256
beta[crisis]:const	0.059	0.098	0.053	0.609	0.543
delta[trend]:const	-0.043	6.428	0.901	-0.007	0.995
delta[trend]:lv1	0.834	1.976	0.393	0.422	0.673
delta[trend]:lv5	1.273	5.975	0.893	0.213	0.831
delta[trend]:lv22	-1.340	9.569	1.336	-0.140	0.889
delta[reversal]:const	-0.417	0.267	0.150	-1.562	0.118
delta[reversal]:lv1	-0.197	0.252	0.108	-0.783	0.434
delta[reversal]:lv5	-0.081	1.102	0.241	-0.073	0.941
delta[reversal]:lv22	0.874	0.575	0.179	1.521	0.128
delta[crisis]:const	1.149	0.105	0.075	10.899	0.000
delta[crisis]:lv1	0.112	0.195	0.073	0.571	0.568
delta[crisis]:lv5	0.059	0.527	0.116	0.111	0.911
delta[crisis]:lv22	0.546	1.118	0.182	0.488	0.625
gamma[trend]:const	-1.262	1.087	0.356	-1.162	0.245
gamma[trend]:g_lvol	0.117	4.280	0.621	0.027	0.978
gamma[trend]:g_mom60	-0.036	1.280	0.278	-0.028	0.978
gamma[trend]:g_dd	0.564	2.331	0.533	0.242	0.809
gamma[reversal]:const	0.026	1.489	0.306	0.017	0.986
gamma[reversal]:g_lvol	-0.372	2.439	0.401	-0.152	0.879
gamma[reversal]:g_mom60	-0.145	0.110	0.099	-1.315	0.189
gamma[reversal]:g_dd	0.203	0.693	0.155	0.293	0.769

Table 57: SMoE estimates: ChiNext (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.016	0.091	0.049	-0.177	0.860
beta[trend]:mom20	0.073	0.034	0.034	2.165	0.030
beta[trend]:mom120	0.066	0.054	0.038	1.207	0.228
beta[reversal]:const	-2.404	1.345	0.661	-1.788	0.074
beta[reversal]:rev1	0.516	0.271	0.267	1.906	0.057
beta[reversal]:rev5	0.585	0.418	0.253	1.400	0.162
beta[crisis]:const	2.865	0.457	0.250	6.265	0.000
delta[trend]:const	0.613	0.145	0.087	4.243	0.000
delta[trend]:lv1	0.057	0.069	0.054	0.828	0.408
delta[trend]:lv5	0.268	0.071	0.073	3.791	0.000
delta[trend]:lv22	0.155	0.139	0.093	1.118	0.263
delta[reversal]:const	1.431	0.235	0.209	6.083	0.000
delta[reversal]:lv1	0.233	0.218	0.196	1.067	0.286
delta[reversal]:lv5	-0.027	0.307	0.239	-0.087	0.931
delta[reversal]:lv22	0.277	0.178	0.198	1.556	0.120
delta[crisis]:const	-0.321	0.338	0.271	-0.951	0.342
delta[crisis]:lv1	0.266	0.197	0.217	1.350	0.177
delta[crisis]:lv5	0.392	0.391	0.328	1.003	0.316
delta[crisis]:lv22	0.206	0.210	0.223	0.981	0.327
gamma[trend]:const	2.763	0.779	0.420	3.546	0.000
gamma[trend]:g_lvol	-1.003	0.237	0.168	-4.236	0.000
gamma[trend]:g_mom60	0.050	0.317	0.194	0.158	0.875
gamma[trend]:g_dd	0.096	0.199	0.156	0.481	0.631
gamma[reversal]:const	0.208	1.194	0.553	0.175	0.861
gamma[reversal]:g_lvol	-0.119	0.306	0.175	-0.388	0.698
gamma[reversal]:g_mom60	0.397	0.347	0.228	1.144	0.252
gamma[reversal]:g_dd	-0.066	0.202	0.177	-0.325	0.745

Table 58: SMoE estimates: S&P 500 (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.609	0.122	0.102	-4.984	0.000
beta[trend]:mom20	0.158	0.060	0.053	2.623	0.009
beta[trend]:mom120	-0.031	0.053	0.050	-0.595	0.552
beta[reversal]:const	0.650	0.037	0.039	17.417	0.000
beta[reversal]:rev1	-0.059	0.021	0.023	-2.873	0.004
beta[reversal]:rev5	-0.166	0.032	0.025	-5.219	0.000
beta[crisis]:const	0.007	0.015	0.016	0.438	0.661
delta[trend]:const	0.132	0.114	0.077	1.156	0.248
delta[trend]:lv1	0.287	0.126	0.104	2.272	0.023
delta[trend]:lv5	0.660	0.177	0.143	3.723	0.000
delta[trend]:lv22	-0.013	0.156	0.115	-0.086	0.931
delta[reversal]:const	-1.004	0.098	0.089	-10.211	0.000
delta[reversal]:lv1	-0.118	0.117	0.112	-1.004	0.315
delta[reversal]:lv5	0.548	0.148	0.145	3.693	0.000
delta[reversal]:lv22	0.776	0.129	0.120	6.007	0.000
delta[crisis]:const	-1.307	0.207	0.148	-6.322	0.000
delta[crisis]:lv1	0.068	0.112	0.104	0.609	0.542
delta[crisis]:lv5	0.827	0.175	0.155	4.716	0.000
delta[crisis]:lv22	0.181	0.208	0.172	0.871	0.384
gamma[trend]:const	-0.050	0.517	0.357	-0.096	0.924
gamma[trend]:g_lvol	0.290	0.226	0.201	1.284	0.199
gamma[trend]:g_mom60	0.153	0.113	0.114	1.346	0.178
gamma[trend]:g_dd	-1.590	0.799	0.608	-1.991	0.046
gamma[reversal]:const	0.078	0.413	0.309	0.188	0.851
gamma[reversal]:g_lvol	0.307	0.209	0.185	1.467	0.142
gamma[reversal]:g_mom60	0.127	0.098	0.107	1.290	0.197
gamma[reversal]:g_dd	-1.619	0.804	0.610	-2.014	0.044

Table 59: SMoE estimates: Nasdaq 100 (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.959	0.091	0.093	10.540	0.000
beta[trend]:mom20	-0.234	0.055	0.046	-4.251	0.000
beta[trend]:mom120	0.002	0.060	0.051	0.041	0.967
beta[reversal]:const	-0.517	0.215	0.127	-2.405	0.016
beta[reversal]:rev1	0.039	0.056	0.049	0.694	0.488
beta[reversal]:rev5	0.050	0.073	0.054	0.693	0.488
beta[crisis]:const	0.085	0.024	0.024	3.586	0.000
delta[trend]:const	-0.530	0.167	0.137	-3.178	0.001
delta[trend]:lv1	0.269	0.186	0.166	1.443	0.149
delta[trend]:lv5	0.110	0.209	0.195	0.524	0.600
delta[trend]:lv22	0.763	0.199	0.154	3.827	0.000
delta[reversal]:const	0.527	0.106	0.066	4.955	0.000
delta[reversal]:lv1	0.211	0.094	0.074	2.250	0.024
delta[reversal]:lv5	0.563	0.123	0.111	4.561	0.000
delta[reversal]:lv22	-0.025	0.107	0.091	-0.235	0.814
delta[crisis]:const	-0.936	0.266	0.186	-3.525	0.000
delta[crisis]:lv1	0.188	0.107	0.101	1.754	0.079
delta[crisis]:lv5	0.770	0.177	0.155	4.343	0.000
delta[crisis]:lv22	-0.094	0.261	0.206	-0.362	0.717
gamma[trend]:const	-0.308	0.466	0.367	-0.661	0.508
gamma[trend]:g_lvol	0.361	0.247	0.218	1.465	0.143
gamma[trend]:g_mom60	0.068	0.136	0.137	0.502	0.615
gamma[trend]:g_dd	-1.121	0.680	0.508	-1.647	0.099
gamma[reversal]:const	0.196	0.574	0.368	0.342	0.733
gamma[reversal]:g_lvol	0.509	0.216	0.189	2.355	0.019
gamma[reversal]:g_mom60	0.118	0.119	0.114	0.991	0.322
gamma[reversal]:g_dd	-0.991	0.682	0.498	-1.452	0.147

Table 60: SMoE estimates: Hang Seng (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.107	0.170	0.082	0.629	0.529
beta[trend]:mom20	0.050	0.114	0.062	0.436	0.663
beta[trend]:mom120	0.029	0.056	0.056	0.513	0.608
beta[reversal]:const	-1.291	0.122	0.118	-10.588	0.000
beta[reversal]:rev1	0.350	0.099	0.089	3.518	0.000
beta[reversal]:rev5	0.326	0.106	0.092	3.093	0.002
beta[crisis]:const	0.164	0.042	0.028	3.914	0.000
delta[trend]:const	0.903	0.445	0.146	2.031	0.042
delta[trend]:lv1	-0.190	0.137	0.101	-1.391	0.164
delta[trend]:lv5	0.069	0.219	0.141	0.317	0.751
delta[trend]:lv22	0.875	0.262	0.149	3.345	0.001
delta[reversal]:const	0.056	0.117	0.134	0.476	0.634
delta[reversal]:lv1	0.746	0.146	0.169	5.115	0.000
delta[reversal]:lv5	0.185	0.265	0.238	0.699	0.484
delta[reversal]:lv22	0.099	0.204	0.181	0.484	0.628
delta[crisis]:const	-0.129	0.280	0.101	-0.462	0.644
delta[crisis]:lv1	0.025	0.164	0.074	0.155	0.877
delta[crisis]:lv5	0.543	0.382	0.137	1.420	0.156
delta[crisis]:lv22	0.376	0.262	0.106	1.436	0.151
gamma[trend]:const	-1.232	1.810	0.605	-0.681	0.496
gamma[trend]:g_lvol	-1.852	0.912	0.442	-2.030	0.042
gamma[trend]:g_mom60	-0.416	0.712	0.274	-0.584	0.559
gamma[trend]:g_dd	-1.517	0.799	0.374	-1.899	0.058
gamma[reversal]:const	-1.766	0.653	0.282	-2.703	0.007
gamma[reversal]:g_lvol	0.124	0.589	0.257	0.210	0.833
gamma[reversal]:g_mom60	0.151	0.187	0.149	0.807	0.420
gamma[reversal]:g_dd	-0.076	0.696	0.357	-0.109	0.913

Table 61: SMoE estimates: Nikkei 225 (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.925	0.156	0.140	-5.930	0.000
beta[trend]:mom20	0.191	0.069	0.060	2.782	0.005
beta[trend]:mom120	-0.180	0.074	0.062	-2.435	0.015
beta[reversal]:const	0.721	0.207	0.148	3.485	0.000
beta[reversal]:rev1	-0.056	0.041	0.039	-1.368	0.171
beta[reversal]:rev5	-0.131	0.056	0.045	-2.320	0.020
beta[crisis]:const	-0.010	0.034	0.032	-0.290	0.772
delta[trend]:const	0.590	0.102	0.084	5.804	0.000
delta[trend]:lv1	0.193	0.101	0.086	1.901	0.057
delta[trend]:lv5	0.747	0.140	0.135	5.345	0.000
delta[trend]:lv22	-0.118	0.134	0.100	-0.882	0.378
delta[reversal]:const	-0.009	0.117	0.099	-0.074	0.941
delta[reversal]:lv1	0.050	0.076	0.079	0.659	0.510
delta[reversal]:lv5	0.120	0.134	0.112	0.895	0.371
delta[reversal]:lv22	0.387	0.119	0.099	3.262	0.001
delta[crisis]:const	-0.622	0.422	0.219	-1.476	0.140
delta[crisis]:lv1	0.435	0.225	0.142	1.938	0.053
delta[crisis]:lv5	0.553	0.227	0.173	2.435	0.015
delta[crisis]:lv22	-0.158	0.541	0.289	-0.292	0.770
gamma[trend]:const	-0.091	0.791	0.441	-0.116	0.908
gamma[trend]:g_lvol	0.227	0.394	0.239	0.575	0.565
gamma[trend]:g_mom60	0.062	0.142	0.127	0.436	0.663
gamma[trend]:g_dd	-1.246	0.376	0.324	-3.313	0.001
gamma[reversal]:const	0.255	0.909	0.527	0.281	0.779
gamma[reversal]:g_lvol	0.329	0.387	0.233	0.850	0.395
gamma[reversal]:g_mom60	0.157	0.140	0.126	1.126	0.260
gamma[reversal]:g_dd	-1.059	0.342	0.306	-3.102	0.002

Table 62: SMoE estimates: DAX (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.400	0.098	0.079	-4.069	0.000
beta[trend]:mom20	0.125	0.053	0.043	2.380	0.017
beta[trend]:mom120	-0.148	0.050	0.044	-2.951	0.003
beta[reversal]:const	0.682	0.162	0.117	4.209	0.000
beta[reversal]:rev1	0.027	0.053	0.048	0.516	0.606
beta[reversal]:rev5	-0.329	0.099	0.072	-3.311	0.001
beta[crisis]:const	0.092	0.026	0.024	3.614	0.000
delta[trend]:const	0.517	0.086	0.056	5.995	0.000
delta[trend]:lv1	0.271	0.107	0.069	2.535	0.011
delta[trend]:lv5	0.419	0.134	0.089	3.132	0.002
delta[trend]:lv22	0.151	0.108	0.071	1.395	0.163
delta[reversal]:const	-0.499	0.248	0.160	-2.006	0.045
delta[reversal]:lv1	0.026	0.179	0.147	0.144	0.885
delta[reversal]:lv5	0.285	0.288	0.210	0.988	0.323
delta[reversal]:lv22	0.224	0.300	0.203	0.745	0.456
delta[crisis]:const	-1.073	0.255	0.192	-4.208	0.000
delta[crisis]:lv1	0.055	0.122	0.119	0.454	0.650
delta[crisis]:lv5	0.784	0.175	0.168	4.478	0.000
delta[crisis]:lv22	0.093	0.180	0.185	0.514	0.607
gamma[trend]:const	0.468	0.453	0.324	1.032	0.302
gamma[trend]:g_lvol	0.082	0.210	0.197	0.391	0.696
gamma[trend]:g_mom60	-0.035	0.127	0.115	-0.279	0.781
gamma[trend]:g_dd	-1.017	0.509	0.381	-1.998	0.046
gamma[reversal]:const	-0.067	0.708	0.480	-0.095	0.924
gamma[reversal]:g_lvol	0.354	0.224	0.218	1.582	0.114
gamma[reversal]:g_mom60	0.162	0.155	0.140	1.046	0.296
gamma[reversal]:g_dd	-0.740	0.598	0.421	-1.238	0.216

Table 63: SMoE estimates: FTSE 100 (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.030	0.042	0.035	0.714	0.475
beta[trend]:mom20	-0.034	0.029	0.022	-1.179	0.238
beta[trend]:mom120	-0.035	0.029	0.023	-1.211	0.226
beta[reversal]:const	0.385	0.101	0.062	3.809	0.000
beta[reversal]:rev1	-0.129	0.045	0.045	-2.831	0.005
beta[reversal]:rev5	-0.184	0.088	0.058	-2.079	0.038
beta[crisis]:const	-0.243	0.077	0.064	-3.153	0.002
delta[trend]:const	-0.616	0.147	0.106	-4.199	0.000
delta[trend]:lv1	0.082	0.110	0.085	0.749	0.454
delta[trend]:lv5	0.380	0.124	0.118	3.072	0.002
delta[trend]:lv22	0.447	0.097	0.099	4.622	0.000
delta[reversal]:const	-1.511	0.225	0.186	-6.718	0.000
delta[reversal]:lv1	0.276	0.420	0.254	0.658	0.511
delta[reversal]:lv5	0.357	0.290	0.261	1.232	0.218
delta[reversal]:lv22	-0.056	0.230	0.214	-0.245	0.807
delta[crisis]:const	0.433	0.103	0.087	4.225	0.000
delta[crisis]:lv1	0.078	0.096	0.081	0.812	0.417
delta[crisis]:lv5	0.679	0.151	0.121	4.495	0.000
delta[crisis]:lv22	0.015	0.110	0.105	0.136	0.892
gamma[trend]:const	0.750	0.412	0.332	1.822	0.068
gamma[trend]:g_lvol	0.179	0.285	0.251	0.629	0.529
gamma[trend]:g_mom60	0.335	0.128	0.121	2.607	0.009
gamma[trend]:g_dd	0.178	0.210	0.199	0.848	0.397
gamma[reversal]:const	-0.918	0.417	0.394	-2.201	0.028
gamma[reversal]:g_lvol	1.034	0.337	0.275	3.066	0.002
gamma[reversal]:g_mom60	-0.040	0.263	0.204	-0.151	0.880
gamma[reversal]:g_dd	2.411	0.661	0.581	3.646	0.000

Table 64: SMoE estimates: CSI 300 futures (IF) (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.274	0.194	0.148	1.412	0.158
beta[trend]:mom20	-0.026	0.090	0.079	-0.293	0.769
beta[trend]:mom120	0.013	0.087	0.069	0.146	0.884
beta[reversal]:const	-0.908	0.264	0.218	-3.446	0.001
beta[reversal]:rev1	0.530	0.217	0.230	2.443	0.015
beta[reversal]:rev5	0.204	0.260	0.209	0.784	0.433
beta[crisis]:const	-0.054	0.091	0.060	-0.592	0.554
delta[trend]:const	0.628	0.233	0.157	2.695	0.007
delta[trend]:lv1	0.096	0.204	0.146	0.472	0.637
delta[trend]:lv5	0.152	0.345	0.208	0.441	0.659
delta[trend]:lv22	0.462	0.420	0.211	1.098	0.272
delta[reversal]:const	0.529	0.679	0.363	0.779	0.436
delta[reversal]:lv1	0.866	0.293	0.292	2.952	0.003
delta[reversal]:lv5	0.719	0.716	0.423	1.004	0.315
delta[reversal]:lv22	-0.324	1.446	0.697	-0.224	0.823
delta[crisis]:const	-0.534	0.252	0.177	-2.123	0.034
delta[crisis]:lv1	-0.077	0.225	0.139	-0.342	0.732
delta[crisis]:lv5	0.279	0.297	0.181	0.939	0.348
delta[crisis]:lv22	0.273	0.253	0.166	1.078	0.281
gamma[trend]:const	-0.413	0.755	0.525	-0.547	0.584
gamma[trend]:g_lvol	-0.231	0.835	0.419	-0.277	0.782
gamma[trend]:g_mom60	0.230	0.299	0.229	0.768	0.442
gamma[trend]:g_dd	-0.216	0.182	0.198	-1.184	0.237
gamma[reversal]:const	-1.919	0.755	0.490	-2.542	0.011
gamma[reversal]:g_lvol	-0.128	0.609	0.377	-0.210	0.834
gamma[reversal]:g_mom60	-0.276	0.283	0.287	-0.973	0.331
gamma[reversal]:g_dd	0.289	0.317	0.343	0.913	0.361

Table 65: SMoE estimates: SHFE gold futures (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.023	0.016	0.017	1.439	0.150
beta[trend]:mom20	-0.121	0.034	0.025	-3.519	0.000
beta[trend]:mom120	0.067	0.022	0.021	3.003	0.003
beta[reversal]:const	-0.042	0.188	0.131	-0.225	0.822
beta[reversal]:rev1	-0.028	0.126	0.103	-0.219	0.827
beta[reversal]:rev5	0.193	0.161	0.104	1.201	0.230
beta[crisis]:const	0.042	0.040	0.035	1.039	0.299
delta[trend]:const	-1.423	0.153	0.100	-9.285	0.000
delta[trend]:lv1	0.292	0.089	0.081	3.264	0.001
delta[trend]:lv5	-0.023	0.135	0.103	-0.173	0.863
delta[trend]:lv22	0.413	0.143	0.108	2.888	0.004
delta[reversal]:const	1.534	0.248	0.228	6.193	0.000
delta[reversal]:lv1	-0.345	0.144	0.134	-2.398	0.016
delta[reversal]:lv5	0.417	0.267	0.282	1.563	0.118
delta[reversal]:lv22	0.417	0.248	0.253	1.686	0.092
delta[crisis]:const	0.139	0.281	0.130	0.496	0.620
delta[crisis]:lv1	0.042	0.051	0.057	0.817	0.414
delta[crisis]:lv5	0.197	0.094	0.087	2.099	0.036
delta[crisis]:lv22	0.553	0.166	0.105	3.342	0.001
gamma[trend]:const	0.086	0.336	0.217	0.256	0.798
gamma[trend]:g_lvol	0.464	0.175	0.158	2.653	0.008
gamma[trend]:g_mom60	-0.723	0.242	0.144	-2.988	0.003
gamma[trend]:g_dd	1.145	0.260	0.185	4.410	0.000
gamma[reversal]:const	-2.334	0.750	0.477	-3.113	0.002
gamma[reversal]:g_lvol	-1.036	0.547	0.355	-1.894	0.058
gamma[reversal]:g_mom60	0.247	0.557	0.219	0.444	0.657
gamma[reversal]:g_dd	-0.207	0.176	0.187	-1.172	0.241

Table 66: SMoE estimates: SHFE copper futures (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.337	0.188	0.149	-1.798	0.072
beta[trend]:mom20	1.326	0.449	0.176	2.952	0.003
beta[trend]:mom120	0.220	0.325	0.163	0.676	0.499
beta[reversal]:const	0.074	0.064	0.031	1.159	0.246
beta[reversal]:rev1	-0.226	0.118	0.046	-1.914	0.056
beta[reversal]:rev5	-0.162	0.149	0.051	-1.088	0.276
beta[crisis]:const	0.005	0.037	0.031	0.142	0.887
delta[trend]:const	0.946	0.471	0.178	2.007	0.045
delta[trend]:lv1	-0.048	0.460	0.201	-0.105	0.916
delta[trend]:lv5	-0.185	0.737	0.322	-0.251	0.802
delta[trend]:lv22	0.169	0.193	0.183	0.876	0.381
delta[reversal]:const	-0.721	0.409	0.144	-1.763	0.078
delta[reversal]:lv1	0.124	0.179	0.094	0.692	0.489
delta[reversal]:lv5	0.328	0.178	0.126	1.847	0.065
delta[reversal]:lv22	0.500	0.493	0.188	1.014	0.311
delta[crisis]:const	0.608	0.301	0.093	2.019	0.043
delta[crisis]:lv1	-0.002	0.112	0.058	-0.020	0.984
delta[crisis]:lv5	0.409	0.108	0.092	3.801	0.000
delta[crisis]:lv22	0.581	0.231	0.096	2.517	0.012
gamma[trend]:const	-1.544	0.436	0.223	-3.540	0.000
gamma[trend]:g_lvol	0.271	0.229	0.170	1.182	0.237
gamma[trend]:g_mom60	-0.232	0.851	0.247	-0.273	0.785
gamma[trend]:g_dd	0.177	0.649	0.227	0.273	0.785
gamma[reversal]:const	-0.282	0.768	0.268	-0.367	0.713
gamma[reversal]:g_lvol	0.776	0.960	0.299	0.808	0.419
gamma[reversal]:g_mom60	-0.742	0.215	0.135	-3.454	0.001
gamma[reversal]:g_dd	1.194	0.313	0.223	3.815	0.000

Table 67: SMoE estimates: DCE soybean meal futures (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.018	0.043	0.038	-0.427	0.669
beta[trend]:mom20	0.090	0.081	0.045	1.115	0.265
beta[trend]:mom120	-0.054	0.055	0.036	-0.979	0.328
beta[reversal]:const	0.045	0.029	0.029	1.560	0.119
beta[reversal]:rev1	-0.086	0.036	0.030	-2.360	0.018
beta[reversal]:rev5	-0.098	0.066	0.039	-1.483	0.138
beta[crisis]:const	0.184	0.231	0.243	0.795	0.427
delta[trend]:const	0.726	0.247	0.121	2.941	0.003
delta[trend]:lv1	0.088	0.061	0.053	1.442	0.149
delta[trend]:lv5	0.154	0.074	0.074	2.072	0.038
delta[trend]:lv22	0.386	0.130	0.093	2.971	0.003
delta[reversal]:const	-0.757	0.474	0.224	-1.599	0.110
delta[reversal]:lv1	0.001	0.081	0.080	0.012	0.990
delta[reversal]:lv5	-0.029	0.141	0.124	-0.208	0.835
delta[reversal]:lv22	0.663	0.346	0.187	1.917	0.055
delta[crisis]:const	3.557	0.360	0.252	9.889	0.000
delta[crisis]:lv1	-0.002	0.196	0.162	-0.009	0.992
delta[crisis]:lv5	0.438	0.399	0.349	1.098	0.272
delta[crisis]:lv22	0.825	0.510	0.382	1.619	0.106
gamma[trend]:const	3.161	0.244	0.225	12.981	0.000
gamma[trend]:g_lvol	1.051	0.438	0.284	2.398	0.016
gamma[trend]:g_mom60	-0.082	0.325	0.194	-0.253	0.801
gamma[trend]:g_dd	0.053	0.329	0.213	0.161	0.872
gamma[reversal]:const	2.965	0.749	0.392	3.958	0.000
gamma[reversal]:g_lvol	1.325	0.462	0.300	2.864	0.004
gamma[reversal]:g_mom60	-0.266	0.283	0.173	-0.941	0.347
gamma[reversal]:g_dd	0.236	0.307	0.200	0.771	0.441

Table 68: SMoE estimates: WTI crude futures (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-2.367	1.423	0.072	-1.663	0.096
beta[trend]:mom20	2.035	0.508	0.219	4.007	0.000
beta[trend]:mom120	-0.755	2.767	0.000	-0.273	0.785
beta[reversal]:const	0.145	0.074	0.051	1.956	0.050
beta[reversal]:rev1	-0.008	0.037	0.035	-0.227	0.820
beta[reversal]:rev5	-0.029	0.036	0.034	-0.803	0.422
beta[crisis]:const	-0.624	0.853	0.183	-0.731	0.465
delta[trend]:const	2.123	5.688	0.000	0.373	0.709
delta[trend]:lv1	-1.654	4.971	0.000	-0.333	0.739
delta[trend]:lv5	0.737	5.016	0.206	0.147	0.883
delta[trend]:lv22	-0.339	1.334	0.726	-0.254	0.799
delta[reversal]:const	1.189	0.077	0.051	15.539	0.000
delta[reversal]:lv1	-0.020	0.072	0.056	-0.283	0.777
delta[reversal]:lv5	0.429	0.123	0.086	3.486	0.000
delta[reversal]:lv22	0.354	0.085	0.067	4.176	0.000
delta[crisis]:const	2.052	0.258	0.128	7.959	0.000
delta[crisis]:lv1	0.585	0.230	0.148	2.544	0.011
delta[crisis]:lv5	-0.160	0.477	0.228	-0.336	0.737
delta[crisis]:lv22	0.651	0.451	0.188	1.442	0.149
gamma[trend]:const	-2.094	4.258	0.000	-0.492	0.623
gamma[trend]:g_lvol	0.866	0.549	0.343	1.577	0.115
gamma[trend]:g_mom60	0.565	3.999	0.000	0.141	0.888
gamma[trend]:g_dd	0.306	0.554	0.351	0.552	0.581
gamma[reversal]:const	1.938	0.970	0.476	1.998	0.046
gamma[reversal]:g_lvol	0.525	0.494	0.222	1.063	0.288
gamma[reversal]:g_mom60	0.511	0.902	0.040	0.566	0.571
gamma[reversal]:g_dd	0.584	0.261	0.198	2.236	0.025

Table 69: SMoE estimates: COMEX gold futures (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.098	0.017	0.017	5.677	0.000
beta[trend]:mom20	-0.019	0.016	0.017	-1.171	0.242
beta[trend]:mom120	0.085	0.019	0.019	4.450	0.000
beta[reversal]:const	-1.638	0.595	0.106	-2.752	0.006
beta[reversal]:rev1	1.095	0.456	0.084	2.399	0.016
beta[reversal]:rev5	-0.169	0.230	0.041	-0.732	0.464
beta[crisis]:const	0.029	0.057	0.040	0.507	0.612
delta[trend]:const	-0.902	0.172	0.087	-5.253	0.000
delta[trend]:lv1	-0.063	0.069	0.057	-0.919	0.358
delta[trend]:lv5	0.055	0.107	0.081	0.511	0.610
delta[trend]:lv22	0.775	0.135	0.081	5.734	0.000
delta[reversal]:const	0.165	3.782	0.712	0.044	0.965
delta[reversal]:lv1	1.880	2.016	0.442	0.933	0.351
delta[reversal]:lv5	-0.297	7.587	1.361	-0.039	0.969
delta[reversal]:lv22	-0.325	2.967	0.595	-0.110	0.913
delta[crisis]:const	0.621	0.366	0.113	1.698	0.089
delta[crisis]:lv1	-0.103	0.085	0.046	-1.207	0.227
delta[crisis]:lv5	0.131	0.130	0.073	1.006	0.314
delta[crisis]:lv22	0.513	0.320	0.100	1.604	0.109
gamma[trend]:const	0.611	0.695	0.251	0.880	0.379
gamma[trend]:g_lvol	0.662	0.256	0.171	2.590	0.010
gamma[trend]:g_mom60	-0.296	0.132	0.095	-2.249	0.025
gamma[trend]:g_dd	0.439	0.116	0.113	3.791	0.000
gamma[reversal]:const	-2.780	1.109	0.439	-2.508	0.012
gamma[reversal]:g_lvol	1.370	0.808	0.296	1.696	0.090
gamma[reversal]:g_mom60	0.518	0.554	0.323	0.934	0.350
gamma[reversal]:g_dd	0.395	0.452	0.316	0.875	0.382

Table 70: SMoE estimates: US 10y T-note futures (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.083	0.078	0.042	-1.058	0.290
beta[trend]:mom20	0.022	0.024	0.018	0.931	0.352
beta[trend]:mom120	-0.017	0.052	0.025	-0.333	0.739
beta[reversal]:const	0.034	0.028	0.014	1.217	0.224
beta[reversal]:rev1	-0.015	0.011	0.008	-1.287	0.198
beta[reversal]:rev5	-0.012	0.008	0.007	-1.569	0.117
beta[crisis]:const	-0.025	0.060	0.036	-0.406	0.685
delta[trend]:const	-1.915	0.684	0.267	-2.800	0.005
delta[trend]:lv1	-0.594	0.267	0.194	-2.222	0.026
delta[trend]:lv5	0.038	0.395	0.238	0.097	0.923
delta[trend]:lv22	1.379	0.455	0.255	3.031	0.002
delta[reversal]:const	-2.580	0.126	0.074	-20.516	0.000
delta[reversal]:lv1	0.070	0.225	0.109	0.312	0.755
delta[reversal]:lv5	0.405	0.148	0.107	2.730	0.006
delta[reversal]:lv22	0.440	0.258	0.129	1.708	0.088
delta[crisis]:const	-1.752	0.161	0.130	-10.876	0.000
delta[crisis]:lv1	0.494	0.217	0.166	2.279	0.023
delta[crisis]:lv5	-0.231	0.274	0.237	-0.844	0.399
delta[crisis]:lv22	0.023	0.326	0.217	0.072	0.943
gamma[trend]:const	-0.145	1.430	0.701	-0.102	0.919
gamma[trend]:g_lvol	-0.776	0.711	0.369	-1.091	0.275
gamma[trend]:g_mom60	-0.150	0.410	0.283	-0.366	0.715
gamma[trend]:g_dd	0.074	0.514	0.327	0.144	0.886
gamma[reversal]:const	1.254	0.525	0.418	2.390	0.017
gamma[reversal]:g_lvol	-0.084	0.538	0.289	-0.156	0.876
gamma[reversal]:g_mom60	-0.081	0.295	0.203	-0.274	0.784
gamma[reversal]:g_dd	0.384	0.294	0.229	1.306	0.192

Table 71: SMoE estimates: Bitcoin (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	0.280	0.063	0.061	4.415	0.000
beta[trend]:mom20	0.308	0.116	0.075	2.645	0.008
beta[trend]:mom120	0.086	0.076	0.055	1.136	0.256
beta[reversal]:const	-0.014	0.041	0.035	-0.352	0.725
beta[reversal]:rev1	-0.268	0.041	0.045	-6.588	0.000
beta[reversal]:rev5	-0.175	0.048	0.046	-3.675	0.000
beta[crisis]:const	-0.262	0.508	0.395	-0.516	0.606
delta[trend]:const	2.041	0.200	0.105	10.203	0.000
delta[trend]:lv1	0.246	0.110	0.086	2.250	0.024
delta[trend]:lv5	0.036	0.125	0.110	0.292	0.770
delta[trend]:lv22	0.761	0.106	0.092	7.185	0.000
delta[reversal]:const	0.040	0.349	0.203	0.114	0.909
delta[reversal]:lv1	0.206	0.222	0.161	0.925	0.355
delta[reversal]:lv5	0.511	0.215	0.191	2.376	0.018
delta[reversal]:lv22	0.787	0.251	0.202	3.137	0.002
delta[crisis]:const	3.848	0.303	0.136	12.717	0.000
delta[crisis]:lv1	-0.023	0.163	0.134	-0.142	0.887
delta[crisis]:lv5	0.614	0.314	0.177	1.957	0.050
delta[crisis]:lv22	-0.252	0.224	0.162	-1.126	0.260
gamma[trend]:const	1.640	0.486	0.256	3.373	0.001
gamma[trend]:g_lvol	0.023	0.224	0.166	0.104	0.917
gamma[trend]:g_mom60	-0.424	0.157	0.146	-2.701	0.007
gamma[trend]:g_dd	0.253	0.126	0.143	2.010	0.044
gamma[reversal]:const	1.062	0.629	0.308	1.688	0.091
gamma[reversal]:g_lvol	0.186	0.262	0.187	0.707	0.480
gamma[reversal]:g_mom60	-0.610	0.155	0.139	-3.945	0.000
gamma[reversal]:g_dd	0.349	0.129	0.133	2.699	0.007

Table 72: SMoE estimates: CSI 300 ETF (510300) (development sample)

Parameter	Estimate	SE (sandwich)	SE (model)	z	p
beta[trend]:const	-0.690	0.307	0.193	-2.245	0.025
beta[trend]:mom20	0.249	0.247	0.167	1.010	0.312
beta[trend]:mom120	0.150	0.352	0.205	0.425	0.671
beta[reversal]:const	0.001	0.047	0.038	0.020	0.984
beta[reversal]:rev1	-0.006	0.027	0.029	-0.226	0.821
beta[reversal]:rev5	-0.094	0.061	0.040	-1.532	0.126
beta[crisis]:const	0.194	0.097	0.069	2.011	0.044
delta[trend]:const	0.300	0.410	0.292	0.732	0.464
delta[trend]:lv1	1.253	0.581	0.427	2.156	0.031
delta[trend]:lv5	0.593	0.560	0.492	1.058	0.290
delta[trend]:lv22	-0.751	0.556	0.450	-1.352	0.176
delta[reversal]:const	-0.798	0.464	0.214	-1.719	0.086
delta[reversal]:lv1	-0.071	0.112	0.105	-0.636	0.525
delta[reversal]:lv5	0.376	0.330	0.190	1.142	0.253
delta[reversal]:lv22	0.644	0.199	0.159	3.243	0.001
delta[crisis]:const	0.794	0.219	0.112	3.626	0.000
delta[crisis]:lv1	0.145	0.098	0.095	1.490	0.136
delta[crisis]:lv5	-0.163	0.145	0.131	-1.128	0.259
delta[crisis]:lv22	1.037	0.232	0.146	4.465	0.000
gamma[trend]:const	-1.449	0.730	0.484	-1.986	0.047
gamma[trend]:g_lvol	0.219	0.349	0.300	0.628	0.530
gamma[trend]:g_mom60	-0.403	0.358	0.269	-1.126	0.260
gamma[trend]:g_dd	0.440	0.296	0.295	1.484	0.138
gamma[reversal]:const	0.157	0.786	0.366	0.200	0.842
gamma[reversal]:g_lvol	0.523	0.513	0.252	1.020	0.308
gamma[reversal]:g_mom60	-0.128	0.168	0.128	-0.764	0.445
gamma[reversal]:g_dd	-0.027	0.161	0.125	-0.166	0.868